

Trench migration, slab bending, and mantle flow at subduction zones

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Abstract

The subduction of dense and stiff tectonic plates into the mantle exerts a fundamental control on plate velocities and flow in the upper mantle. On Earth, a broad spectrum of subduction styles are apparent, both in terms of slab morphology (e.g., slab dip angles) and slab kinematics (e.g., trench migration rates). The development of a dynamic framework to explain the origin of various types of slab kinematics is required both to understand the temporal evolution of the integrated plate tectonic system, and to use plate kinematic observations to infer the mechanical properties of subduction systems (e.g., lithospheric plate rheology). We use dynamically evolving numerical models of regional subduction to isolate the first order physical processes governing the subducting slab dynamics, particularly the lateral migration of slabs, and the bending of plates during subduction. We focus particularly on the role of the overriding plate, and mantle rheology, on dictating the slab kinematics observed: two features often neglected in dynamically evolving subduction models. Chapter 1 explores the control that the mechanical properties of the overriding plate can exert on subducting slab evolution, and introduces the general numerical modeling strategy adopted throughout. We find that the structure of the overriding plate exerts a strong control on subducting slab evolution, and therefore suggest that the wide range of overriding plate styles on Earth potentially induces a significant amount of the observed variability in slab rollback rates and slab morphologies. In chapter 2, we utilize global estimates of subducting plate curvature, in conjunction with 2-D numerical models, to investigate subducting plate rheology. We conclude that plastic weakening in the subducting plate, in conjunction with overriding plate structure, is required to generate the slab curvature systematics on Earth. In chapter 3, we turn our attention from lithospheric to asthenospheric rheology, in order to address the role of a power-law mantle viscosity on rates of trench migration. A power-law mantle rheology reduces the mantle viscosity directly adjacent to the plates, where mantle shear stress is greatest, an effect which reduces the relative importance of trench retreat. In addition to

the inclusion of an overriding plate, this provides a mechanism for reducing the high average retreat rates, relative to Earth, often observed in subduction modeling studies. In chapter 4, we extend our subduction models to move beyond passive overriding plates, and consider far-field forces transmitted from proximal subduction zones. We substantiate the suggestion that slab interactions at “double slab” systems can induce non-standard plate kinematics, particularly advancing trench motion and rapid plate convergence. In addition, we lay out the subduction configurations within which conventional “single slab” behavior, or a variant of it, is exhibited, and conversely where such single slab physics is not applicable. Developing an understanding of the components of single slab subduction dynamics that are applicable is an important step towards a physics-based parameterization of such complex subduction systems in plate tectonic reconstructions, where geological constraints are often sparse.

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Introduction

The subduction of dense and stiff oceanic plates into the sublithospheric mantle constitutes the primary driving force for plate tectonics on Earth. Within the global tectonic system, subduction embodies the local interplay between the negative buoyancy of the slab, and subduction-resisting forces associated with mantle flow, slab bending, and the subduction interface. Thus, for a particular slab, the mode of subduction depends on the relative magnitudes of these force components, and how they evolve through time. This thesis is comprised of a suite of studies which utilize time-evolving, numerical models to untangle the relative magnitudes of these forces, and understand how they manifest in plate kinematics (e.g., lateral trench motions). Such an understanding is required in order to use plate kinematic observations to inform our assessment of the dynamics and mechanical properties (e.g., plate strength) of the subduction system.

We focus particularly on trench migration (chapters 1, 3, and 4) and slab curvature (chapter 2): kinematic expressions of complex subduction processes that have first order importance in dictating the global tectonic system. Trench migration constitutes the lateral migration of subduction zone trenches, which has direct consequences for the style of back-arc deformation, and slab curvature exerts a key control on the amount of gravitational energy dissipated during plate bending. In this collection of studies, we have both identified additional dynamic controls on these processes, and refined our understanding of mechanisms previously proposed. In particular, much of this work argues for the importance of the overriding plate in dictating these aspects of subduction (chapters 1, 2, and 4). Over the last ~40 years, the development of increasingly complex geodynamic subduction models has enabled significant progress to be made towards a holistic understanding of the subduction process. However, only during the last decade or so have dynamic subduction models began to consider “two-plate” subduction, where the overriding plate is included and deforms according to the subduction slab and the mantle flow it induces, and only during the last few years has this become routine in 3-D. We therefore utilize these advances in all of our studies, and devote significant effort to characterizing the dynamic role of an overriding plate in the subduction processes. By modifying the magnitude of the lifting force associated with negative pressure in the mantle wedge, we find that the properties of the overriding plate (e.g., lithospheric thickness) can exert a strong control on both rollback rates and slab curvature. Additionally, the overriding plate acts as a stress-guide within which tectonic forces can propagate

between plate margins, and therefore acts as a dynamic link between multiple subduction zones (chapter 4). A long-term goal involves developing a physical basis for understanding how multiple subduction zones can interact, and evolve in unison, in a global setting. Building on the work presented in the previous chapters, chapter 4 extends our 3-D models to consider multiple subduction zones, which is a key step prior to moving to a global network of plate boundaries. While the four studies presented have the overarching themes discussed, they are somewhat distinct in terms of the specific aspect of the subduction system that they target, and so they are summarized separately in the following text.

Chapter 1: Motivated by the wide range of overriding plate styles on Earth, we use 2-D numerical models to explore the control that the mechanical properties of the overriding plate can exert on trench migration. We additionally probe the relationship between trench migration and the overriding plate stress state, an analog for overriding plate deformation (e.g., mountain building, rifting). We find that a significant amount of the observed variability in slab rollback rates and morphologies observed on Earth can potentially be attributed to overriding plate structure. Additionally, we highlight the temporal correspondence between variation in the rates of trench migration induced by slab interaction with a strong lower mantle, and overriding plate stress state. This wide-ranging subduction dynamics study sheds light on the (not so simple) relationship between trench motions and overriding plate stress state, and will aid researchers in determining the physical validity of proposed styles of slab/trench evolution.

Chapter 2: This study investigates the evolution of subducting plate curvature. In striving to understand the balance of forces acting on an individual slab, subducting plate curvature is a key component because it exerts a strong control on the magnitude of the subduction-resisting bending force. As the forces associated with bending take on variable functional forms for different plate rheologies, and curvature is a manifestation of slab bending, observations of slab curvature should contain information about subducting plate rheology. By augmenting our 2-D numerical models with global correlations of previously published data-sets, we show that rheological weakening of the slab bending region, in conjunction with overriding plate structure, is required to generate the slab curvature systematics on Earth. We propose that, through controlling the depth extent of the mantle wedge pressure induced “lifting force”, the overriding plate thickness is an important component, and that

a sufficiently weak slab is required to respond to such lifting force variation. This is in line with the apparent lack of a subducting plate thickness control on slab curvature, and further points toward a prominent role for the overriding plate in the global subduction system.

Chapter 3: This chapter addresses the effect that a power-law mantle rheology has on rates of trench migration. 3-D, numerical modeling studies that investigate the factors which dictate the rates, and style, of trench motion typically consider an isoviscous mantle rheology. However, laboratory experiments and shear wave splitting measurements indicate that large regions of the upper mantle deform within the dislocation creep regime, where stress and strain rate follow a power-law relationship. A power-law mantle rheology induces relative viscosity reductions within the asthenosphere, particularly directly adjacent to the lithospheric plates where the mantle is under high shear stress. For a range of plate geometries, we find that this acts to reduce the rate of trench retreat, relative to subducting plate motion. We show that this plate motion “preference” can be understood, to first order, by considering how the magnitudes of forces that act to resist both trench retreat and plate motion change with the inclusion of this additional rheological complexity. We suggest that the inclusion of a power-law mantle viscosity may be of importance in reducing the high relative trench retreat rates often observed in numerical models to levels more in line with present day Earth.

Chapter 4: Lastly, we consider the influence of far-field forces propagated from one subduction zone to another. Having established the importance of even a passive overriding plate in terms of subducting slab dynamics, we now explore the dynamic effects of an overriding plate which is attached to an additional subduction zone. In addition to this configuration within which both slabs dip in the same direction, which is applicable to the Izu-Bonin-Marianas and Ryuku double slab subduction system, we consider alternative dip polarity configurations. We lay out the subduction configurations within which conventional “single slab” behavior, or a variant of it, is exhibited, and conversely where such single slab physics is not applicable. Such single slab behavior breaks down for the rear slabs in double slab systems with equivalent directions, within which enigmatic plate kinematics such as trench advance (present day: Izu-Bonin-Marianas) and anomalously high convergence rates (late Cretaceous: India-Eurasia) have been attributed to slab-slab interactions. In our models, we find that such phenomena can be attributed to the transmission of slab pull across the

plate boundary, and we augment our fully dynamic models with instantaneous computations in order to isolate the key controls on slab-slab force transmission.

Chapter 1

Trench migration and overriding plate stress in dynamic subduction models

1.1 Summary

On Earth, oceanic plates subduct beneath a variety of overriding plate (OP) styles, from relatively thin and negatively buoyant oceanic OPs to thick and neutrally/positively buoyant continental OPs. The inclusion of an OP in numerical models of self-consistent subduction has been shown to reduce the rate that subducting slabs roll back relative to the equivalent single plate models. We use dynamic, 2-D subduction models to investigate how the mechanical properties, namely viscosity, thickness, and density, of the OP modify the slab rollback rate and the state of stress of the OP. In addition, we examine the role of the subducting plate (SP) viscosity. Because OP deformation accommodates the difference between the slab rollback rate and the far-field OP velocity, we find that the temporal variations in the rollback rate results in temporal variations in OP stress. The slabs in our models roll back rapidly until they reach the lower mantle viscosity increase, at which point the rollback velocity decreases. Concurrent with this reduction in rollback rate is a switch from an OP dominated by extensional stresses to a compressional OP. As in single plate models, the vis-

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cosity of the SP exerts a strong control on subducting slab kinematics; weaker slabs exhibit elevated sinking velocities and rollback rates. The SP viscosity also exerts a strong control on the OP stress regime. Weak slabs, either due to reduced bulk viscosity or stress-dependent weakening, have compressional OPs, while strong slabs have dominantly extensional OPs. While varying the viscosity of the OP alone does not substantially affect the OP stress state, we find that the OP thickness and buoyancy plays a substantial role in dictating the rate of slab rollback and OP stress state. Models with thick and/or negatively buoyant OPs have reduced rollback rates, and increased slab dip angles, relative to slabs with thin and/or positively buoyant OPs. Such elevated trench rollback for models with positively buoyant OPs induces extensional stresses in the OP, while OPs that are strongly negatively buoyant are under compression. While rollback is driven by the negative buoyancy of the subducting slab in such models of free subduction, we conclude that the physical properties of the OP potentially play a significant role in modulating both rollback rates and OP deformation style on Earth.

1.2 Introduction

The subduction of a negatively buoyant lithospheric plate below an overriding plate (OP) and into the sub-lithospheric mantle is one of the fundamental components of plate tectonics and yet the degree of control that the mechanical properties of the OP have on the dynamics of the subducting plate (SP) is not entirely clear. Subduction of oceanic plates of various ages occurs beneath OPs with a broad range of mechanical properties, with continental OPs typically thicker and more positively buoyant than their oceanic counterparts. Tomographic images of subduction zones show that slabs subduct with a wide range of upper-mantle dip angles and exhibit a wide variety of deformation styles upon reaching the transition zone, from slabs that appear to flatten or fold (e.g. Japan) to slabs that appear to extend deep into the lower mantle with a wide variety of dip angles (e.g. Central America) (e.g. Li et al., 2008). Dynamic modeling studies that consider the SP in isolation show that the viscosity of the subducting slab, relative to that of the surrounding mantle, exerts a strong control on slab evolution, particularly the rate at which the slab hinge rolls back (e.g. Funicello et al., 2003b; Schellart, 2004; Bellahsen et al., 2005; Enns et al., 2005; Stegman et al.,

2006; Di Giuseppe et al., 2008; Ribe, 2010). The rollback rate significantly affects subducting slab morphology, with slabs that rapidly roll back typically having reduced dips (Uyeda and Kanamori, 1979; Garfunkel et al., 1986; Kincaid and Olson, 1987; Griffiths et al., 1995; Christensen, 1996). If subducting slabs have sufficiently low viscosity and density, such reduced dips result in an elevated propensity for slab flattening in the transition zone (e.g. Davies, 1995; Christensen, 1996; Čížková et al., 2002; Enns et al., 2005; Ribe, 2010). While rollback is driven by the negative buoyancy of the subducting slab in free subduction models, it has been shown that the presence of an OP significantly reduces slab rollback velocities (e.g. Yamato et al., 2009; Butterworth et al., 2012). Motivated by the wide range of OPs present on Earth, we investigate the degree to which variable OP properties (density, thickness, viscosity) can contribute to explaining the wide range of rollback rates, and slabs dips, observed.

In single plate models, relatively high viscosity slabs exhibit elevated rollback, lower viscosity slabs have near-stationary trenches, and intermediate viscosity slabs (factor ~ 100 to 10,000 more viscous than mantle), such as those modeled here, exhibit a wide range of rollback styles/rates (e.g. Funiciello et al., 2003b; Schellart, 2004; Bellahsen et al., 2005; Enns et al., 2005; Stegman et al., 2006; Di Giuseppe et al., 2008; Funiciello et al., 2008; Faccenna et al., 2007; Ribe, 2010). The dip of the subducting slab as it impinges on the base of the model domain or a viscosity discontinuity, which is related to the amount of rollback that occurs before the slab impinges on the lower mantle, is then thought to dictate the subsequent style of slab evolution (Di Giuseppe et al., 2008; Schellart, 2008a; Ribe, 2010). In addition to explicitly examining the how the OP mechanical properties affects subduction dynamics, we investigate how lithospheric viscosity affects subduction dynamics in two-plate models, thereby taking into account the viscosity of both the SP and OP.

After examining subducting slab dynamics for uniformly strong (factor 2000 plate-mantle viscosity contrast) and weak (factor 100 viscosity contrast) lithospheric plates, we investigate the role of stress-dependent viscosities. Laboratory experiments predict that deformation in high stress regions is dominated by dislocation creep, in which the effective viscosity has a power law dependence on stress (e.g. Karato and Wu, 1993; Hirth and Kohlstedt, 2003). Models that do include an overriding plate, yet restrict trench motion, show that the inclusion of a stress-dependent rheology decreases the viscosity of the mantle wedge which reduces hydrodynamic stresses on the upper surface of the slab, and so increases the dip angle

(Billen and Hirth, 2005, 2007). Additionally, the inclusion of pseudo-plastic behavior, with a yield stress that is sufficiently low to weaken the portion of the slab that bends at the 660 km viscosity jump, has been shown to prohibit the penetration of a slab into the lower mantle (Čížková et al., 2002, 2007; Čížková and Bina, 2013). Because we allow the trench to migrate freely, this study allows the stress-dependent rheology to exert a control on trench migration, and so the migration-related component of slab dip.

The inclusion of an overriding plate, coupled to the subducting plate by a weak crust and mantle flow tractions, enables an investigation into how the lithospheric properties affect the stresses induced in the OP by subduction (e.g. Clark et al., 2008; Capitanio et al., 2010a, 2011; Schellart and Moresi, 2013). On Earth, large scale OP deformation of the overriding plate ranges from the compressional “Chilean type” to the extensional “Marianas type” (Uyeda and Kanamori, 1979). A link between the OP stress regime and slab rollback has been suggested based on kinematic constraints, in that OP deformation must accommodate the disparity between rollback and far-field OP motion (e.g. Jarrard, 1986). OP extension is expected in the case of purely slab-buoyancy driven trench retreat with a passive OP (e.g. Elsasser, 1971; Molnar and Atwater, 1978; Schellart, 2008b), whereas a component of OP compression is expected where the OP is driving, or at least contributing to, the trench retreat (e.g. Jarrard, 1986; Heuret and Lallemand, 2005; Heuret et al., 2007). Because there are exceptions to both end-members on Earth, it appears that the complexity of subduction zones prohibits such a single overarching mechanism for large scale subduction-induced OP deformation (e.g. Carlson and Melia, 1984; Sdrolias and Müller, 2006; Schellart, 2008b). Even in free subduction models, in which the OP is not externally pushed towards the SP, near-trench OP compression is observed Capitanio et al. (2010a, 2011); Schellart and Moresi (2013). This is due to subduction-induced flow in the mantle wedge, and suggests a departure from either the slab-driven or OP-driven idealized models, even when far-field forcing effects are neglected. We examine how lithospheric rheology and OP mechanical properties can, even in the absence of external forces, influence the slab-driven OP stress regime, and contribute to the generation of the range of OP deformation regimes observed on Earth.

We investigate the controls on subducting slab evolution, and OP stress, using fully dynamic, 2-D subduction models. Time-progressive numerical modeling studies of subduction with an OP often kinematically constrain the subducting slab by imposing plate/mantle

velocities (e.g. van Hunen et al., 2000; Čížková et al., 2002; Billen and Hirth, 2005; van Dinther et al., 2010; Ghazian and Buiter, 2013) and/or fix the position or shape of the weak zone/fault which decouples the two plates (e.g. Gurnis and Hager, 1988; King and Hager, 1994; Čížková et al., 2002; Běhouňková and Čížková, 2008; van Hunen and Allen, 2011; Bottrill et al., 2012). Fixing the trench position inhibits slab rollback/advance and so places a non-physical constraint on the evolution of the slab, and prescribing a surface velocity can add artificial energy to the system if the imposed velocity/viscosity is not consistent with that derived from the purely buoyancy-driven problem (Han and Gurnis, 1999). This study aims to improve our understanding of the role of the OP using numerical models in which flow is driven entirely by buoyancy forces. The free-trench models of Zhong and Gurnis (1995) demonstrate that the rate of trench rollback decreases as slabs penetrate into the lower mantle, and the dynamic models of Sharples et al. (2014) show that slabs with thicker OPs resist trench retreat more than those with thin OPs. Recent work by Garel et al. (2014) shows that, using dynamic two-plate thermo-mechanical models, the wide range of slab morphologies observed on Earth can be reproduced with various SP-OP age combinations and a upper-lower mantle viscosity increase of factor 30. We use a simplified rheological setup relative to Garel et al. (2014) and, instead of testing the role of SP/OP plate age (i.e. all mechanical properties dictated by age-dependent thermal structure), we vary the lithospheric viscosity, thickness, and density of the OP individually, in order to illuminate the various mechanisms potentially operating on Earth.

1.3 Model description

1.3.1 Numerical method and model setup

We use the finite-element code CitcomCU to solve the equations governing convection in an incompressible (Boussinesq approximation), viscous fluid with negligible inertia and, in our case, zero internal heating (e.g. Zhong, 2006; Moresi and Gurnis, 1996). The equations solved – the conservation of mass, momentum and energy – are given in non-dimensional form below:

$$\nabla \cdot \mathbf{u} = 0 \quad (1.1)$$

$$\nabla p + \nabla \cdot (\eta(\nabla \mathbf{u} + \nabla \mathbf{u}^T)) + RaT\mathbf{e}_z = 0 \quad (1.2)$$

$$\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T = \nabla^2 T \quad (1.3)$$

where $\mathbf{u}$ is velocity, p is dynamic pressure, η is viscosity, T is temperature, $\mathbf{e}_z$ is a unit vector in the vertical direction, and Ra is the Rayleigh number given by:

$$Ra = \frac{\alpha \rho_0 g \Delta T h^3}{\kappa \eta_0}. \quad (1.4)$$

We use $Ra = 1.13 \times 10^7$ for all models which corresponds to a temperature contrast of 1300 K, a reference viscosity of 1×10^{20} Pa s, and the material parameters given in Table 1.1. Because the lithospheric plates are defined by a constant temperature, as opposed to half-space cooling, we use a reduced thermal expansivity (α) value of $1.5 \times 10^{-5} \text{ K}^{-1}$ in order to give a reasonable bulk lithosphere-mantle density contrast of 65 kg/m^3 . The domain size is $7920 \text{ km} \times 1320 \text{ km} \times 15 \text{ km}$ (i.e. the computation is effectively 2-D). Both the subducting and overriding plates have initial uniform thicknesses of 80 km, and uniform initial temperature of 273 K (T_0) (Fig. 1.1). In all models, subduction is initiated by prescribing an asymmetric lithospheric geometry in the trench region which provides the initial instability required for the lithosphere to freely subduct. This initial proto-slab extends to a depth of 150 km and has a radius of curvature of 200 km. The trench is initially located at $x = 5000 \text{ km}$ and the plates extend to a distance of 500 km from the sidewalls. We also investigate slab dynamics in models where the overriding plate (OP) trailing edge is attached to the sidewall. Such models are termed “fixed OP” models throughout the manuscript.

To facilitate decoupling of the subducting plate (SP) from the OP, and the free-slip boundary, we insert a 15 km thick, weak crustal layer within the SP which eradicates the need for plastic weakening at the trench (although the dynamic consequences of Byerlee yielding are examined in section 1.4.4) (e.g. Běhounková and Čížková, 2008; Stegman et al., 2010; Quinquis et al., 2011). The crust extends along the whole length of the SP and has a constant viscosity of $0.2\eta_{mant}$ ($= 2 \times 10^{19} \text{ Pa s}$). By holding the crustal properties constant throughout this study, we interpret variations in slab behavior to be a result of the other parameters/rheologies explored (e.g. Androvičová et al., 2013). We have tested crustal viscosities below ($\eta_c = \eta_{mant}/25$) and above ($\eta_c = \eta_{mant}$) the chosen value, and find that reducing the viscosity increases both the subducting plate and trench retreat velocities, but does not sig-

Table 1.1: Model parameters

Symbol	Parameter	Reference (range)	Units
h_l	lithospheric thickness	80 (40-120)	km
h_c	crustal thickness	15	km
w_u	overriding plate length	2200	km
w_s	subducting plate length	4750	km
h	domain height	1320	km
w	domain width	7920	km
T_0	surface/lithospheric temperature	273	K
T_m	mantle temperature	1573	K
ρ_0	reference density	3300	kg/m ³
α	thermal expansivity	1.5×10^{-5}	K ⁻¹
κ	thermal diffusivity	10^{-6}	m ² /s
η_0 (η_{mant})	reference/mantle viscosity	10^{20}	Pa s
η_c	crust viscosity	2×10^{19}	Pa s
E	Frank-Kamenetskii parameter	6.21 (4.61-7.60)	-
n	power law exponent	1 (1, 3.5)	-
σ_T	power law transition stress	100 (50-150)	MPa
b	Byerlee cohesion	60	MPa
a	Byerlee friction coefficient	0.6	-
λ	Byerlee pore pressure factor	0.15	-

nificantly modify the relative partitioning of subduction between these two components. Additionally, the morphological evolution of the slab is comparable for the crustal viscosities tested. We do not include a crust within the OP as we find that this typically gives rise to two-sided subduction (i.e. the OP detaches from the upper boundary). In the few models with an OP crust that do subduct in a one-sided fashion, we find that the slab dynamics, and large-scale OP stress state, are unchanged relative to the models with only a SP crust. Therefore, this approximation allows us focus on the processes associated with mature subduction, as opposed to the setup necessary to create the one-sided subduction observed on Earth.

The crust is defined using compositional tracers. Crustal properties are attributed to nodes that are above a certain compositional threshold, C_t (i.e. if $C > C_t = 0.2$). Below 200 km depth, the crust is linearly tapered to zero thickness at mid-mantle depths. This tapering, as opposed to a sudden cutoff, is done to limit artificial stress discontinuities in the subducting lithosphere and is implemented by linearly increasing C_t from 0.2 at 200 km depth, to 1 at 660 km depth. The compositional material has no additional density component and is

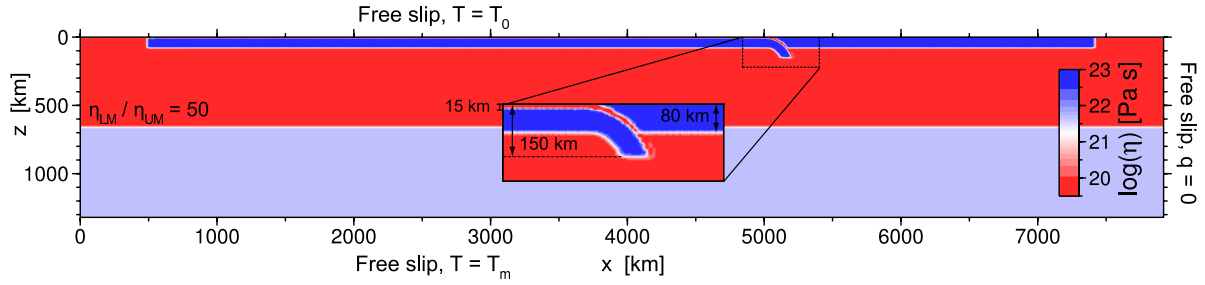


Figure 1.1: Initial viscosity field for the reference model (see Table 1.1 for reference parameters).

advected, using a marker tracing method, in the induced flow field:

$$\frac{\partial C}{\partial t} + \mathbf{u} \cdot \nabla C = 0. \quad (1.5)$$

40 compositional markers are initially contained within each of the cubic elements, and element dimensions vary from 5 km in the upper 330 km of the domain to 15 km elsewhere. We have tested that this element size is sufficient by ensuring that reducing the element size, by a factor of 2, does not modify the observed systematics, and has only a minor effect on the maximum sinking velocity (4% increase).

The upper boundary has a constant temperature of 273 K (T_0) and the base and sides have zero heat flux. Mechanically, all boundaries are free slip (reflecting). Such sidewall boundary conditions cause lateral confinement of the mantle flow which has been shown to promote reduced trench rollback, and slab folding atop a viscosity discontinuity, relative to models with periodic (Enns et al., 2005) or open side boundary conditions (Chertova et al., 2012). The use of a free slip upper boundary is an approximation to the true deformable free surface on Earth, which simplifies the numerics. This is likely to have only a minimal effect on slab dynamics as it has been shown that the use of a significantly weak crust allows the subducting lithosphere to decouple from the surface, and bend, and so exhibit a similar behavior to the equivalent model with a free surface (Quinquis et al., 2011).

1.3.2 Rheology

The first section of this work considers subduction in a setting where viscosity is solely a function of temperature. While experimental work (e.g. Karato and Wu, 1993; Hirth

and Kohlstedt, 2003) suggests a complex dependence of viscosity on many parameters including pressure, water content, and grain size, we use a simple rheology in order to attempt to isolate and identify first-order effects on slab dynamics. We approximate the viscosity temperature-dependence as Frank-Kamenetskii (1969),

$$\eta_N = \eta_0 \exp(E(T_R - T)), \quad (1.6)$$

where T is the non-dimensional temperature, which varies between 0 and 1, and T_R is the non-dimensional reference temperature of 1. Our reference models use $E = 6.2$, which gives a maximum viscosity contrast, between the hottest and coldest material, of 500. Initially the lithosphere has a uniform temperature and so is isoviscous, as in compositional models. As the slab subducts, thermal diffusion causes the edges of the subducting slab to heat, and so weaken, while the core remains cool and retains its initial strength. The effective viscosity contrast during the free-sinking phase (e.g. Fig. 1.2c), taken as the viscosity contrast averaged over all nodes with $T < 0.5$, is reduced to ~ 350 which is compatible with the range of effective viscosity contrasts of 100-500 estimated from observables such as minimum radii of curvature, and plate-trench kinematics (Conrad and Hager, 1999; Becker et al., 1999; Funiciello et al., 2008; Wu et al., 2008; Stegman et al., 2010).

The use of a Frank-Kamenetskii rheology with such a low E is a significant approximation to laboratory-derived Arrhenius rheologies which would suggest that lithospheric viscosities are ~ 20 orders of magnitude greater than the underlying mantle (equivalent $E \sim 45$) (e.g. Frank-Kamenetskii, 1969; Karato and Wu, 1993; Hirth and Kohlstedt, 2003). Such high lithospheric strengths are, however, likely limited by plastic failure. In our models, including a rheology with an equivalently strong temperature-dependence, and a maximum viscosity cut-off, gives rise to issues regarding the obtainment of stable subduction. The subducting slab cools the region of the mantle wedge adjacent to the subducting crust which, for a strongly temperature-dependent viscosity, results in strong material above weak crust connecting the two plates. This results in sluggish plate motions, and an OP stress state that is affected by the gravitational pull of the additional material located above the crust. At the expense of limiting rheological realism, we avoid these non-physical effects by reducing the temperature-dependence of the viscosity. Our simplified rheology allows us to easily probe the first order effects of various lithospheric properties, without slab stability issues.

We also examine the dynamic effects of a stress-dependent viscosity, in which there is a power law relationship between stress and strain rate in the upper mantle ($z < 660$ km). For simplicity, we assume that the power law viscosity has an exponential temperature dependence with an exponential factor equivalent to that of the Newtonian viscosity (E) divided by the power law exponent (n). This allows the power law viscosity (η_P) to be written in terms of the Newtonian viscosity (η_N):

$$\eta_P = k_P \dot{\epsilon}_{II}^{\frac{1-n}{n}} \eta_N^{\frac{1}{n}}. \quad (1.7)$$

Here, $\dot{\epsilon}_{II}$ is the second invariant of the strain rate tensor, and k_P is a constant pre-factor. We use $n = 3.5$ to approximate the stress-dependence of viscosity in the dislocation creep regime. Prescribing a transition stress, σ_T , at which the power law and Newtonian strain rates are equal allows the constant in eq. 1.7, k_P , to be expressed as a function of σ_T and n . Assuming the total strain rate is the sum of the Newtonian and power law components, a joint Newtonian/power law viscosity is then computed as follows:

$$\frac{1}{\eta_{eff}} = \frac{1}{\eta_N} + \frac{1}{\eta_P}. \quad (1.8)$$

Inserting the two viscosity components into eq. 1.8 then gives an effective viscosity (η_{eff}) which is dominantly Newtonian eq. 1.6 at stresses below σ_T , and power law (eq. 1.7) at stresses above σ_T :

$$\eta_{eff} = \frac{\eta_N}{1 + \left(\frac{2\dot{\epsilon}\eta_N}{\sigma_T} \right)^{1-\frac{1}{n}}}. \quad (1.9)$$

We also test how the inclusion of a visco-plastic rheology, which approximates brittle deformation in the lithosphere (e.g. Schott and Schmeling, 1998), modifies slab dynamics relative to the purely viscous reference. In this case, the effective viscosity is composed of a Newtonian (eq. 1.6) and a plastic term (eq. 1.11):

$$\frac{1}{\eta_{eff}} = \frac{1}{\eta_N} + \frac{1}{\eta_{yield}}, \quad (1.10)$$

where the plastic viscosity is calculated as follows:

$$\eta_{yield} = \frac{\tau_{yield}}{2\dot{\epsilon}_{II}}. \quad (1.11)$$

The yield stress, τ_{yield} , is defined by Byerlee's friction relationship (Byerlee, 1968):

$$\tau_{yield} = (a\sigma_l + b)\lambda, \quad (1.12)$$

where σ_l is the lithostatic pressure ($= \rho_0 g z$), b and a are constants (see Table 1.1), and λ is a pore pressure pre-factor which is not well constrained. We restrict plastic yielding to lithospheric depths ($z < 80$ km), and use $\lambda = 0.15$. This value of the pre-factor results in significant plastic weakening, yet does not weaken the trench region completely, and is comparable to that favored in previous studies (e.g. Enns et al. (2005): 0.1, Di Giuseppe et al. (2008): 0.08).

1.4 Results

Firstly, we present our reference model with a Newtonian rheology and characterize the effect of the overriding plate (OP), and upper-lower mantle viscosity contrast on the evolution of the subducting slab. We then explore the role of the lithospheric-sublithospheric mantle viscosity contrast, and stress-dependent rheologies, on our reference model with a mobile OP. Lastly, for the Newtonian reference model, we focus on the role of the OP mechanical properties for models with both free and fixed OPs.

1.4.1 Reference model and role of viscosity stratification

The morphological evolution of the subducting slab is affected by the vertical viscosity structure. Figure 1.2 shows the reference model with a uniform viscosity mantle (top panel) compared with two other models in which the viscosity increases by a factor of 50 (middle panel) and 100 (lower panel) at the transition between the upper and lower mantle ($z = 660$ km). The temporal evolution of the slab sinking (V_Z), trench retreat (V_T), and subducting plate velocities (V_P) are shown in Figure 1.3, for the models with mid-mantle viscosity increases of factors 50 (green line) and 100 (blue). The uniform viscosity mantle model is omitted from the kinematics plot, as it requires a vastly stretched velocity axis (i.e. maximum $V_Z = 33$ cm/yr). Also shown in Figure 1.3 are the kinematics for models with fixed (black) and absent OPs (red), which are discussed in the subsequent section. V_T is computed by tracking the rightmost location of the crustal material at the surface, and V_P is computed by tracking the position of the trailing edge of the subducting plate.

Irrespective of the vertical viscosity structure of the mantle (Fig. 1.2), the slab initially sinks freely and the lithosphere unbends to increase the radius of curvature initially imposed. The sinking velocity increases due to the increasing slab pull force, until the slab begins to

feel the viscosity increase/lower boundary (Fig. 1.3a). At this time, the slab begins to slow and curve upwards due to the viscous drag exerted by the lower mantle/boundary, which causes a reduction in the upper-mantle slab dip angle. In the interest of conciseness, we subsequently refer to the average upper mantle slab dip angle as simply the slab ‘dip’. The imposed lithosphere-mantle viscosity contrast is too low to favor a subduction mode in which the trench advances, and so the trench retreats throughout subduction (Ribe, 2010; Stegman et al., 2010). Subduction is partitioned approximately equally between trench retreat and plate advance, with both components increasing as the magnitude of the slab pull force increases during the free-sinking phase (Fig. 1.3b). Because the slab is retreating rapidly during its descent through the upper mantle ($V_T \sim 5$ cm/yr), it has a substantially inclined slab dip (Fig. 1.2c, e). In addition to the slab’s reasonably low viscosity, and the absence of phase transition-related density effects, this low dip angle causes the slab to flatten horizontally above the viscosity discontinuity, and then slowly sink into the lower mantle (Fig. 1.2d) (e.g. Christensen, 1996; Čížková et al., 2002; Enns et al., 2005). Upon interaction of the slab tip with the viscosity increase/lower boundary, both the plate advance and trench retreat velocities decrease.

The evolution of the OP, and SP bending region, stress field for models with a uniform viscosity mantle, and with a factor 50 viscosity increase, are shown in the top four rows of Figure 1.4. As we mainly focus on the stress state of the flat-lying OP, we plot the horizontal deviatoric stress. While the amplitudes and spatial positions of the highly stressed regions vary, the style of OP stress is comparable for the free OP models with and without a mid-mantle viscosity increase (Figs. 1.4a-g). The highest stresses, of >50 MPa, are found within the lithosphere, particularly within the trench bending region where there is extension above, and compression below, the horizontal mid-plane of the subducting lithosphere. Comparably large stresses also occur below the trench where down-dip extension occurs due to the slab pull force. In the OP, stresses are maximum in the fore-arc region where horizontal compression occurs, and at distances of greater than ~ 300 km from the plate interface where extension occurs (e.g. Tagawa et al., 2007a; Capitanio et al., 2010a; Nakakuki et al., 2010; Schellart and Moresi, 2013).

At the time-step prior to interaction with the viscosity discontinuity, both models have fore-arc compression (layered model: ~ 10 MPa, non-layered: ~ 20 MPa) in the OP at

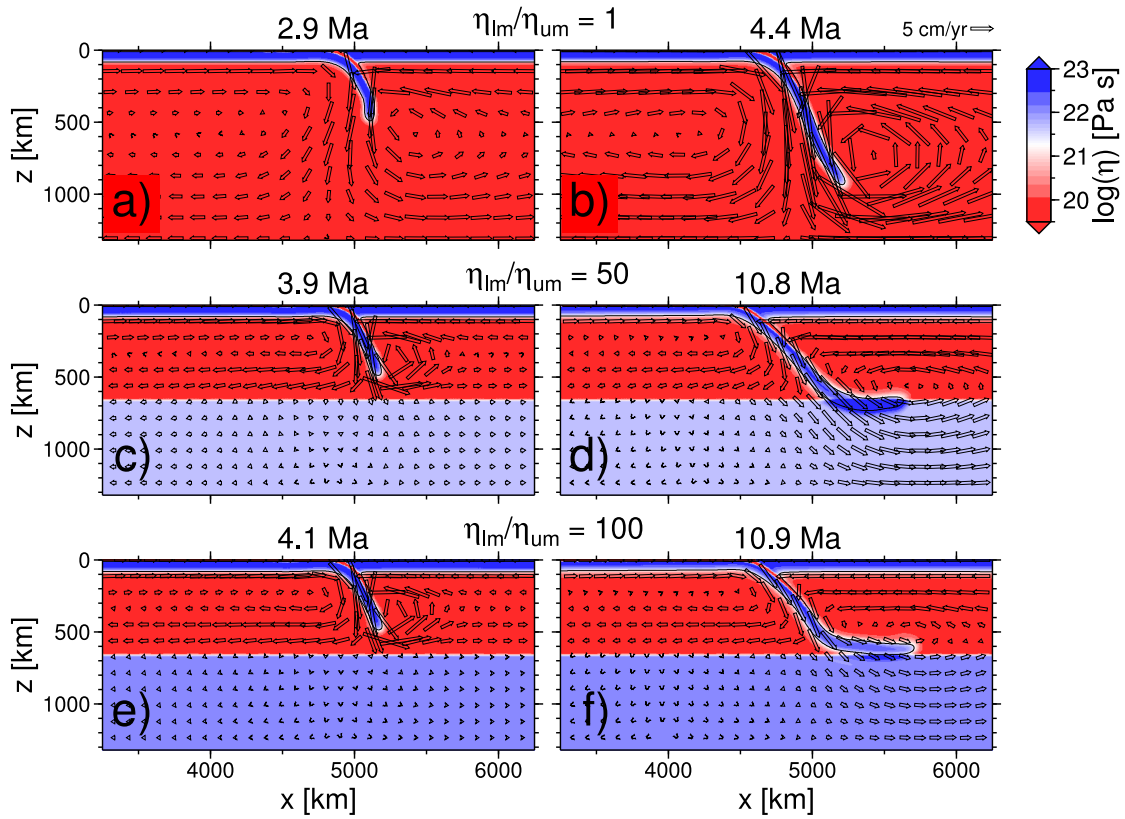


Figure 1.2: Viscosity field showing the evolution of the model without an upper-lower mantle viscosity contrast (a, b), and with viscosity contrasts of 50 (reference model: c, d), and 100 (e, f). All models have free OPs. The viscosity field shown here is a zoom in to the slab region. Overlain are velocity vectors.

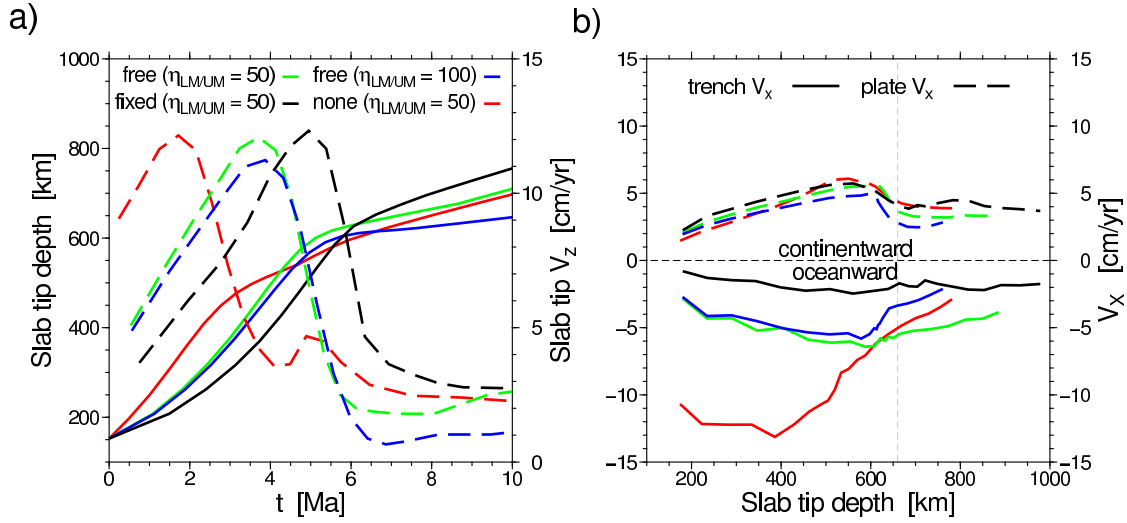


Figure 1.3: Kinematic evolution of models with variable mid-mantle viscosity contrasts (green $\eta_{LM/UM} = 50$, blue: $\eta_{LM/UM} = 100$), model without an overriding plate (red), and model with a fixed overriding plate (black). Shown is, (a), slab tip depth (solid line) and vertical velocity (dashed) versus time and, (b), subducting plate (dashed line) and trench velocity (solid) as a function of slab tip depth.

close proximity to the trench (Fig. 1.4a, e). While fore-arc compression is located at equivalent positions, extensional stresses are focused at different distances away from the trench in the models with and without a stratified mantle. Extensional stresses of ~ 20 MPa are centred at ~ 750 km from the trench for the layered viscosity case, and ~ 1000 km from the trench in the model without a stratified mantle. As well as being centered at a larger distance from the trench, the region of significant extensional stress is also broader in the model without a viscosity increase. As the slab descends in the model without a viscosity jump, the amplitude of both compression and extension increases (both to ~ 40 MPa) as the vigor of mantle flow increases due to increased subduction velocities, yet they do not vary in terms of relative magnitude. In contrast to this, the OP stresses in the layered viscosity model have a strong time dependence. As the slab interacts with the viscosity increase and rollback velocity decreases, a shift from dominantly OP extension to dominantly OP compression occurs (Fig. 1.4e, g). This spatial variability in the degree of OP extension between models, and how it evolves temporally, correlates with the style of mantle flow below the OP, suggesting a basal traction origin for the observed extension (see sec. 1.5.1).

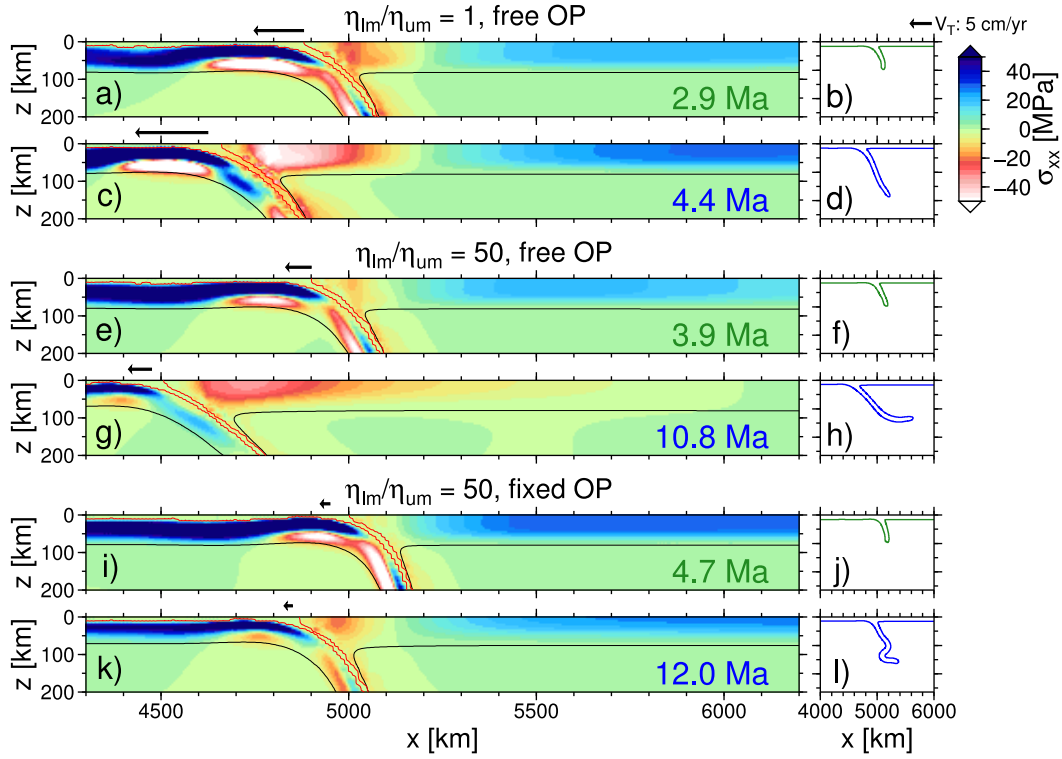


Figure 1.4: Horizontal deviatoric normal stress in the trench region and OP for model with a free OP and no upper-lower mantle viscosity contrast (a to d), reference model with a free OP and upper-lower mantle viscosity contrast of 50 (e to h), and model with a fixed OP and upper-lower mantle viscosity contrast of 50 (i to l). Negative values (red) correspond to compression, and positive values to tension (blue). Panels to the right show slab isotherms to illustrate the slab morphology, and a vector scaled by the trench retreat velocity is plotted above the trench.

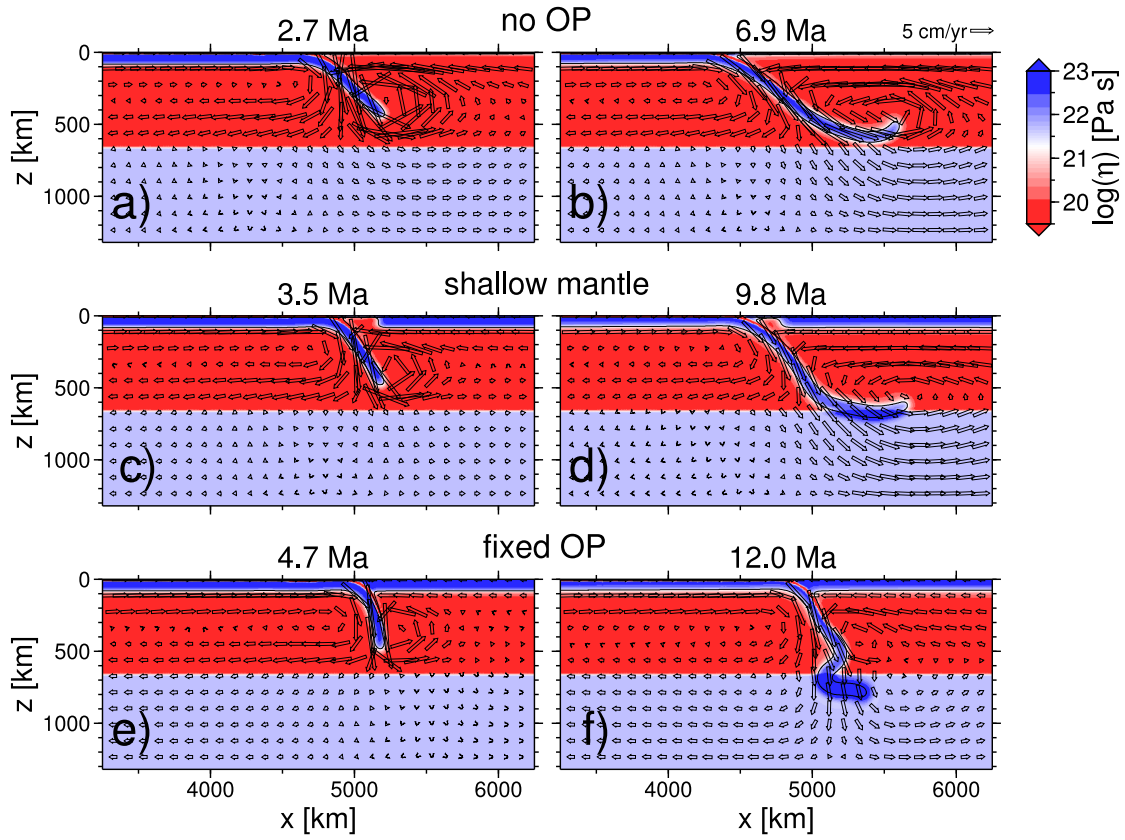


Figure 1.5: Viscosity field showing the evolution of a model without an overriding plate (a, b), with a free overriding plate separated from the subducting plate by an initially 300 km wide region of mantle material (c, d), and with an overriding plate fixed to the side of the domain (e, f).

1.4.2 Effect of overriding plate

The presence of an overriding plate (OP) limits the vertical extent of the poloidal flow cell and increases the depth at which the fastest, trench-directed horizontal mantle flow occurs. The top row of Figure 1.5 shows the slab morphology in the model without an OP. Relative to the model with a free OP, the slab dip angle is reduced in the single plate case (cf. Figs. 1.2c, 1.5a). This reduced dip occurs in part because the trench retreat velocity is greater during the free-sinking stages for the single plate model, relative to the model with an OP (Fig. 1.3b) (e.g. Yamato et al., 2009; Butterworth et al., 2012). In the model without an OP, subduction occurs dominantly via trench retreat (i.e. $V_T \gg V_P$). Inclusion of a free OP results in subduction that is partitioned approximately equally between plate advance and trench retreat (Fig. 1.3b).

The inclusion of an OP limits the vertical extent of large scale mantle wedge flow, and places strong and dense material adjacent to the subducting plate (SP) hinge. In order to determine which of these OP effects causes slabs that are subducting beneath OPs to behave so differently from those in isolation, we examine models in which the OP is separated from the subducting plate (SP) by a weak crust and an additional region of mantle material (e.g. Yamato et al., 2009), relative to the reference model where the plates are separated by only the weak crust (e.g. Capitanio et al., 2010a). We refer to the model with mantle material between the SP and OP as the “shallow mantle wedge” model (Figs. 1.5c,d). The upper mantle slab dip in the shallow mantle wedge model is comparable to that in the reference model with an OP (cf. Figs. 1.2c, 1.5c). In align with this, the average trench retreat velocity in the shallow wedge model (not shown) is comparable to that of the reference model with an OP, albeit slightly elevated (~ 5 cm/yr relative to ~ 4 cm/yr). Thus, while trench retreat is driven by the slab’s negative buoyancy, the presence of an OP reduces the tendency of the slab to retreat at its high, single plate rate whether or not the OP is directly adjacent to the SP.

However, whether the SP is in direct contact with the OP or not does exert a strong control on the OP stress regime. Significant OP fore-arc compression (≥ 20 MPa) only occurs when the OP is directly adjacent to the SP, as return flow-induced basal tractions drag the OP into the SP. In the decoupled models, compression is concentrated in the thermal boundary layer which develops above the region of shallow mantle. The basal traction-driven OP

extension is, however, comparable in both extent and amplitude (~ 20 MPa).

Models in which the OP trailing edge is attached to the side of the domain, termed “fixed” OP models here, exhibit trench retreat velocities that are further reduced ($V_T = 2$ cm/yr). This is because rollback requires thinning of the OP. This gives rise to slabs with large dip angles which fold and penetrate into the lower mantle (Fig. 1.5f). The OP stress state in the fixed OP model is dominated by high extensional stresses (~ 35 MPa), relative to the free OP case (~ 20 MPa), due to the rollback-driven OP stretching (cf. Fig. 1.4e, i). As observed in previous work, the OP stress state oscillates between periods of intense and more minor extension as the slab folds, and the rollback rate oscillates (Clark et al., 2008; Capitanio et al., 2010a,b). While the partitioning of subduction between V_T and V_P is strongly modified by the presence and mobility of the OP, it is found that the maximum sinking velocity ($V_Z \sim 12$ cm/yr) does not depend on how the OP is modeled (i.e. absent, free, or fixed).

1.4.3 Plate viscosity

We now investigate the role of lithospheric viscosity in our two-plate Newtonian models by analyzing models with plates that have lower ($\eta' = \eta_{\text{slab}}/\eta_{\text{mantle}} = 100$) and higher viscosities ($\eta' = 2000$) than in the reference models ($\eta' = 500$). The relatively low viscosity plate models are often referred to as “weak” and the relatively high viscosity models as “strong” throughout the text. Figure 1.6 illustrates the slab morphologies of the lithospheric viscosity end-member models for both free (left) and fixed overriding plates (right), and the associated subducting plate kinematics are shown in Figure 1.7. Here, for both the free (black) and fixed OP (red) models, characteristic velocities are plotted as a function of lithospheric strength. The maximum trench retreat, plate, and slab tip sinking velocities are extracted, for periods before ($z_{\text{slab}} < 630$ km) and after ($z_{\text{slab}} > 690$ km) the slab has reached the upper-lower mantle viscosity discontinuity, from time series such as those shown in Figure 1.3. Additionally, the maximum trench-plate velocity ratio, which relates to how the subduction velocity is partitioned between slab rollback and plate advance, is plotted as a function of plate strength. We have verified that using the mean velocity, as an alternate metric, does not change the systematics observed.

We find that the slab in the intermediate viscosity reference model ($\eta' = 500$; Fig. 1.2d) has a lower dip angle than both the strong and weak slabs (Fig. 1.6a, c), a trend also observed

in the equivalent models without an OP (not shown). In the strong plate model, the slab can penetrate into the high viscosity lower mantle, which results in a relatively high dip angle throughout subduction. The slab in the weak plate model has an increased dip angle during the later phase of subduction due to the strong reduction in rollback that the slab experiences as it impinges on the mid-mantle viscosity jump (e.g. Zhong and Gurnis, 1995; Enns et al., 2005). Figure 1.7a shows that, during the free sinking phase (filled circles), rollback velocity increases with decreasing plate viscosity. As mentioned above, there is a significant reduction in the retreat velocity in the lower mantle for the weak slabs. However, the trench retreat velocity of the stronger slabs, which penetrate more easily through the viscosity contrast, is only slightly reduced (Fig. 1.7a).

Slab rollback in the models with fixed OP occurs at the expense of OP thinning and so trench retreat is slower than in the free models for all plate viscosities (Fig. 1.7a). There is a greater disparity in rollback rates, and slab dips, between the free and fixed OP cases for the strong plate models ($\Delta V_T/V_{T,free} \approx 0.70$), relative to the weak models ($\Delta V_T/V_{T,free} \approx 0.45$) (Fig. 1.7a). The subducting plate velocity (V_P) shows less variability (Fig. 1.7b). The anomalously high V_P data points, during the latter phase ($z_{slab} > 690$ km) for the weak plate model ($\eta' = 100$), occur because subduction begins to initiate at the SP trailing edge. For both the free and fixed OP cases, the strong plate models are observed to have significantly reduced slab sinking velocities (Fig. 1.7d). The maximum sinking velocity, of ≈ 16 cm/yr for $\eta' = 100$ and ≈ 8 cm/yr for $\eta' = 2000$, is unaffected by whether the OP is free to move (e.g. Capitanio et al., 2010a; Meyer and Schellart, 2013).

The lithospheric viscosity exerts a strong control on the stress regime of the OP (Fig. 1.8). In addition to the general trend of increased fore-arc compression as the slab reaches the mid-mantle viscosity jump for all models, there is a switch from OPs dominated by fore-arc compression (~ 30 MPa) in the weak plate model, to OPs that are dominantly extensional (~ 40 MPa) in the strong plate model (Fig. 1.8). The amplitude of the convergence velocity between the trench and the trailing edge of the OP ($V_{op} - V_T$), which corresponds to either net OP extension (negative convergence velocity) or net OP compression (positive), is plotted for the free OP models in Figure 1.9a. For the weaker plate models, the rate of trench retreat decreases significantly as the slab feels the viscosity increase mid-mantle depths, while the trenchward OP velocity decreases at a much slower rate. This results in net convergence, as

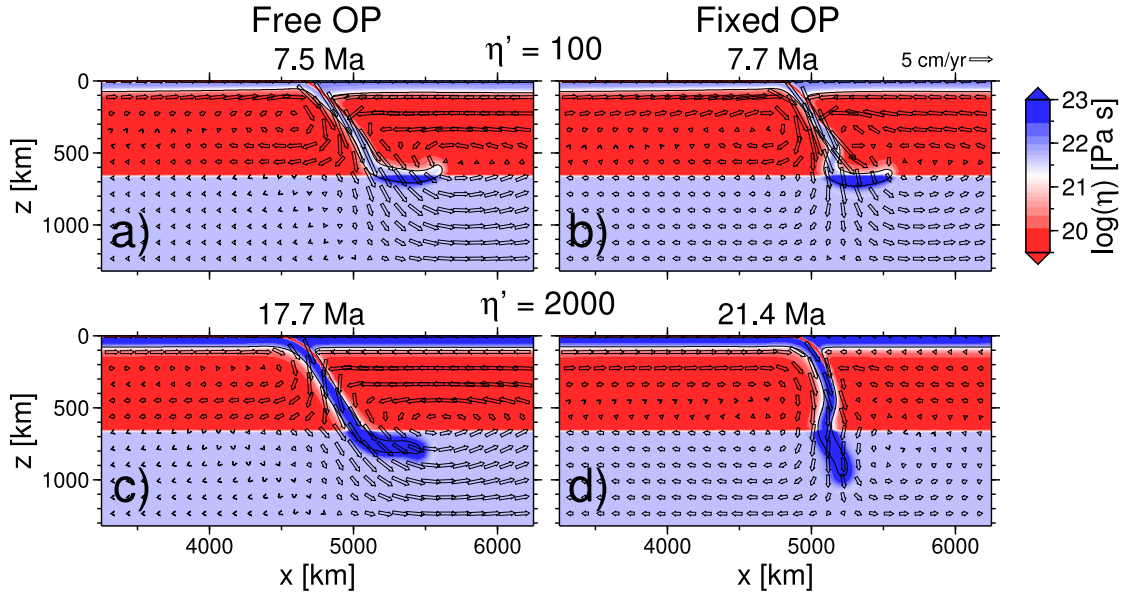


Figure 1.6: Viscosity field, during the latter stages of subduction, for models with variable plate-mantle viscosity contrasts (η'). Left panels show cases in which the OP is free (a, c), and right panels show fixed OP models (b, d).

is seen in the stress field (Fig. 1.8c), during the later stages of subduction. In contrast to this, the trench retreat velocity in the strong plate model continues to increase as the slab sinks into the lower mantle. This results in an OP trailing edge with net motion away from the trench (i.e. negative $V_{op} - V_T$) for the duration of subduction, which gives rise to net OP extension (Fig. 1.9a). Despite having a much lower amplitude convergence velocity (0.2 to 0.6 cm/yr) than the weak plate model (up to 4.5 cm/yr), the strong plate model has comparable OP stresses because the convergence velocity scales with horizontal strain rate.

In order to determine the degree of control that the OP viscosity has on the subducting slab dynamics and OP stress regime, relative to the role of the SP viscosity, we have additionally tested the effect of varying solely the OP viscosity ($\eta'_{op} = 100, 2000$). For the free OP models, reducing the OP viscosity does not have a strong effect on the subducting slab morphology. Slabs in the low viscosity OP model do rollback slightly faster ($\Delta V_T \approx 1$ cm/yr), which results in a minor reduction in the dip angle (not shown). As for the models in which the viscosity of both plates is varied, the variability in slab morphology between the free and fixed OP models is much greater for the high viscosity OP models. In contrast to the switch from compression to extension observed when the viscosity of the subducting plate (SP) is

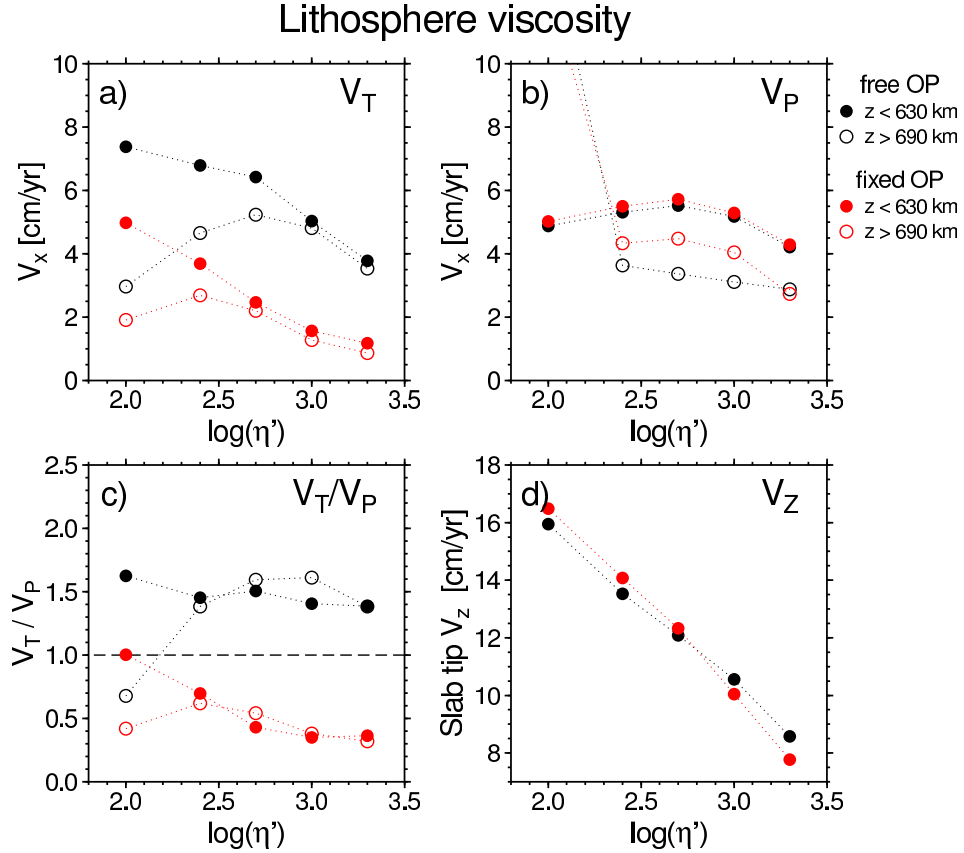


Figure 1.7: Dependence of subducting plate kinematics on plate-mantle viscosity contrast (η'). Diagnostic velocities are taken as the maximum upper mantle (slab tip $z_{slab} < 630$ km: filled circles) and lower mantle velocities ($z_{slab} > 690$ km: hollow circles), for both the free (black circles) and fixed overriding plates (red circles) (see Fig. 1.6). Shown is, a), the maximum upper and lower mantle trench velocity, b), plate velocity, c), trench-plate velocity ratio, and d), vertical slab tip velocity. The reference model corresponds to $\log(\eta') = 2.69$.

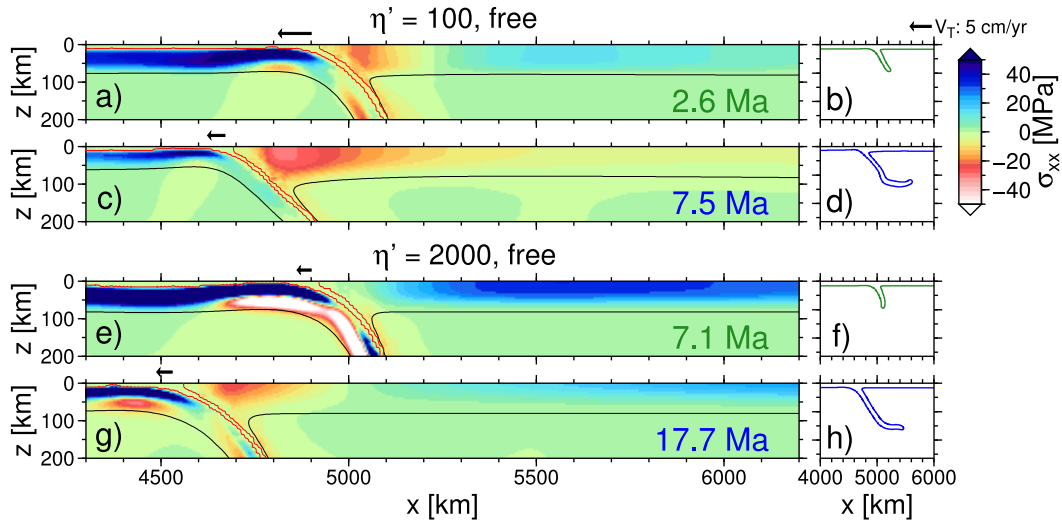


Figure 1.8: Evolution of the horizontal deviatoric normal stress in the trench region and OP for models with variable plate-mantle viscosity contrasts and a free OP (see Fig. 1.6). Upper panels (a to d) show the weak plate model ($\eta' = 100$), and lower panels (e to h) show the strong plate model ($\eta' = 2000$). For detailed caption, see Fig. 1.4.

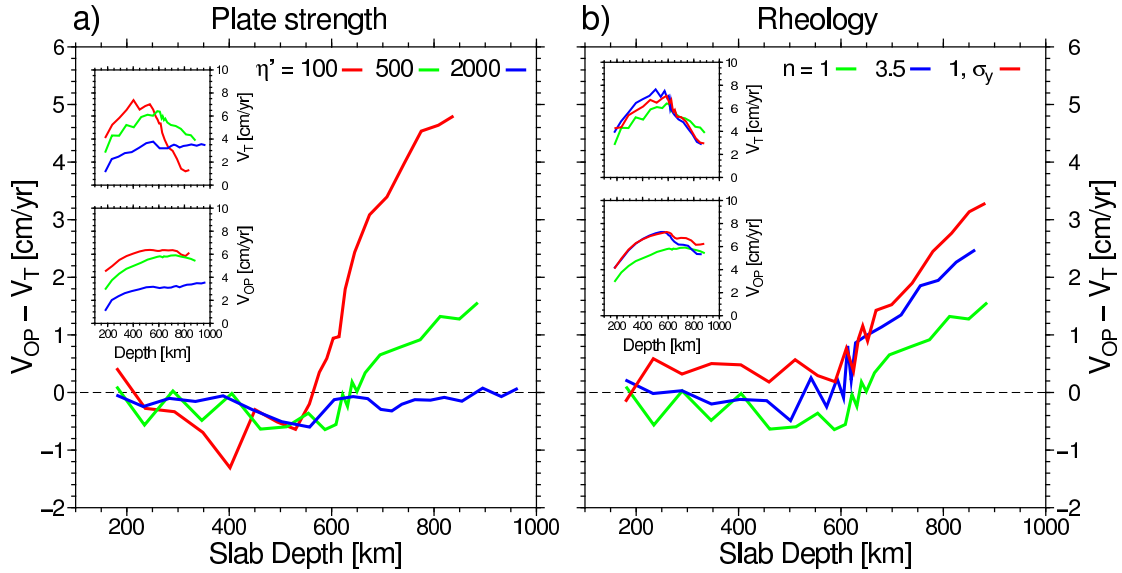


Figure 1.9: Convergence velocity between the trench (V_T) and the trailing edge of the overriding plate (V_{OP}) plotted as a function of slab tip depth for free OP models with, a), variable plate-mantle viscosity contrasts (see Fig. 1.6) and, b), rheologies (see Fig. 1.10). Small inset panels show the evolution of the two individual components, V_T and V_{OP} . Negative convergence velocity corresponds to net OP extension, and positive convergence velocity to net OP compression.

also varied, the viscosity of the OP does not significantly modify the OP stress state during either the free-sinking or flattening phases. This suggests that it is the viscosity of the SP which exerts the strongest control on the OP stress state for the free OP models.

1.4.4 Stress-dependent rheologies

Having examined the effects of varying the viscosity of the entire lithosphere, we now investigate how the inclusion of stress-dependent rheologies, which focus viscosity weakening in regions of high deformation, modify subducting slab kinematics and overriding plate (OP) stress in our free OP model. We explore the effect of a power law rheology (eqs. 1.7-1.9) with a power law exponent (n) of 3.5, and a Newtonian rheology with Byerlee plasticity (eqs. 1.10-1.12). We use a transition stress (σ_T) of 100 MPa, which puts the low stress, ambient mantle within the diffusion creep regime, and the deforming lithosphere in the dislocation creep regime. While a value of 100 MPa is comparable to that used in previous studies which parameterize the Newtonian-power law transition by a constant stress (e.g. McNamara et al. (2001): 70 MPa), there is considerable uncertainty regarding the appropriate value of this tuning parameter, and its dependence on temperature, pressure and composition. We therefore examine its effect at the end of this section. In these models, the mantle immediately surrounding the slab is dominantly in the Newtonian regime. This is not the case for studies that use more complex, laboratory derived rheologies (e.g. Billen and Hirth, 2005, 2007; Nakakuki and Mura, 2013; Garel et al., 2014), and the dynamic implications of this are discussed in sec. 1.5.2.

The morphological effect of including a power law rheology is shown in Figure 1.10 (a, b). The greatest stresses occur within the strong lithosphere, and so here the model is in the power law regime. Maximum weakening, relative to the Newtonian viscosity, of a factor of ~ 10 occurs at the trench where the bending stresses are maximum. Additionally, triggered by extensional stresses, the surface portion of the subducting plate is in the power law regime with relative viscosity reductions extending from the bending region and monotonically decreasing towards the trailing edge. As the slab reaches the mid-mantle (Fig. 1.10b), the slab pull force is, in part, supported by the high viscosity lower mantle, and so the degree of relative weakening is reduced in the subducting slab/plate. In accordance with the evolution of the OP stress state (Fig. 1.12a, c), minor weakening occurs in the extensional back-arc during

the free-sinking phase, and more significant weakening occurs in the compressional fore-arc once the slab has reached the mid-mantle. In the visco-plastic model (Fig. 1.10e-f), viscosity weakening is of greater magnitude, more spatially localized and, due to the linear increase in τ_{yield} with depth, only at shallow lithospheric depths (e.g. Čížková et al., 2007). As seen in single plate models, the strongest plastic weakening is triggered by extensional bending stresses, and occurs in the surface portion of the subducting plate at a distance of ~ 150 km from the trench (e.g. Enns et al., 2005; Di Giuseppe et al., 2008).

The peak sinking velocity (V_Z) of the models with viscosity weakening is substantially higher (15-16 cm/yr) than for the equivalent Newtonian reference model (≈ 12 cm/yr) (Fig. 1.11a). Because viscosity reductions in the surrounding mantle are minor, such elevated V_Z , as in the model with a weak subducting plate (Fig. 1.7d: for $\eta'_{op} = 100$, $V_Z \approx 16.5$ cm/yr), is due to a reduction in the bending region viscosity, which reduces the subduction-resisting plate bending force. Within the upper mantle, subducting plate velocity (V_P) and, to a greater degree, trench retreat velocity (V_T) are elevated with respect to the reference model ($\Delta V_T \approx 1.5$ cm/yr, $\Delta V_P \approx 0.75$ cm/yr). This results in slab dips that are slightly reduced relative to the Newtonian reference (cf. Figs. 1.10a, 1.2c). After interaction with the high viscosity lower mantle, as in the weak slab model (Fig. 1.7a), the initially elevated rollback rate is reduced to below that of the Newtonian reference, particularly for the model with a power law rheology (Fig. 1.11b). This gives rise to an increase in slab dip in the power law model, relative to the Newtonian reference (cf. Figs. 1.10b, 1.2d).

The inclusion of stress-dependent weakening modifies the OP stress regime relative to the Newtonian reference. In both the power law and visco-plastic models, the OP is under greater fore-arc compression (15-25 MPa), and there is a reduced region of significant back-arc extension (250 km), relative to the Newtonian model (compression of ~ 10 MPa, and extension over ~ 700 km) (cf. Figs. 1.4, 1.12). This trend persists when slabs reach the mid-mantle viscosity discontinuity, with significant ≥ 10 MPa compression extending over a wider region (> 1600 km) than in the reference model (~ 900 km). This tendency towards a compressive OP for models with stress-dependent rheologies is concurrent with elevated trench-OP convergence throughout the duration of subduction (Fig. 1.9b). The model with Byerlee plasticity exhibits the greatest amplitude fore-arc compression (~ 40 MPa), which follows from the elevated convergence velocity (Fig. 1.9b). Additional features in the OP stress

field include reduced bending stresses in the subducting plate relative to the Newtonian models, and the focusing of highly stressed regions at greater depths for the model with plasticity (Fig. 1.12g) (e.g. Čížková et al., 2007).

We now consider the transition stress, σ_T , which is used to tune the regions of the model which undergo power law deformation. We have examined the effect of reducing σ_T to 50 MPa (Fig. 1.10c, d) and increasing σ_T to 150 MPa for models with $n = 3.5$ and a free OP. Reducing σ_T accentuates power law deformation in the bending region, where the degree of viscosity weakening is greater in magnitude, but less localized in space. During the free sinking phase, reducing the transition stress to 50 MPa enables OP weakening, relative to the Newtonian case, in the back-arc and fore-arc simultaneously. Reducing σ_T also promotes a reduction in slab rollback, and slab buckling, upon slab interaction with the lower mantle. Conversely, increasing σ_T to 150 MPa results in an OP that undergoes only very minor weakening throughout the duration of subduction, and has a slab morphology similar to the Newtonian reference.

1.4.5 Overriding plate thickness

We now return to the Newtonian rheology models to investigate the role of overriding plate (OP) thickness and density, on the dynamics of the subducting slab and the OP stress regime. In order to target the role of the OP, the mechanical properties of the subducting plate (SP) are kept constant throughout this section ($\eta'_{sp} = 500$, $h_{sp} = 80$ km, $\rho_{sp} - \rho_m = 65$ kg/m³).

Firstly, we examine the effect of OP thickness (h_{op}) on the subducting slab. Figure 1.13 (a-d) shows slabs with relatively thin (40 km) and thick OPs (120 km), during the slab flattening phase. For the free OP case, subduction below a thinner OP results in a decrease in slab dip (Fig. 1.13a). In analogy to the single plate case, this reduction in slab dip stems from an increase in slab rollback that occurs because a thin OP suppresses the “free” rate of trench retreat less than a thick OP. For the fixed OP models, we also find that increasing h_{op} reduces V_T and increases slab dip. As is the case for increasing the OP viscosity, this effect is stronger when the OP is fixed, as V_T is controlled by the OPs resistance to uniaxial stretching. Therefore, models with high viscosity or thick OPs have similar slab morphologies, with near-vertical dip angles and penetration into the lower mantle (Fig. 1.13d).

This strong dependence of trench retreat on OP thickness is shown in Figure 1.14a,

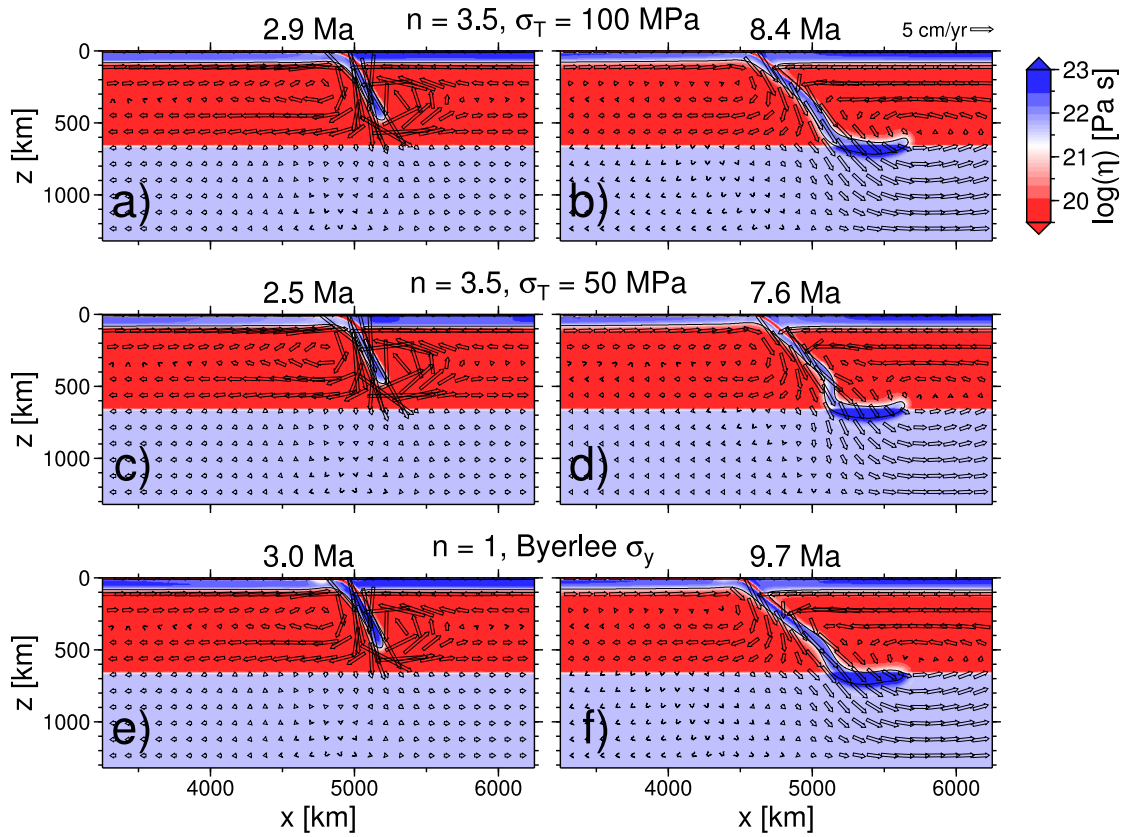


Figure 1.10: Slab evolution for free OP models with power law (a to d) and visco-plastic (e, f) rheologies. The top two rows show power law models with an exponent (n) of 3.5, and transition stresses of 100 MPa (a, c) and 50 MPa (c, d). The bottom row shows the visco-plastic model with a depth-increasing Byerlee yield stress (e, f).

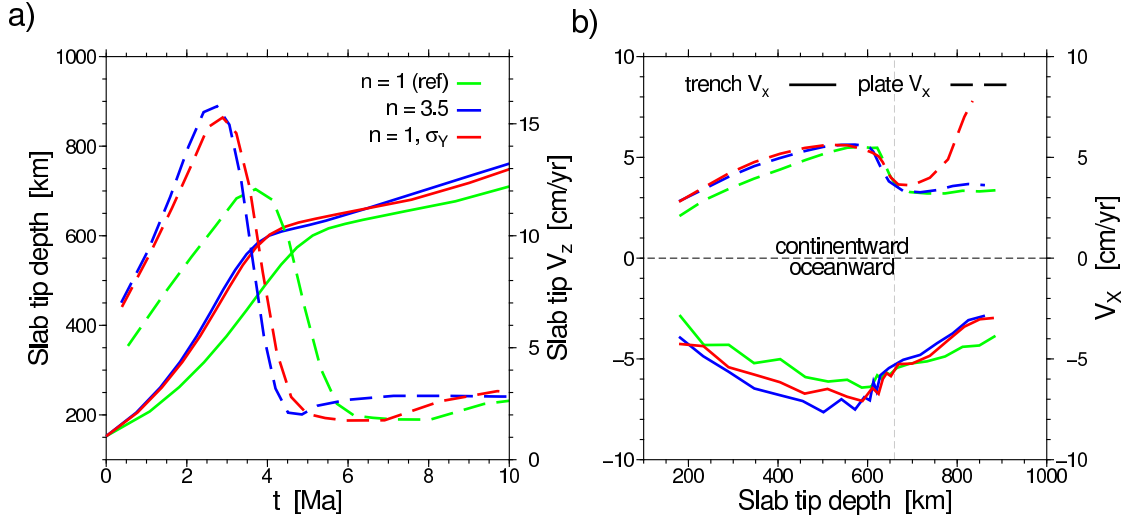


Figure 1.11: Kinematic evolution of free OP models with stress-dependent rheologies (see Fig. 1.10). Shown is, a), slab tip depth (solid line) and vertical velocity (dashed) versus time and, b), subducting plate (dashed line) and trench velocity (solid) as a function of slab tip depth.

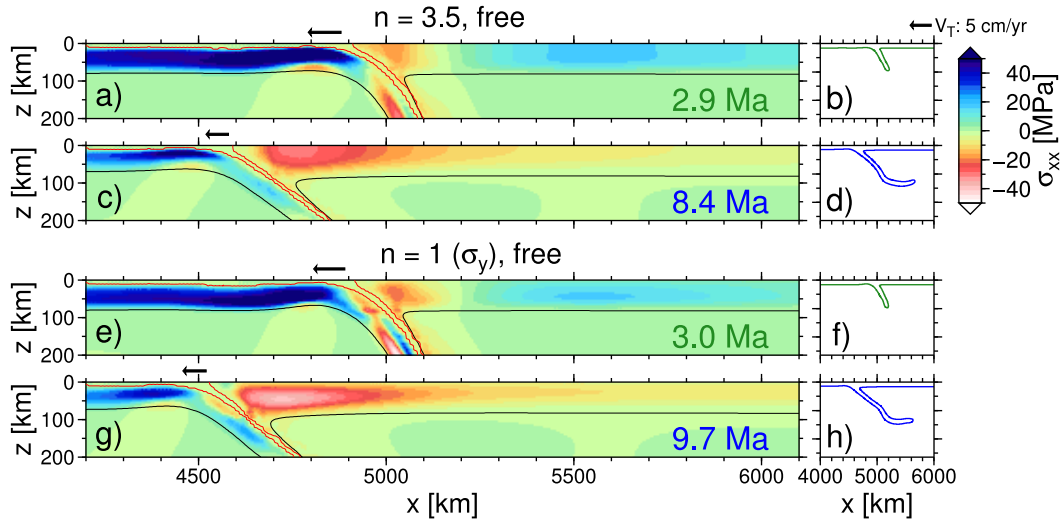


Figure 1.12: Evolution of the horizontal deviatoric normal stress in the trench region and overriding plate for models with stress-dependent rheologies and a free OP (see Fig. 1.10). Upper panels (a to d) show the $n = 3.5$, power law model ($\sigma_T = 100$ MPa), and the lower panels (e to h) show the visco-plastic model with a depth-increasing Byerlee yield stress. For detailed caption, see Fig. 1.4.

for both free and fixed OPs. The maximum trench retreat rate decreases with increasing OP thickness (free OP in upper mantle, for $h_{op} = 40$ km; $V_T \approx 8.0$ cm/yr, for $h_{op} = 120$ km; $V_T \approx 5.2$ cm/yr), except during the latter stages of the free OP models, where V_T is strongly affected by slab interaction with the viscosity discontinuity. The subducting plate velocity (V_P) is insensitive to overriding plate thickness and so V_T/V_P generally decreases with increasing h_{op} , owing to the reduction in V_T . It is found that, despite significant variation in the partitioning of subduction between V_T and V_P , the maximum slab sinking velocity of ~ 12 cm/yr is not affected by h_{op} (e.g. Capitanio et al., 2010a; Duarte et al., 2013; Meyer and Schellart, 2013). In agreement with previous work, the amplitude of overriding plate stresses depends strongly on the thickness of the OP. The thick OP models have reduced maximum extensional and compressional stress amplitudes because trench rollback is preferentially accommodated by OP motion/translation, rather than thinning/thickening as in the thin OP model (Capitanio et al., 2010a, 2011; Meyer and Schellart, 2013).

1.4.6 Overriding plate density

We now examine the role of overriding plate (OP) buoyancy, the density contrast between the OP and surrounding mantle ($\Delta\rho_{op} = \rho_{op} - \rho_m$), for a fixed SP buoyancy ($\Delta\rho_{sp} = \rho_{sp} - \rho_m = 65$ kg/m³). Figure 1.13 (e-h) shows slabs with positively buoyant ($\Delta\rho_{op} = -130$ kg/m³), neutrally buoyant ($\Delta\rho_{op} = 0$), and negatively buoyant ($\Delta\rho_{op} = 130$ kg/m³) OPs. It is apparent that decreasing the buoyancy of the OP significantly reduces the dip angle of the subducting slab, particularly for the case in which the OP is fixed at its lateral edge. Upon examining the kinematics, it can be seen that for all models, and during both phases of subduction (i.e., lower and upper mantle), maximum trench retreat velocity increases as slab buoyancy decreases (e.g., free OP model in upper mantle: For $\Delta\rho_{op} = 130$ kg/m³; $V_T \approx 6.4$ cm/yr, and for $\Delta\rho_{op} = -130$ kg/m³; $V_T \approx 8.7$ cm/yr) (Fig. 1.14c, d). The subducting plate velocity remains relatively constant as it is controlled by the buoyancy and strength of the subducting slab, which is held constant here (i.e. $V_P \approx 5.5$ cm/yr for all $\Delta\rho_{op}$). The increase in rollback velocity for positively buoyant OPs occurs because the lateral density increase, across the trench from OP to SP, results in a pressure gradient directed in the SP direction. Such elevated slab rollback reduces the dip angle of the slab (cf. Figs. 1.13e, g). Conversely, the reduced V_T for slabs with negatively buoyant OPs results in an increased slab dip angle.

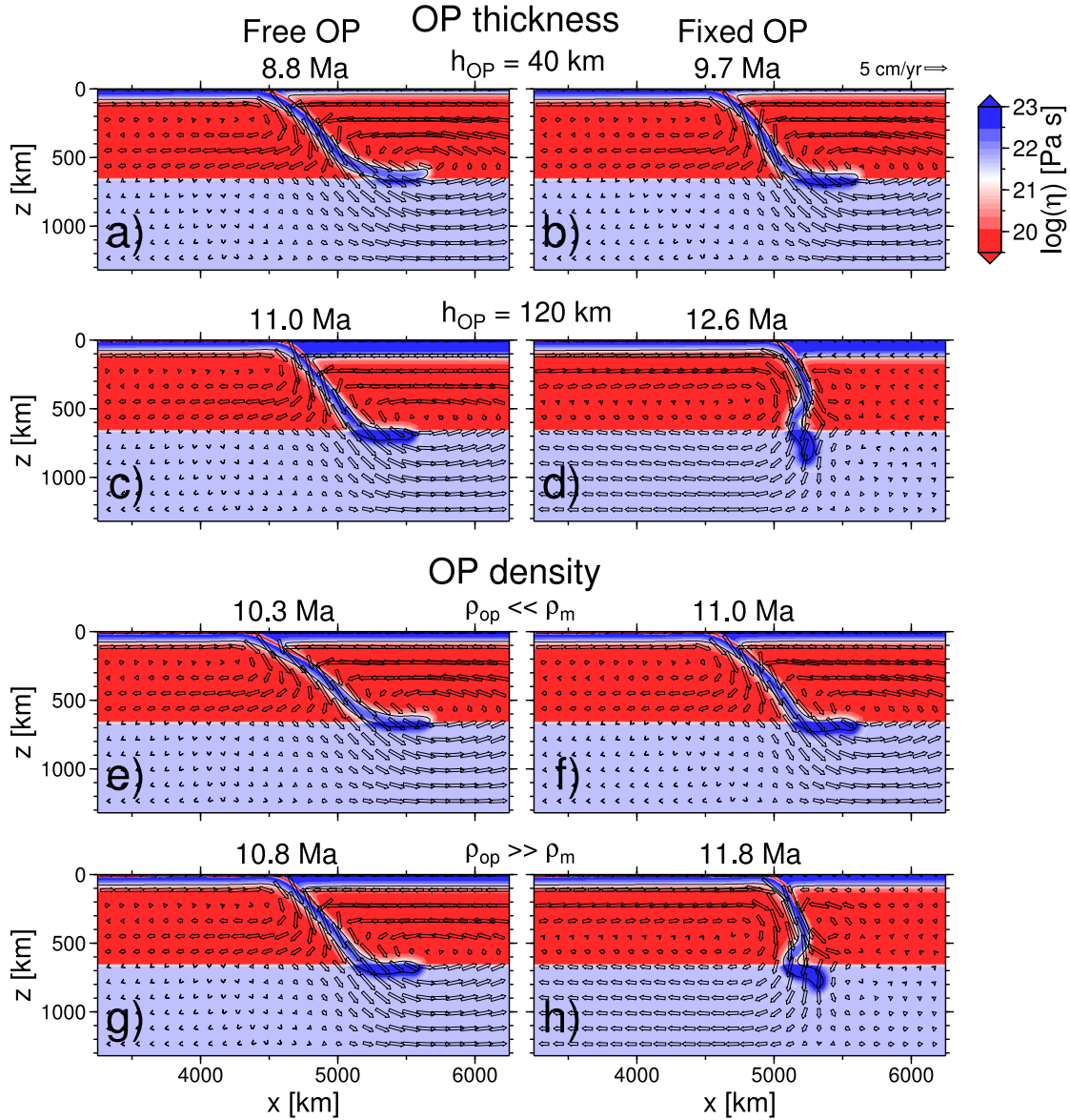


Figure 1.13: Viscosity field, during the latter stages of subduction, for models with variable overriding plate thickness (h_{op}) and overriding plate buoyancy ($\rho_{op} - \rho_m$). Shown are models with thin OPs ($h_{op} = 40$ km: a, b), thick OPs ($h_{op} = 120$ km: c, d), positively buoyant OPs ($\rho_{op} - \rho_m = -130$ kg/m³: e, f), and negatively buoyant OPs ($\rho_{op} - \rho_m = 130$ kg/m³: g, h). Left panels show cases in which the OP is free (a, c, e, g), and right panels show fixed OP models (b, d, f, h).

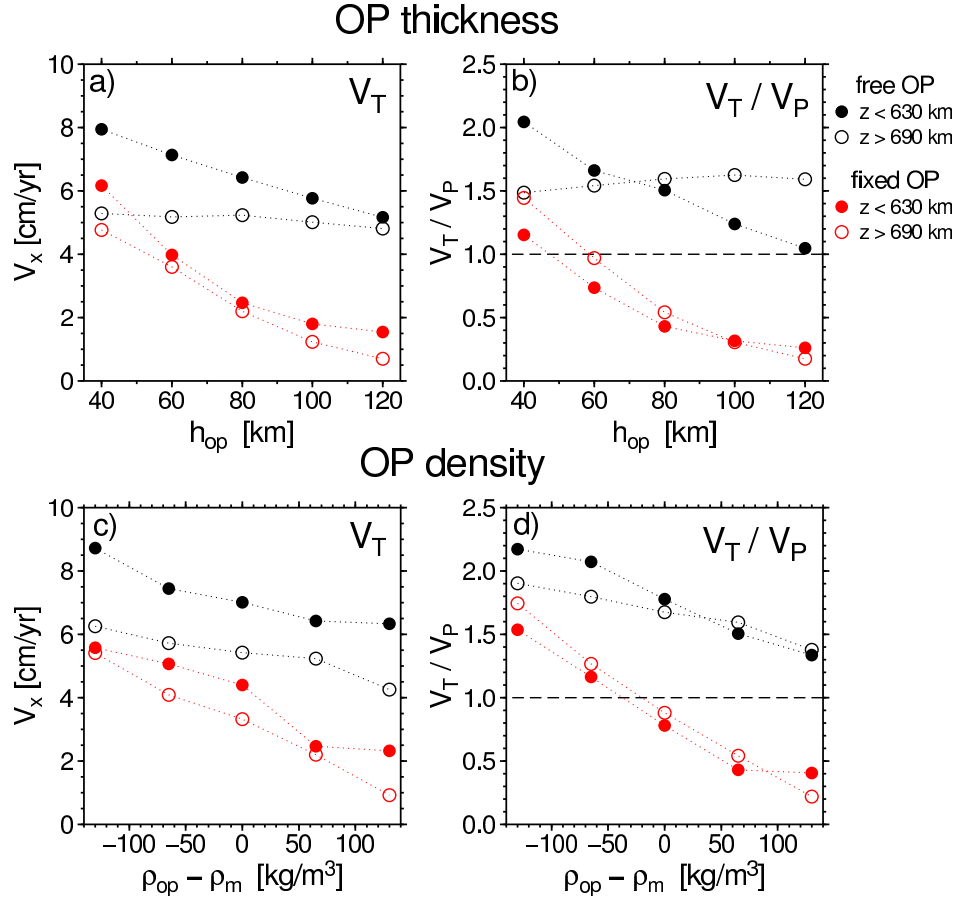


Figure 1.14: Dependence of trench retreat on overriding plate thickness (h_{op}) and overriding plate density ($\Delta\rho_{op} = \rho_{op} - \rho_m$) (see Fig. 1.13). Shown is the maximum upper (filled circles) and lower mantle (hollow circles) trench velocity (a, c), and trench-plate velocity ratio (b, d). Note, reference model corresponds to $h_{op} = 80 \text{ km} = h_{op,r}$ and $\Delta\rho_{op} = 65 \text{ kg/m}^3$.

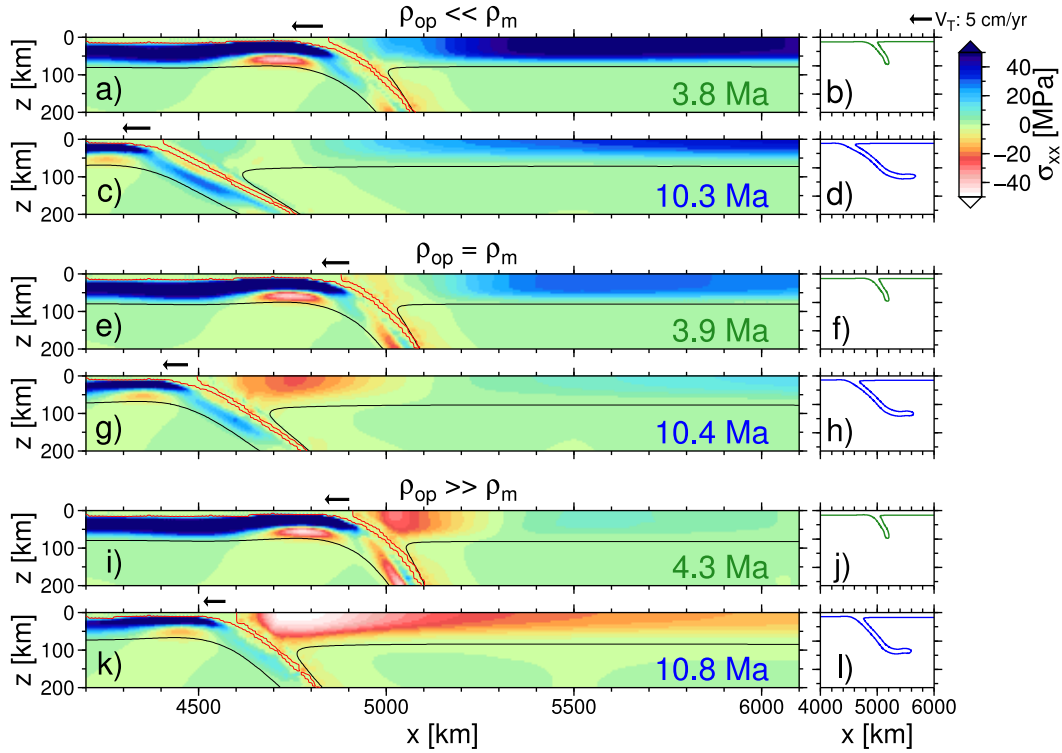


Figure 1.15: Evolution of the horizontal deviatoric normal stress in the trench region and overriding plate for models with variable overriding plate buoyancy and a free OP (see Fig. 1.13). Upper panels (a to d) show the positively buoyant overriding plate model ($\rho_{op} - \rho_m = -130 \text{ kg/m}^3$), center panels (e to h) show the neutrally buoyant OP model ($\rho_{op} = \rho_m$), and lower panels (i to l) show the negatively buoyant OP model ($\rho_{op} - \rho_m = 130 \text{ kg/m}^3$). For detailed caption, see Fig. 1.4.

This effect is more prominent for the fixed OP models, where the slab dip transitions from $\sim 45^\circ$ in the positively buoyant OP model to near vertical in the negatively buoyant OP model (Fig. 1.13f, h).

The OP buoyancy has a very strong effect on the OP stress regime, as is shown in Figure 1.15 for the free OP models. There is a transition from purely overriding plate extension in the positively buoyant OP model, to concurrent fore-arc compression and back-arc extension in the neutrally buoyant OP model, and then purely compression in the negatively buoyant OP model. The reference model, with $\Delta\rho_{op} = 65 \text{ kg/m}^3$, has an OP density, and so OP stress regime and slab rollback velocity (Fig. 1.4e-h) midway between the negatively buoyant and neutrally buoyant OP models shown (Fig. 1.15).

1.5 Discussion

1.5.1 General features

Role of overriding plate

We observe slab rollback in all of our models and find that the presence of an OP reduces the maximum trench retreat velocity by a factor of ~ 2 . This observation is in agreement with previous modeling work by Yamato et al. (2009) and Butterworth et al. (2012). Previous dynamic subduction models with an OP either consider two plates that are separated by only a thin, weak crust (e.g. Capitanio et al., 2010a; Schellart and Moresi, 2013; Garel et al., 2014) or a larger region containing mantle material (e.g. Yamato et al., 2009; Butterworth et al., 2012). Whether the OP and SP are separated only by crust, or crust and a region of mantle material (i.e. “shallow mantle wedge” models), does not significantly modify the trench retreat rate or slab dip angle. This suggests that the reduced slab dip angle in the single plate models is a result of the restriction of large-scale vertical flow, and not the presence of lithospheric plate adjacent to the SP/crust. The presence of mantle material adjacent to the SP hinge in the shallow mantle wedge model does, however, result in trench retreat that is slightly elevated relative to the reference OP model ($\Delta V_T \approx 1$ cm/yr). This is due to the SP-directed pressure gradient arising from the lateral density increase from wedge to SP. This effect is of importance in models with variable ρ_{op} , and is discussed in more detail in sec. 1.5.2, but is of secondary importance relative to the vertical restriction of large-scale mantle flow.

We find that the OP does not significantly affect plate advance velocity (V_P), and so the presence of an OP alters the partitioning of subduction between plate advance and trench retreat by reducing V_T to a value comparable to V_P (i.e., reduces V_T / V_P). This is consistent with previous modeling work (Capitanio et al., 2010a; Schellart and Moresi, 2013), and the observation that, in nature, trench motion is typically smaller than plate motions (e.g. Heuret and Lallemand, 2005). Thus, while rollback is “slab-driven” in our models of free subduction (e.g. Elsasser, 1971; Kincaid and Olson, 1987; Stegman et al., 2006; Schellart, 2008b), it is affected by the OP and its mechanical properties even in models where the OP is free to move. In contrast, Capitanio et al. (2010a) find that the presence of a free OP does not modify how subduction is partitioned between rollback and plate advance. As other parameter values are

comparable, this is potentially due to the very strong subducting plate core ($3000\eta_{\text{mantle}}$) used by Capitanio et al. (2010a), which increases the stiffness of the subducting plate hinge. As is observed in previous two-plate modeling studies (e.g. Capitanio et al., 2010a; Meyer and Schellart, 2013), the vertical sinking velocity (V_Z) is unaffected by the style of OP. This suggests that the mechanical properties of the SP, namely the negative buoyancy and hinge stiffness ($\propto \text{plate viscosity} \times \text{thickness}^3$, for an isoviscous plate), control slab sinking (Conrad and Hager, 1999).

When the slab subducts beneath an overriding plate that is fixed to the edge of the box, which is perhaps analogous to a large continental OP with low plate velocity (Forsyth and Uyeda, 1975), we find that rate of trench retreat is significantly reduced (e.g. Capitanio et al., 2010a,b). This occurs because, in models with fixed OPs, slab rollback is inhibited by the tensile strength of the OP (Ribe, 2001: $\propto \eta'_{op} h_{op}$), as the OP must be stretched for rollback to occur. Therefore, a fixed OP exerts a much stronger control on slab rollback than a free OP, further reducing V_T from its single plate rate (e.g. Capitanio et al., 2010a,b).

Role of viscosity stratification

In our reference model, we include a factor 50 increase in the background viscosity at a depth of 660 km by increasing the prefactor in our viscosity law (eq. 1.6) (e.g. Enns et al., 2005; Garel et al., 2014). For simplicity, we do not include other phase transitions, and the associated density/buoyancy effects (e.g. Tagawa et al., 2007a,b; Nakakuki et al., 2010; Nakakuki and Mura, 2013), which have been shown to promote slab folding atop 660-km (Čížková and Bina, 2013). Even without density effects, which tend to inhibit penetration through the mid-mantle discontinuity, we find the a $\times 50$ viscosity increase causes the subducting slab in our reference free OP model to flatten in the mid-mantle. The recent two-plate, thermo-mechanical free subduction models of Garel et al. (2014) show that, with a mid-mantle viscosity jump of factor 30 and variable overriding plate (OP) and subducting plate (SP) ages, the wide range of slab morphologies observed on Earth can be generated. While we mostly focus on the OP mechanical properties while holding the SP properties fixed, and so a narrower parameter space than Garel et al. (2014), we find that our models produce at least two of the modes of deformation observed. Our free OP reference model flattens above 660 km and then stagnates, analogous to Garel et al. (2014) “horizontally deflecting” mode, and our fixed

OP subducts near-vertically with a near-stagnant trench and folds at the viscosity discontinuity, analogous to their “vertically folding” mode. We find that the rate of trench rollback increases as the slab sinks through the upper mantle, and then begins to reduce once the slab feels the effect of the viscosity discontinuity (e.g. Zhong and Gurnis, 1995). This reduction in rollback rate is greater for models in which the mid-mantle viscosity increase is greater (Fig. 1.3b). This observation of reduced rollback is important in subsequent discussions of OP stress state, and is consistent with two-plate modeling studies that parameterize the discontinuity similarly (e.g. Zhong and Gurnis, 1995; Enns et al., 2005; Garel et al., 2014), and studies that treat the discontinuity as an impenetrable free/no slip boundary (e.g. Schellart and Moresi, 2013; Duarte et al., 2013).

Overriding plate stress state

Recent 3-D, dynamic subduction models with an OP have suggested that toroidal flow due to slab rollback is the main driver of OP extension (Duarte et al., 2013; Meyer and Schellart, 2013; Schellart and Moresi, 2013). In our 2-D models, toroidal flow is, by definition, absent. However, we still generate significant OP extensional normal stresses for models with (≥ 20 MPa) and without (≥ 30 MPa) a mid-mantle viscosity jump, and earth-like mechanical and rheological parameters. Because toroidal flow is absent, our results are therefore most directly applicable to subduction segments towards the center of very wide subduction zones where poloidal flow is likely dominant (e.g. South America, Melanesia). We observe elevated extensional stresses over a broad (~ 900 km) region, and at large distances (≥ 350 km) from the plate boundary in our models. This suggests that these extensional stresses are related to basal tractions (e.g. Sleep and Toksoz, 1971), as opposed to rollback induced trench suction which would be expected to occur in the region near the plate interface (e.g. Shemenda, 1993).

Extensional stresses in the OP are induced by a trench directed increase in horizontal basal traction as the upwelling component of the return flow becomes horizontal below the lithosphere (e.g. Sleep and Toksoz, 1971; Toksoz and Hsui, 1978; Schellart and Moresi, 2013). A basal traction origin is also supported by the increase in trench, “back-arc” distance that occurs when the mid-mantle viscosity jump is absent and the return flow cell is widened (cf. Fig. 1.4a, e), and the occurrence of significant extensional stresses even when the OP is separated from the SP by mantle material (i.e. sec. 1.4.2: “shallow mantle wedge” models).

Horizontal compression (≥ 20 MPa) occurs near to the plate boundary, and is a result of rapid mantle wedge flow dragging the OP towards the plate interface at a rate more rapid than trench retreat (Capitanio et al., 2010a; Schellart and Moresi, 2013; Nakakuki and Mura, 2013). While this study focuses on the mechanical properties of the lithosphere, previous modeling results show that the OP stress state is strongly affected by viscosity and buoyancy anomalies within the mantle wedge (Billen et al., 2003).

The coexistence of a compressional fore-arc and extensional back-arc is in agreement with previous numerical modeling studies, both in 3-D (Schellart and Moresi, 2013) and 2-D domains (Capitanio et al., 2010a; Nakakuki et al., 2010; Nakakuki and Mura, 2013). This suggests a departure from either of the two idealized models of either slab or OP-driven trench motion, as the OP would be expected to be in pure extension or compression if the trench retreat was, respectively, purely slab or OP-driven (e.g. Schellart et al., 2007; Heuret et al., 2007; Jarrard, 1986). Additionally, the OP stress regime in our reference model has a strong time dependence (e.g. Capitanio et al., 2010a), with a shift towards a more compressive OP once the slab has reached the mid-mantle viscosity increase (e.g. Fig. 1.4e, g), a feature not observed in the model without a mid-mantle viscosity jump (Fig. 1.4a, c). Slabs are more steeply dipping during the earlier, extensional phase, which is consistent with the observation that extensional back-arc basins typically occur above steeply dipping slabs (Uyeda and Kanamori, 1979; Lallemand and Heuret, 2005). The shift, from an extensional to a compressional OP, occurs due to the reduction in the rate of trench retreat that occurs as the relatively weak slab is obstructed by the mid-mantle viscosity discontinuity (Zhong and Gurnis, 1995; Christensen, 1996; Čížková et al., 2002), which allows the tractions in the mantle wedge nose to drag the OP into the SP hinge. The amplitude of back-arc extension decreases because the interaction of the slab with the viscosity discontinuity causes a change in the style of return flow from fast, localized return flow cell to more sluggish, longer wavelength return flow which does not produce the basal traction gradients required to induce significant extensional stress. Therefore, the physical nature of the phase transitions on Earth, and how they are parameterized in such geodynamic models, potentially plays a significant role in dictating OP stress state and its temporal evolution.

As observed by Capitanio et al. (2010a,b) and Nakakuki and Mura (2013), fixing the OP subjects it to elevated extension (~ 35 MPa) as a result of plate thinning. There is still a

small region of low amplitude (≥ 5 MPa) fore-arc compression in the fixed OP models (slab-flattening phase), suggesting that moderate near-trench compressional stresses, due to rapid mantle flow in the nose of the mantle wedge, may be a ubiquitous feature irregardless of the OP mobility.

1.5.2 Subducting slab kinematics and overriding plate stress

Despite rollback being “slab driven” in the case of free subduction, we find that the physical properties of the OP have a substantial effect on the partitioning of subduction between plate advance and rollback, and therefore slab dip (Uyeda and Kanamori, 1979; Kincaid and Olson, 1987; Griffiths et al., 1995; Christensen, 1996). While our models do not include the far-field forcing that is necessary to test the hypothesis that trench rollback is controlled by the motion of the OP (e.g. Jarrard, 1986; Heuret and Lallemand, 2005; Heuret et al., 2007), previous modeling work of van Dinther et al. (2010) demonstrates that an externally forced OP can modify trench motion. Our discussion therefore focuses on the role that the lithospheric properties, namely viscosity, thickness and density, have in modulating the “free” rate of slab rollback and the OP stress state.

Plate viscosity

Varying the lithospheric viscosity, of both the SP and the OP, exerts a strong control on both SP kinematics (Fig. 1.7) and OP stress (Fig. 1.8). During the free sinking phase, our strong slab models have the lowest slab rollback rates (V_T), and the steepest dips due to an increased stiffness in the bending region which resists unbending during slab descent (Conrad and Hager, 1999; Enns et al., 2005; Di Giuseppe et al., 2008; Ribe, 2010). The slab in the weak plate slab model has a lower dip angle during the free-sinking phase (cf. Figs. 1.8b, f). This is compatible with single-plate compositional models that observe steep upper mantle dips and subsequent slab rollover or folding above the discontinuity for strong slabs (e.g. Enns et al., 2005; Di Giuseppe et al., 2008; Ribe, 2010), and observations of old oceanic slabs dipping near-vertically in the western pacific subduction zones (e.g. Di Giuseppe et al., 2009).

As the slab tip touches the mid-mantle viscosity increase, there is a significant decrease in V_T for the slab in the weak plate model, while V_T in the strong slab model is unaf-

fects (Fig. 1.7a) (e.g. Zhong and Gurnis, 1995; Enns et al., 2005). This reduction in V_T in the weak slab model causes the net OP compression to become strongly positive (Fig. 1.9a), giving rise to an OP dominated by compressional stresses (Fig. 1.8c). Our models therefore confirm that, as observed previously (Capitanio et al., 2010a; Schellart and Moresi, 2013; Nakakuki and Mura, 2013), OP compression occurs in models in which deformation and rollback is “slab-driven”. This contradicts the classic model of slab-driven rollback, in which solely OP plate extension accompanies rollback. The net OP convergence in the strong slab model also increases slightly during the latter stages of subduction but never becomes positive (Fig. 1.9a), resulting in an OP that is dominated by extensional stresses throughout subduction (Fig. 1.8). The lithospheric viscosity is, therefore, a key parameter in determining the OP stress state in such two-plate models. The slab-driven model predicts that an increase in OP extension is due to increased slab rollback (e.g. Elsasser, 1971; Schellart, 2008b). However, for the strong lithosphere model, the rollback velocity is reduced (relative to the weak plate model), while the OP becomes more extensional. This suggests that, due to the variable motion of the OP trailing edge, a simple correlation between V_T and OP deformation style may not be expected even in the case of “slab-driven” rollback and deformation.

The inclusion of a stress-dependent viscosity reduces the lithospheric viscosity in localized regions, particularly in the hinge region of the SP where bending stresses are large (e.g. Enns et al., 2005; Billen and Hirth, 2005; Stegman et al., 2006; Di Giuseppe et al., 2008). The resulting SP kinematics and OP stress regimes, for all rheologies (power law viscosity, and Byerlee plasticity), are comparable to that of the weak plate model discussed above, in that the mid-mantle viscosity increase causes a reduction in V_T which results in strong OP compression. This indicates that the strength of the SP is a key control on the OP stress regime in free OP models and that weakening the slab, either using a stress-dependent rheology or a reduced bulk viscosity, promotes OP compression.

Our simplified power law rheology gives rise to only minor viscosity reductions in mantle surrounding the slab. Therefore, we do not expect to observe an increase in plate velocity due to reduced mantle tractions, or an increase in slab dip associated with reduced hydrodynamic suction as seen in previous fixed-trench models (Tovish et al., 1978; Billen and Hirth, 2005, 2007; Billen, 2008). However, we suggest that an increase in rollback during the free-sinking phase, due to a reduced strength SP hinge, may override this dip effect in

models with mobile trenches. The dominant control of the SP viscosity, over the OP viscosity, is supported by models in which only the OP viscosity is varied. Neither the subducting slab shape or the OP stress regime are strongly affected by the OP viscosity for the free OP models.

In addition to V_T and the OP stress state, the SP viscosity strongly affects the vertical slab sinking velocity (V_z). A reduction in SP viscosity, either due to stress-dependent weakening or a reduced bulk viscosity, increases V_z . As the SP density is equivalent in these models, this increase in (V_z) can be attributed to the decrease in hinge stiffness. For a low viscosity SP, a low hinge stiffness ($\propto$ plate viscosity $\times$ thickness³, for an isoviscous plate) causes a reduction in the dissipation of gravitational potential energy as the SP bends at the trench (Conrad and Hager, 1999; Ribe, 2010).

Overriding plate thickness and density

The OP thickness (h_{op}) and buoyancy ($\Delta\rho_{op} = \rho_{op} - \rho_m$) both have strong effects on V_T and the OP stress regime. As observed in the dynamic two-plate models of Garel et al. (2014) and Sharples et al. (2014), and the experiments of Meyer and Schellart (2013), increasing h_{op} reduces the slab rollback velocity resulting in decreased V_T / V_P and greater dip angles. This reduced rollback is a more significant effect on the slab dip than the elevated hydrodynamic suction that, due to a larger high viscosity region (OP) above the subduction interface, tends to flatten the slab (e.g. Dvorkin et al., 1993). Rodríguez-González et al. (2012) find that slabs subducting beneath older, thicker OPs have reduced dips due to an increased hydrodynamic suction, but these models have fixed trenches and so do not take into account variation in slab rollback which has a strong effect on slab dip (e.g. Kincaid and Olson, 1987; Griffiths et al., 1995).

Additionally, varying $\Delta\rho_{op}$ has a strong effect on the rate of trench retreat (V_T) with slabs with positively buoyant OPs ($\Delta\rho_{op} = -130 \text{ kg/m}^3$) exhibiting faster rollback than those with negatively buoyant OPs ($\Delta\rho_{op} = 130 \text{ kg/m}^3$) (Fig. 1.14c). Increased rollback for slabs subducting beneath positively buoyant OPs, analogous to continental OPs or oceanic lithosphere with thick crust (Shemenda, 1993), occurs due to the lateral hydrostatic pressure gradient ($=\Delta\rho gh$) which is directed from the OP to the SP. This pressure gradient acts to push the SP backwards, thereby enhancing rollback (and reducing slab dip). This enhanced rollback is accommodated by an extensional, spreading OP (Fig. 1.15a, c) (e.g. Shemenda, 1993). When

the OP is negatively buoyant, the opposite is true, and the pressure gradient is directed towards the OP which reduces rollback and enhances OP compression. Such an increase in slab rollback, and reduction in slab dip angle, for subduction beneath a positively buoyant continental OP could potentially offset the decrease in rollback that we observe to be associated with subduction beneath a thick OP. Capitanio et al. (2010b) find that subduction partitioning, between rollback and plate advance, is unaffected by $\Delta\rho_{op}$. As mentioned previously, this is potentially due to the stiff subducting plate core used by Capitanio et al. (2010a) placing a stronger constraint on SP evolution than the lateral pressure gradients induced by an OP-SP density contrast.

Properties of fixed overriding plates

While the previous discussion has focused on subduction beneath free OPs (i.e. not attached to model sidewall), it is found that the OP mechanical properties play a stronger role in dictating slab rollback rates for models with fixed OPs. This is because rollback is forced to occur at the expense of OP thinning (e.g. Capitanio et al., 2010a,b), which is dictated by the OP's tensile strength (Ribe, 2001: $\propto \eta'_{op} h_{op}$). Increasing OP strength, either by increasing h_{op} or η'_{op} , drastically reduces slab rollback which, in turn, increases the dip of the slab and its penetrative power through the viscosity discontinuity (Griffiths et al., 1995; Christensen, 1996; Capitanio et al., 2010b). The OP density, $\Delta\rho_{op}$, also exerts a stronger control on V_T in the fixed OP models. This is because heavy, fixed OPs resist the gravitational spreading required to facilitate rollback (Shemenda, 1993), while rollback can occur via OP translation in the free OP models.

Therefore, one may expect the mechanical properties of OPs that are relatively immobile (e.g. large continental OPs) to exert a stronger control on the rollback, and dip angle, of the subducting slab, relative to OPs that are more mobile (e.g. with ridge/basin proximal to the trench). As there is only a weak correlation between oceanic SP age and slab dip (Jarrard, 1986; Lallemand and Heuret, 2005), the variability in overriding plate density, thickness, and in the case of a fixed OP, viscosity, potentially plays a significant role in explaining the wide range of slab rollback rates, and dip angles, observed on Earth. The slab sinking velocity is, however, independent of whether the OP is free or fixed. As pointed out previously (e.g. Conrad and Hager, 1999; Capitanio et al., 2010a; Meyer and Schellart, 2013), the suggests

that the properties of the subducting slab, namely the negative buoyancy and hinge stiffness, controls vertical slab sinking.

1.6 Conclusions

We use 2-D, dynamic models of subduction with an overriding plate (OP) to investigate the controls on the subducting slab, and stress state of the OP. In all of the models, with OPs that are both free to move and attached to the sidewall, the subducting slab rolls back. Removal of the OP elevates the slab rollback velocity to a value that is significantly faster than the subducting plate velocity, which results in reduced slab dip angles (e.g. Yamato et al., 2009). In contrast, fixing the position of the trailing edge of the OP greatly reduces the rate of slab rollback, which gives to an increased slab dip angle (e.g. Capitanio et al., 2010a).

Despite the absence of toroidal flow, which has been shown to drive OP extension in 3-D models (e.g. Meyer and Schellart, 2013; Schellart and Moresi, 2013), basal tractions due to poloidal mantle return flow are sufficiently large to generate significant OP extensional stresses (> 20 MPa) in our 2-D models. Simultaneously, the region nearer to the trench is under compression, due to rapid mantle wedge flow dragging the OP into the subducting plate (SP), at a velocity faster than the rate of rollback. Additionally, the OP stress state is strongly time dependent. As the subducting slab impinges on the high viscosity lower mantle, the slab rollback rate is reduced which, because the reduction in the velocity of the OP trailing edge is more minor, causes a shift towards stronger OP compression. It is found that the SP viscosity exerts a strong control on the OP stress state. Low viscosity subducting slabs, either due to a reduced bulk viscosity or a stress-dependent rheology, give rise to compressional OPs. Lower viscosity slabs also have elevated sinking velocities due to a reduction in the stiffness, and so bending resistance, of the SP hinge.

While the mechanical properties of solely the OP have only a minor effect on the slab sinking and SP velocities, we find that they exert a substantial control on slab rollback, and the OP stress state. Reducing the thickness of the OP gives rise to increased slab rollback rates and elevated OP stress, both compressional and extensional. Reducing the density of the OP, so that it is neutrally or positively buoyant, enhances slab rollback which results in reduced slab dips and a shift from simultaneous OP compression and extension, to solely extension.

In addition, fixing the lateral edge of the OP strengthens the dependence of rollback rate, and OP stress regime, on OP mechanical properties (e.g. Capitanio et al., 2010a,b). While rollback and OP deformation is driven by the negative buoyancy of the SP in such models of free-subduction, we therefore suggest that the physical properties and mobility of the OP potentially play a significant role in modulating both slab rollback and OP deformation.

Chapter 2

Overriding plate thickness control on subducting plate curvature

2.1 Summary

Subducting plate (SP) curvature exerts a key control on the amount of bending dissipation associated with subduction, and the magnitude of the subduction resisting bending force. However, the factors controlling the development of SP curvature are not well understood. We use numerical models to quantify the role of SP rheology on the minimum radius of curvature, R_{min} . We find that R_{min} depends strongly on the SP thickness when the rheology is viscous. This dependence is substantially reduced when the SP behaves plastically, in line with the lack of correlation between R_{min} and SP thickness on Earth. In contrast, plasticity leads to a strong positive correlation between R_{min} and the overriding plate (OP) thickness. Using an analysis of R_{min} vs. OP thickness, we show that such a positive correlation exists on Earth. This suggests that OP structure, in conjunction with SP plasticity, is crucial in generating slab curvature systematics on Earth.

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2.2 Introduction

Subducting slabs have long been recognized as the major driver of plate motions on Earth (Forsyth and Uyeda, 1975). The change of gravitational potential energy due to lithospheric plates sinking into the mantle is balanced by viscous dissipation in both the plates and the surrounding mantle. The curvature of the subducting plate (SP) at the trench exerts a key control on the amount of dissipation due to plate bending, and the size of the subduction-resisting bending force (e.g. Conrad and Hager, 1999; Becker et al., 1999; Buffett, 2006; Wu et al., 2008; Ribe, 2010).

Significant attention has been given to the influence of SP parameters on slab curvature: Assuming that bending dissipates a constant proportion of the work done by sinking slabs, a positive scaling is expected between SP thickness and radius of curvature (Buffett, 2006; Buffett and Heuret, 2011). Such an increase in SP curvature for thick SPs is indeed observed in laboratory (Bellahsen et al., 2005; Schellart, 2009; Irvine and Schellart, 2012) and numerical (Capitanio et al., 2009) experiments with viscous slabs. However, a positive correlation between SP age, which controls plate thickness, and R_{min} , estimated from earthquake hypocenter distributions, appears to be absent on Earth (Buffett and Heuret, 2011; Fourrel et al., 2014). It has been suggested that a plastic rheology component, in which the SP hinge viscosity is reduced due to stresses induced by the SP's negative buoyancy, could explain the discrepancy between the radius of curvature systematics observed on Earth, and that predicted for a viscous plate (Buffett and Heuret, 2011; Buffett and Becker, 2012). Such plastic weakening of the SP bending region, e.g. in the form of deep crustal faulting, is expected to occur on Earth (e.g. Ranero et al., 2003, 2005), driven by the bending stresses of the negatively buoyant SP (e.g. Billen and Gurnis, 2001). We therefore consider how the dependence of R_{min} on SP thickness is modified for visco-plastic SPs, relative to the isoviscous SPs typically considered in such studies.

In addition to the physical properties of the SP, recent numerical experiments have shown that overriding plate (OP) structure can exert a strong control on SP dynamics/morphology (e.g. Yamato et al., 2009; Sharples et al., 2014; Garel et al., 2014). Particularly, thick OPs result in SPs with large mantle-scale dips. However, the possibility that OP structure exerts a control on R_{min} , and so slab bending, remains unexplored as experiments investigating slab curva-

ture have so far been constrained to single plate studies, thus neglecting the possible role of the OP. For plastic plates, Buffett and Becker (2012) showed that the bending moment may saturate during subduction, because hinge stresses can be limited by the plastic yield stress. A consequence of this saturation is that the bending stresses may not be large enough to balance the opposing gravitational torque (Ribe, 2010; Buffett and Becker, 2012). Thus, additional forces associated with the OP or plate interface/weak zone have been hypothesized to play a role (Buffett and Becker, 2012), and may exert a control on the development of slab curvature. Using dynamically self-consistent, two plate models, similar to those presented in chapter 1 (Holt et al., 2015a), we therefore investigate the effect of OP thickness, h_{OP} , on the curvature of the SP (R_{min}). In addition to moving towards more Earth-like numerical models, incorporating interactions between the two plates allows us to directly investigate how SP rheology, in conjunction with OP structure, affects the dynamics of the subducting slab.

Lastly, after characterizing the dependence of slab curvature on lithospheric plate thickness for various simple slab rheologies, we compare the model systematics to those derived from slab curvature estimates on Earth. While previous compilations of subduction zone parameters suggest a correlation between slab dip and OP velocity and/or deformation regime (e.g. Lallemand and Heuret, 2005; Heuret et al., 2007), a potential OP thickness control on slab curvature has not been investigated. We use the R_{min} estimates of Buffett and Heuret (2011) and the lithospheric thickness estimates of Bird et al. (2008) to firstly verify the lack of a correlation between R_{min} and SP thickness, and then probe subduction zones on Earth for a potential correlation between R_{min} and OP thickness.

2.3 Numerical Method and Model Setup

We use the finite-element code CitcomCU (Moresi and Gurnis, 1996; Zhong, 2006) to model subduction in a Cartesian domain. CitcomCU solves the equations governing convection in an incompressible viscous fluid (Boussinesq approximation) with negligible inertia. The domain size is 7920 km $\times$ 1320 km $\times$ 13.2 km (i.e. the computation is effectively 2-D), and all boundaries are mechanically free slip. The initial subducting and overriding plate lengths are 4500 km and 2400 km, respectively. Thermally, the upper boundary has a constant temperature of 273 K, while the base and sides have zero heat flux. In order to target first

order dynamics, we use a simple subduction setup, both mechanically and rheologically. The lithosphere-asthenosphere temperature contrast is set to 1200 K. Because we use thick, constant temperature lithospheric plates, as opposed to a half-space cooling profiles, we use a reduced thermal expansivity, of $1.25 \times 10^{-5} \text{ K}^{-1}$, to give a reasonable bulk density contrast of 50 kg/m^3 . The sublithospheric mantle has a reference viscosity of 10^{20} Pa s and density of 3300 kg/m^3 . With a thermal diffusivity, κ , of $10^{-6} \text{ m}^2/\text{s}$, this corresponds to a Rayleigh number of 1.13×10^7 .

In the constant viscosity SP models (“viscous” models), lithospheric plates are a factor of 500 more viscous than the sub-lithospheric mantle. Viscosity is defined according to Frank-Kamenetskii (1969), using a high temperature dependence ($E = 10$) and is limited by a maximum cut-off ($500\eta_0$),

$$\eta_N = \min(\eta_0 \exp(E(1 - T)), 500\eta_0), \quad (2.1)$$

where T is the non-dimensional temperature. In the visco-plastic SP models, an additional plastic viscosity is added to the rheological formulation. The effective viscosity, η_{eff} , is then taken as the minimum of η_N and the plastic viscosity, which limits the allowable stress to the value of the yield stress, τ_y (e.g. Schott and Schmeling, 1998; Enns et al., 2005):

$$\eta_{eff} = \min\left(\eta_N, \frac{\tau_{yield}}{2\dot{\epsilon}_{II}}\right), \quad (2.2)$$

where $\dot{\epsilon}_{II} = \sqrt{\dot{\epsilon}_{ij}\dot{\epsilon}_{ij}/2}$ is the second invariant of the deviatoric strain rate (Einstein summation convention). To focus on the role of the SP rheology, we confine plastic yielding to within the SP. This ensures that the strength of the OP is constant throughout models, and so removes the possibility that differential rates of OP deformation contribute to the observed curvature variations. For simplicity, we consider a constant τ_y . In order to capture varying degrees of plastic yielding, we test three τ_y values of 50, 75 and 100 MPa. As in many other subduction modeling studies (e.g. Schott and Schmeling, 1998; Enns et al., 2005; Di Giuseppe et al., 2008), our τ_y values are significantly lower than experimental estimates of the depth-averaged yield stress in oceanic lithosphere (Mei et al. (2010): 300 MPa). While this discrepancy remains enigmatic, in the current work we do not aim to replicate the exact mechanics of subduction on Earth, but probe potential mechanisms responsible for first-order trends ob-

served. Our τ_y values have been chosen empirically, in order to be high enough to prevent the SP hinge from completely weakening, and low enough to place a significant proportion of the hinge in the plastic regime.

Subduction is initiated by allowing the SP to extend below the OP to a depth of 200 km with a radius of curvature (R_{init}) of 250 km. We examine how this initial condition affects the dynamics observed by testing other values of R_{init} (200 km and 300 km). To facilitate decoupling of the SP from the OP, we insert a 15 km thick, weak ($0.5\eta_{mant}$) crustal layer within the SP (e.g. Běhouňková and Čížková, 2008; Stegman et al., 2010). More details about the implementation of this compositional crust, and resolution tests of similar models are described in chapter 1 (Holt et al., 2015a). Finite element dimensions vary from 2.5 km in the upper 330 km of the domain to 7.5 km elsewhere. In order to investigate the dependence of slab curvature on the SP and OP thicknesses (h_{SP} , h_{OP}), we vary h_{SP} between 75 and 125 km (with $h_{OP} = 50$ km), and h_{OP} between 25 and 125 km (with $h_{SP} = 100$ km).

To compute the radius of curvature, we track the location of the subducting plate centerline, i.e. halfway between the top and base of the lithosphere (Fig. 2.1). This is done by placing passive tracers at the, initially horizontal, plate center and allowing them to advect in the induced flow field. The minimum radius of curvature, R_{min} , is then computed by fitting cubic splines to the SP centerline, according to the methodology of Buffett and Heuret (2011). While R_{min} is strongly time-dependent, we choose to compute the typical R_{min} at an equivalent slab-tip depth of 760 km in each of the models. We do this by linearly interpolating R_{min} to this depth using the two adjacent numerical timesteps, as illustrated in Figure 2.2. This depth corresponds to halfway between the initial slab depth (200 km) and the base of the model (1320 km), and gives the bending region enough time to evolve from the geometry initially imposed while being far enough from the base to not be significantly affected by the bottom of the box. We additionally examine the temporal evolution of R_{min} .

2.4 Plate Thickness Control on Radius of Curvature

Initially, we focus on the effect of the subducting plate thickness, h_{SP} , on radius of curvature for a constant overriding plate thickness, h_{OP} , of 50 km (Fig. 2.3b). For these experiments, the OP is thinner than the SP, in order to focus on the dynamics of the SP. As

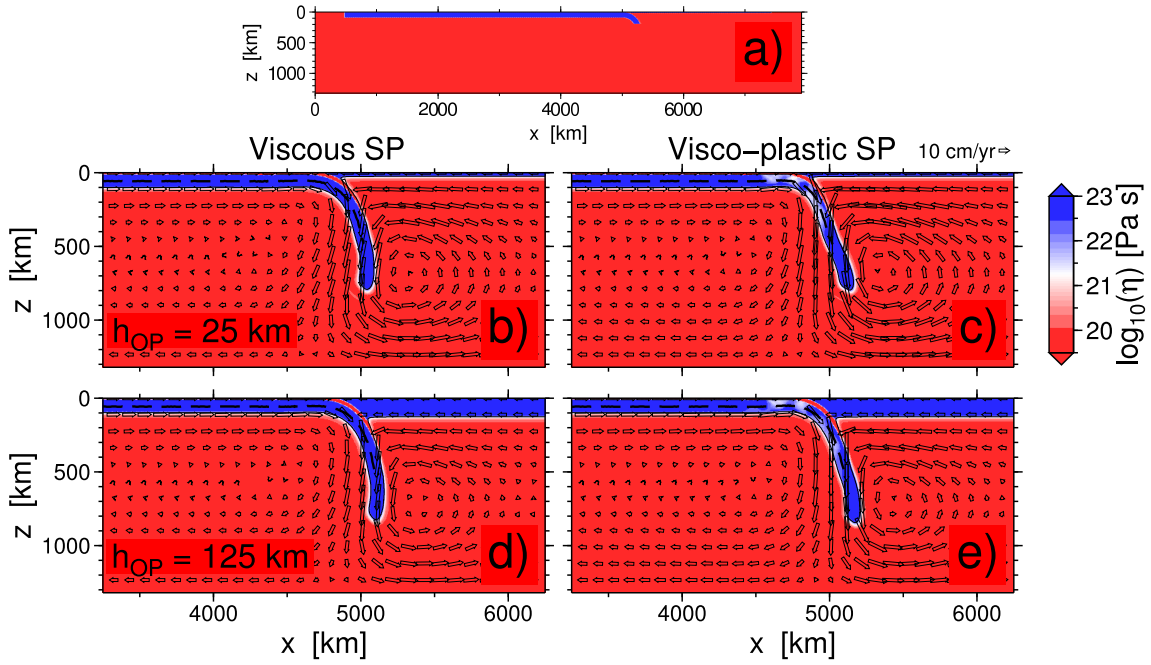


Figure 2.1: Model viscosity field of, a) initial model setup for viscous case with $h_{SP} = 100$ km and $h_{OP} = 25$ km, and, b-e) at evolved timesteps for a selection of models with variable h_{OP} . The left column shows models with a viscous subducting plate rheology (b: $h_{OP} = 25$ km, d: 125 km), and the right column shows models with a visco-plastic subducting plate rheology with $\tau_y = 75$ MPa (c: $h_{OP} = 25$ km, e: 125 km). $h_{SP} = 100$ km in all models shown. The subducting plate center-lines used to computed R_{min} are overlain in black.

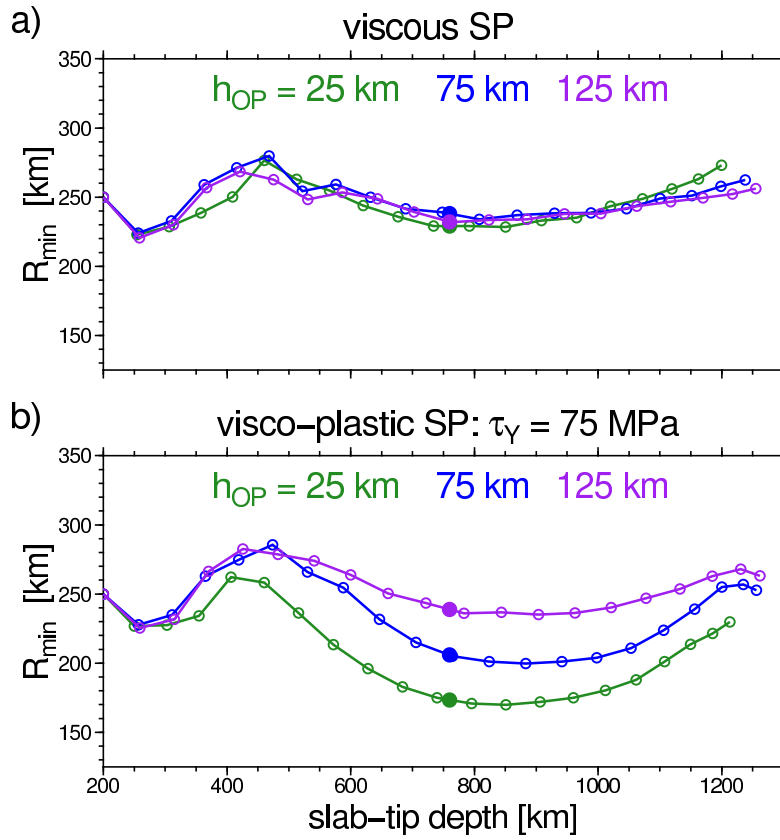


Figure 2.2: Temporal evolution of R_{min} for, a) viscous and, b) visco-plastic subducting plates ($\tau_Y = 75$ MPa) with variable h_{OP} . R_{min} is plotted as a function of the depth of the slab-tip. The points indicated by enlarged, filled circles correspond to those used for Fig. 2.1. All models have $h_{SP} = 100$ km and $R_{init} = 250$ km.

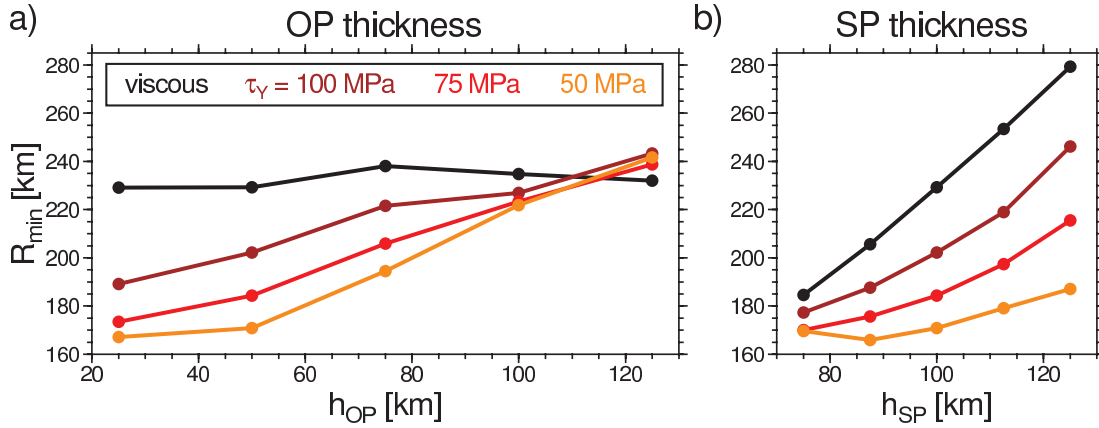


Figure 2.3: Dependence of R_{min} on, a) h_{OP} and, b) h_{SP} . R_{min} is computed at an equivalent slab-tip in each model ($z_{slab} = 760$ km). For the models with variable h_{OP} , $h_{SP} = 100$ km, and for the models with variable h_{SP} , $h_{OP} = 50$ km. All models have equivalent initial curvature, $R_{init} = 250$ km.

observed previously for single plate models (Bellahsen et al., 2005; Schellart, 2009; Irvine and Schellart, 2012), the radius of curvature (R_{min}) for viscous SPs increases with increasing h_{SP} . R_{min} increases from 185 to 280 km when h_{SP} is increased from 75 to 125 km (Fig. 2.3b). The dependence is near-linear with an average slope, computed from the two end-points, of 1.9. In detail, it appears that the relationship is slightly non-linear with a power-law exponent, computed from a linear regression of a log-log plot, of 0.81 (Fig. 2.4a).

The strength of the h_{SP} control on R_{min} significantly weakens with the addition of a plastic viscosity component in the SP, and the dependence becomes more strongly non-linear (Fig. 2.3b). For a yield stress of 100 MPa, R_{min} increases by 69 km when increasing h_{SP} from 75 to 125 km. This change in R_{min} is $\sim 70\%$ of that observed in the viscous plate models. Reducing τ_y , which increases the area of the bending region deforming plastically, reduces the strength of this positive trend further still. In the models with τ_y of 50 MPa, R_{min} increases by only 17 km, or $\sim 20\%$ of the change for the viscous plates. In addition, for all values of h_{SP} , visco-plastic SPs have reduced R_{min} relative to the equivalent viscous SP models.

We next examine the control of the overriding plate thickness, h_{OP} , on R_{min} , for a constant $h_{SP} = 100$ km (Fig. 2.3a). The radius of curvature of viscous SPs is found to be largely independent of h_{OP} , with a relatively constant R_{min} of ~ 230 km for a wide range of

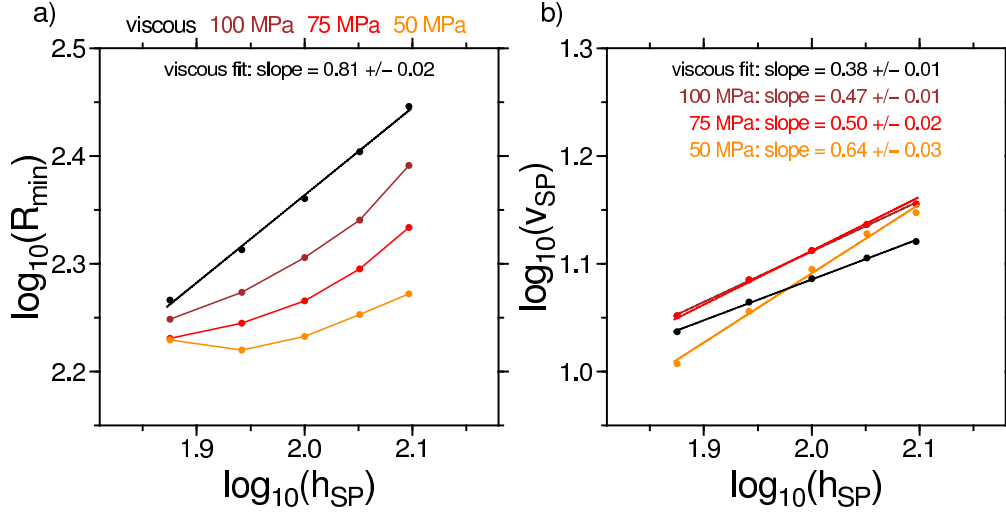


Figure 2.4: Log-log plots of, a) radius of curvature (R_{min}) vs. subducting plate thickness (h_{SP}) and, b) subducting plate velocity (v_{SP}) vs. subducting plate thickness (h_{SP}), for the slab-tip depths shown in Figure 2.3b ($z_{slab} = 760$ km). For the rheologies which plot linearly on the log-log plot (i.e. power-law functions), linear regression fits are shown. Quoted are the log-log slopes (i.e. power-law exponents) and the slope uncertainty. For the models with variable h_{OP} , $h_{SP} = 100$ km, and for the models with variable h_{SP} , $h_{OP} = 50$ km. All models have equivalent initial curvature, $R_{init} = 250$ km.

h_{OP} , from 25 to 125 km. Conversely, SPs with a plastic viscosity component exhibit a strong positive correlation between R_{min} and h_{OP} . The trends are approximately linear, albeit much less uniform than the trend between h_{SP} and R_{min} for viscous SPs (Fig. 2.3b), and so we use the slope computed from the end points at $R(h_{OP} = 25 \text{ km})$ and $R(h_{OP} = 125 \text{ km})$ to compare the strength of the relationship for variable yield stresses. For τ_y of 50, 75, and 100 MPa, the slope is 1.5, 1.3, and 1.1, respectively. Therefore, the radius of curvature has a stronger dependence on h_{OP} when the size of the region of plastic failure increases. As observed when examining h_{SP} , R_{min} is generally reduced for visco-plastic SPs relative to the equivalent viscous models. In addition, for the thickest overriding plates, R_{min} converges to a relatively constant ~ 240 km.

In order to test whether the positive correlation between R_{min} and h_{OP} is representative of the entire model runs, we additionally examine the time dependence of R_{min} (Fig. 2.2). In Figure 2.2, R_{min} is plotted as a function of slab-tip depth (z_{slab}) for both viscous models and the intermediate yield stress visco-plastic models ($\tau_y = 75$ MPa) with variable h_{OP} . As observed previously (e.g. Becker et al., 1999; Capitanio et al., 2007; Irvine and Schellart, 2012),

R_{min} is strongly time dependent. However, the positive correlation between R_{min} and h_{OP} for the visco-plastic models is prevalent throughout the majority of the model run, while the R_{min} trajectory clearly shows no such dependence for the viscous SP models. Only during the initial period, during which the SP is dynamically adjusting away from the morphology initially imposed ($200 \text{ km} < z_{slab} < 550 \text{ km}$), are these trends less developed. In addition, as the slab tip gets close to the base of the box ($z = 1320 \text{ km}$), R_{min} increases as the slab dip reduces. We therefore consider the trends shown in Figure 2.3 to be representative of the free-sinking phase of the dynamic models.

2.5 Correlations on Earth

Following the numerical modeling observations that a plastic subducting plate rheology gives rise to a positive correlation between R_{min} and h_{OP} and a reduced correlation between R_{min} and h_{SP} , we now turn our attention to subduction zones on Earth to test this hypothesis. We use the R_{min} estimates of Buffett and Heuret (2011), computed from spline fits to 133 earthquake hypocenter profiles along 12 major subduction zone systems. Additionally, as in Lallemand and Heuret (2005), we remove 12 anomalous SP profiles that contain ridges, plateaus, or continental lithosphere. However, our findings do not depend on this selection (see below).

We derive h_{OP} estimates from the Bird et al. (2008) model of lithospheric thickness which is itself derived from the S20RTS model of global shear-velocity variations (Ritsema et al., 2004), and other geophysical constraints. The h_{OP} data point is picked at 500 km in-board of the subduction zone trench along the trench-perpendicular azimuth of the profiles used to compute R_{min} (Fig. 2.5a). We observe a positive correlation between R_{min} and h_{OP} (Fig. 2.5b), as observed for visco-plastic SPs in the numerical modeling (Fig. 2.3a). We compute a linear (Pearson) correlation coefficient of 0.66 and a Spearman's rank correlation coefficient of 0.70 (Fig. 2.5b). While such correlation coefficient values are encouraging, there is undoubtedly strong uncertainty in these data-sets, particularly the tomographically derived h_{OP} estimates. Also, the appropriate distance from the trench to pick the overriding plate thickness likely varies between different subduction zone segments, due to the wedge geometry and lateral extent of overriding plate deformation. However, using a constant trench-OP

distance appears appropriate for examining global trends. While we find that 500 km gives the highest correlation coefficients, we observe positive correlation coefficients, ranging from 0.52 to 0.69 for distances of 250 and 750 km (Fig. 2.6). In addition, using the full Buffett and Heuret (2011) dataset (i.e., not removing profiles containing ridges/continental material) reduces both the linear (Pearson) and Spearman's rank correlation coefficients by only 0.03 (Fig. 2.6c, d).

Inspecting these data-sets geographically it is apparent that this positive correlation, on a global scale, is largely due to SPs subducting beneath thick continental lithosphere in the Eastern Pacific having large R_{min} (typically $250 \text{ km} < R_{min} < 500 \text{ km}$) and SPs subducting beneath thinner, oceanic OPs in the Western Pacific having smaller R_{min} (typically $75 \text{ km} < R_{min} < 250 \text{ km}$) (Fig. 2.5a). A more local region which appears to follow the hypothesized positive $R_{min} - h_{OP}$ correlation appears is Central America/Mexico. In Central America, R_{min} appears to be locally reduced in the region of low h_{OP} , relative to Mexico where a thicker OP gives rise to greater R_{min} . In contrast, the subducting plate is younger/thinner in the Northern portion of the Cocos plate where R_{min} is greatest. It therefore appears that, at least along this subduction zone, the OP mechanical properties dominate over those of the SP in controlling slab curvature. There are, however, subduction zones within which this positive correlation between h_{OP} and R_{min} does not appear to be present. Most striking are some of the profiles in the Sunda-Banda arc, where SPs with large R_{min} of 300–400 km subduct beneath OPs that appear to be relatively thin (~ 50 – 125 km). Despite removing profiles with obvious SP ridges/plateaus, there are likely still significant buoyancy anomalies in the subducting plate due to the highly heterogeneous nature of this arc (e.g. McBride and Karig, 1987). In such locations, and other regions with significant SP buoyancy heterogeneity, local R_{min} variation is likely dominated by SP structure. However, on a global scale it appears that OP structure is playing a key role in dictating SP curvature. In addition to OP thickness, the density structure of the OP may also impact SP curvature by controlling the lithostatic pressure gradient exerted across the plate interface. The Banda-Sunda arc is a candidate for such an effect, as the OP transitions from continental in the northwest to oceanic in the southeast.

Lastly, we verify that there is indeed a poor correlation between R_{min} and subducting plate thickness (h_{SP}). We extract h_{SP} from the same lithospheric thickness map, at a

distance of 250 km outboard of the trench (Fig. 2.5c). As reported in previous work (Buffett and Heuret, 2011), we compute very low, and negative, correlation coefficient values of -0.14 (Pearson) and -0.13 (Spearman). The lack of a positive correlation between R_{min} and h_{SP} is incompatible with the systematics observed for purely viscous SPs in the numerical models (Fig. 2.3b), and provides further support for the importance of SP plasticity.

2.6 Discussion

Previous modeling studies have investigated how h_{SP} affects R_{min} for isoviscous slabs (e.g. Bellahsen et al., 2005; Schellart, 2009; Irvine and Schellart, 2012). However, the derived scaling relationships appear to be absent on Earth (Buffett and Heuret, 2011; Fourel et al., 2014). This led Buffett and Heuret (2011) to speculate that plasticity may be important in producing the curvature systematics observed on Earth. In addition to examining how plasticity can modify the curvature systematics, we consider the possible role of overriding plate structure, which has thus far remained unexplored. We are able to investigate this interaction because our numerical models self-consistently couple the dynamics of the two plates, without imposing kinematic constraints.

However, we do simplify many aspects of the physical system. Most notably, we adopt a simplified lithospheric viscosity structure, thus neglecting the layered rheological structure inferred from experimental work (e.g. Goetze and Evans, 1979; Karato and Wu, 1993; Kohlstedt et al., 1995), and omit density/viscosity contrasts associated with mid-mantle phase transitions. In addition, our simulations are 2-D and so we neglect the both role of toroidal flow (e.g. Funiciello et al., 2003b; Kincaid and Griffiths, 2003) and dissipation due to lithospheric bending in the 3rd dimension. Lastly, we do not include a deformable free surface, instead prescribing a free slip upper boundary. Despite such simplifications, our numerical setup allows us to identify the first-order effects of OP structure, and SP hinge weakening, on slab curvature, which is a crucial step prior to moving on to more Earth-like simulations.

2.6.1 Curvature as a function of SP thickness

A key observation in the numerical models is that R_{min} increases with h_{SP} for viscous SPs (Fig. 2.3b). While this relationship has not been explored in models with a coupled

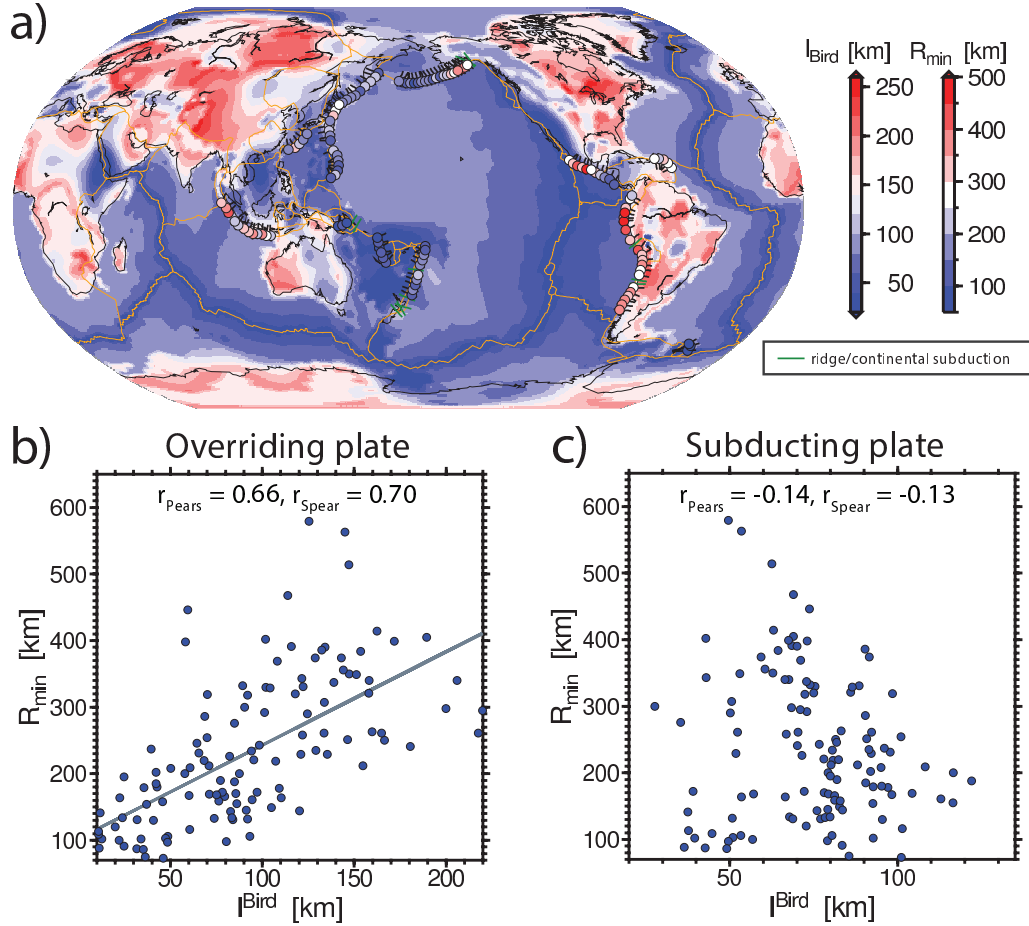


Figure 2.5: Correlation between radius of curvature, R_{min} , and lithospheric thickness extracted from the Bird et al. (2008) model. a) Lithospheric thickness map overlain by R_{min} estimates (points) located at the profile locations used for the spline fit (Buffett and Heuret, 2011). Green profiles indicate anomalous subduction segments omitted from the correlations. b) Scatter plot of R_{min} against the corresponding overriding plate thickness, extracted from the lithospheric thickness map at a position denoted by the on-board extent of the black ticks (i.e. 500 km away from the trench). Grey line shows the linear regression fit and the legend specifies linear (Pearson) and Spearman rank correlation values. c) Scatter plot of R_{min} against the corresponding subducting plate thickness, extracted from the lithospheric thickness map at a position denoted by the out-board extent of the black ticks (i.e. 250 km away from trench).

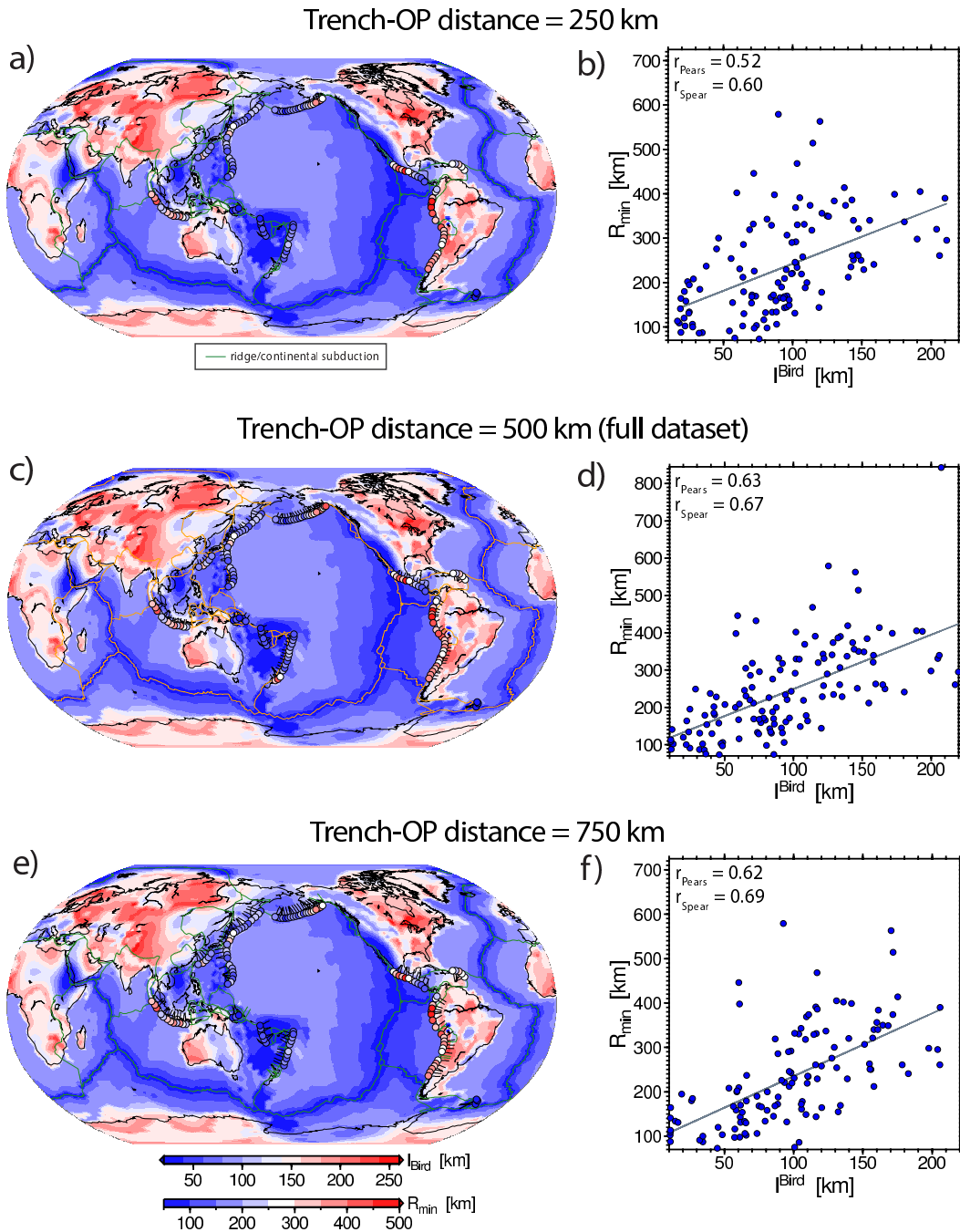


Figure 2.6: Dependence of correlation between radius of curvature, R_{min} , and OP thickness on data correlation settings/parameters. Panels a, b, e, and f illustrate the effect of varying the trench-OP sampling distance, and panels c and d show the data correlation for the reference trench-OP distance (500 km) but without the removal of ridge/continental subduction segments. Left panels show Bird et al. (2008) lithospheric thickness map overlain by Buffett and Heuret (2011) R_{min} estimates, with black ticks connecting the trench to the OP thickness sampling points. Right panels show the corresponding scatter plots, with correlation coefficient values.

OP, previous single-plate studies also find that a thicker, isoviscous SP results in a greater R_{min} (Bellahsen et al., 2005; Schellart, 2009; Irvine and Schellart, 2012). A linear dependence of R_{min} on h_{SP} is reported in the studies aforementioned. The linear h_{SP} - R_{min} trends have variable slopes, presumably due to variable SP rheologies/strengths (Irvine and Schellart (2012): 2.8, Bellahsen et al. (2005): 4.5), relative to the reduced average slope of 1.9 found here. While the trend observed here is near linear, the power-law exponent deviates slightly from 1, with a value of 0.81 (Fig. 2.4a). To first order, this basic positive trend can be understood by assuming that SP bending dissipates a constant proportion of the buoyancy flux energy source (e.g. Conrad and Hager, 1999). A proportionality between the buoyancy flux (left hand side) and a thin-sheet approximation of the bending dissipation (right hand side) gives,

$$\Delta\rho g h_{SP} D v_{SP} \propto \left(\frac{h_{SP}}{R_{min}} \right)^3 \eta' v_{SP}^2, \quad (2.3)$$

where $\Delta\rho$ is the slab-mantle density contrast, g the acceleration due to gravity, D the slab depth, v_{SP} the subducting plate velocity, and η' the slab-mantle viscosity ratio. Rearranging for R_{min} yields:

$$R_{min} \propto h_{SP}^{2/3} v_{SP}^{1/3}. \quad (2.4)$$

We find that v_{SP} has a power-law dependence on h_{SP} , with a power-law exponent of 0.38 (Fig. 2.4b). Substituting this dependence in for v_{SP} into equation 2.4 gives $R_{min} \propto h_{SP}^{0.79}$. Considering the significant data scatter around the $\log(R_{min})$ - $\log(h_{SP})$ regression, this is a reasonable match to the power-law exponent computed directly from the curvature results (Fig. 2.4a: 0.81), which suggests that the viscous subducting plates satisfy the above energy balance. However, as discussed previously and shown in Figure 2.5a, there does not appear to be a positive correlation between h_{SP} and R_{min} on Earth (Buffett and Heuret, 2011; Fourrel et al., 2014).

Plasticity reduces the absolute value of R_{min} , relative to that of the viscous SPs. The effective viscosity of the bending region is reduced in visco-plastic SPs, and so this observation of reduced R_{min} can be understood by previous work (Bellahsen et al., 2005; Di Giuseppe et al., 2008; Schellart, 2008a, 2009; Capitanio et al., 2009) which demonstrates that viscous SPs with reduced bulk η' have reduced R_{min} . This is in-line with the suggestion of Capitanio

et al. (2007, 2009) that bending dissipation is controlled by the magnitude of slab pull, with R adjusting to produce a particular bending dissipation which is independent of SP rheology.

A plastic viscosity component also significantly reduces the strength of the dependence of R_{min} on h_{SP} (Fig. 2.3b), more in-line with the apparent absence of a positive trend on Earth. If one assumes a similar proportionality between buoyancy flux and bending dissipation for a plastic plate with constant yield stress, then $R_{min} \propto h_{SP}$ is expected (Buffett, 2006). Since the observed relationship is not linear (Fig. 2.3b), the assumption that bending dissipation is a constant fraction of the buoyancy flux is likely violated. While the inclusion of a laboratory derived composite rheology with pressure-dependent yielding may improve the agreement between observations and model systematics, we use our simplified modeling approach to propose that weakening of the SP bending region is an important component in diminishing the positive correlation between R_{min} and h_{SP} expected for viscous SPs. For a plastic SP, the bending stresses in the hinge can become insufficient to balance the torque due to buoyancy (Buffett and Becker, 2012). Thus, Buffett and Becker (2012) propose that forces on the surface of the slab, potentially associated with the low viscosity crustal channel and/or the mantle wedge, must play an essential role in balancing the gravitational torque.

2.6.2 Curvature as a function of OP thickness

We find that the overriding plate thickness, h_{OP} , plays indeed a significant role in dictating SP curvature. As h_{OP} increases, the length of the plate interface, composed of weak “crustal” material, increases. This weak channel, as well as the adjacent portion of the OP/mantle wedge, is under low dynamic pressure (Fig. 2.7), thus preventing the SP from detaching from the OP. For large h_{OP} , this prohibits the curvature, R_{min} , from reducing significantly below that initially imposed (R_{init}). On the other hand, because the OP is relatively strong and contains no rheological weakening, R_{min} cannot increase above R_{init} for thick OPs, as this requires deforming the OP. Therefore, for $h_{OP} = 125$ km, all SP rheologies result in comparable R_{min} . On Earth, this is analogous to the initial geometry of a subduction zone interface/thrust controlling the subsequent curvature evolution for thick OPs, as initially proposed by Jarrard (1986).

While R_{min} is strongly affected by R_{init} for thick OPs, this is not the case for thin OPs. As h_{OP} is reduced, the plate interface control reduces, and slab curvature tends towards

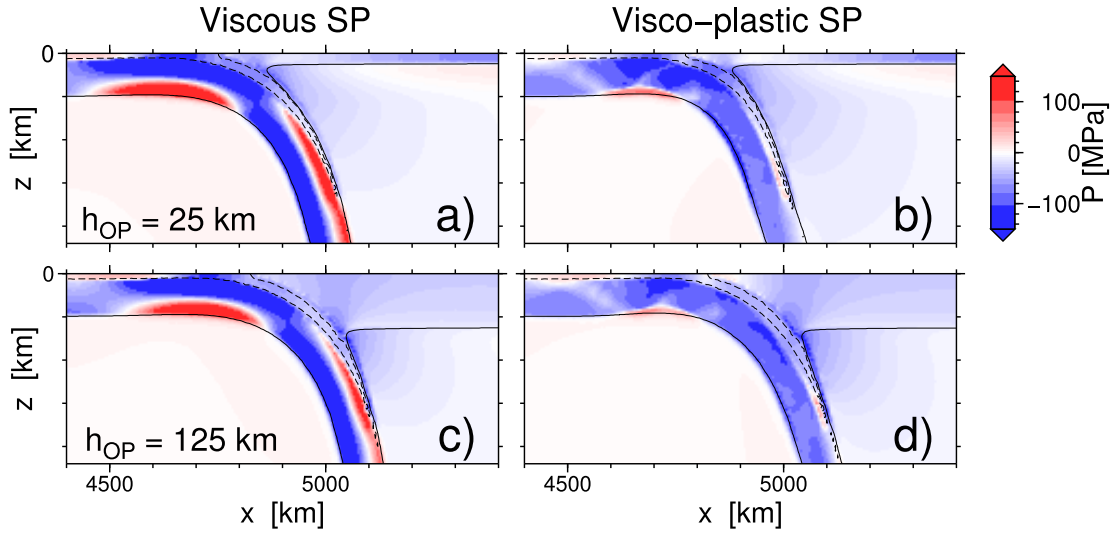


Figure 2.7: Dynamic pressure field for model time-steps shown in Figure 2.1: The left column shows models with a viscous subducting plate rheology (a: $h_{OP} = 25$ km, c: 125 km), and the right column shows models with a visco-plastic subducting plate rheology with $\tau_y = 75$ MPa (b: $h_{OP} = 25$ km, d: 125 km). $h_{SP} = 100$ km in all models shown. The weak crustal layer is outlined with a black dashed line.

the single-plate R_{min} (i.e., without OP influence). For visco-plastic models, as discussed in the previous section, slabs have a naturally reduced R_{min} relative to the isoviscous SPs. Thus, plastic models have reduced curvature for low h_{OP} which, because R_{min} is relatively constant between different rheologies for high h_{OP} , results in a strong positive trend between R_{min} and h_{OP} , relative to that observed for isoviscous SPs. The strength of this trend is dependent on the interface geometry initially imposed, R_{init} (Fig. 2.8). Visco-plastic models with a lower R_{init} of 200 km have a weaker positive trend because R_{min} is reduced for the high h_{OP} values, where it is strongly influenced by the plate interface geometry, and only slightly modified for low h_{OP} (25 km). However, for all R_{init} explored (200 km to 300 km), the positive trend is always stronger in models with visco-plastic SPs relative to purely viscous SPs (Fig. 2.8). In addition, an R_{init} of 300 km is required to produce a positive h_{OP} - R_{min} trend for the viscous SP models, which is significantly greater than the average R_{min} of ≈ 200 km estimated for Earth (Buffett and Heuret, 2011).

We infer a positive trend between h_{OP} and R_{min} in natural subduction zones (Fig. 2.5b). Informed by the modeling work, we suggest that this is due to a combination SP hinge weakening (e.g. Ranero et al., 2003, 2005), and the pre-existing geometry of the plate

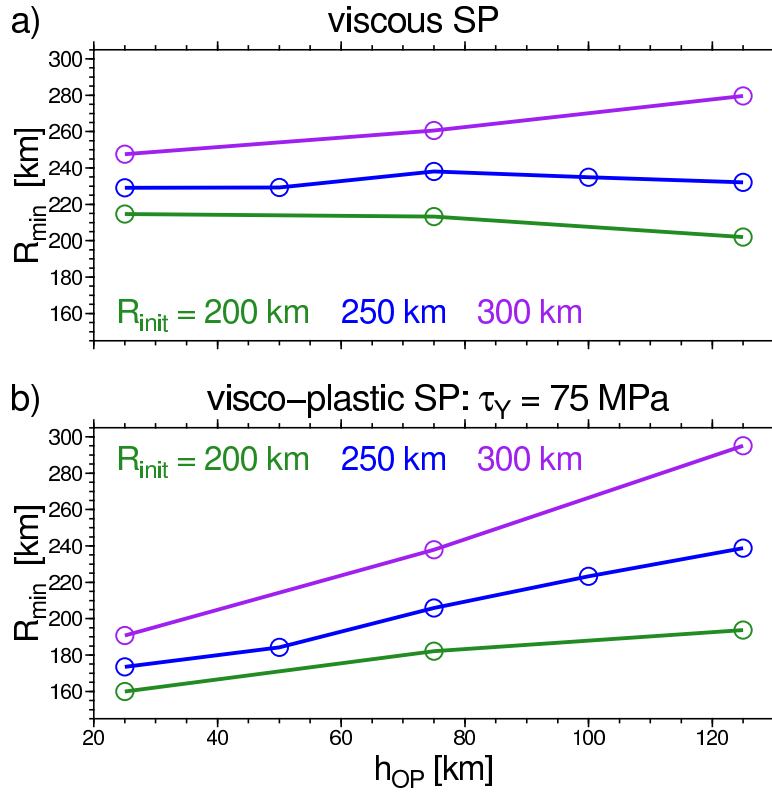


Figure 2.8: Dependence of R_{min} on h_{OP} for variable initial (center-line) radius of curvature (R_{init}) for both viscous (a) and visco-plastic SPs (b). All models have $h_{SP} = 100$ km. R_{min} is computed at an equivalent slab-tip in each model ($z_{slab} = 760$ km). Note, the reference R_{init} is 250 km (blue).

interface (e.g. Jarrard, 1986). Due to the large uncertainty inherent in estimating lithospheric thickness from global tomographic models (e.g. Fischer et al., 2010), the data compilation serves only to investigate the first-order nature of the relationship between h_{OP} and R_{min} on Earth, and so we have not attempted to interpret the specific functional form of the regression (Fig. 2.5b). In addition, we expect contributions from other OP mechanical properties that are coupled to h_{OP} , such as the OP density which exerts a control on the lithostatic pressure gradient between the two plates. However, the correlation of R_{min} and h_{OP} is significantly stronger than that observed between R_{min} and h_{SP} . It therefore appears that the OP structure is playing a key role in dictating curvature of the SP, which is compatible with the positive correlation between R_{min} and h_{OP} observed only for visco-plastic SPs (Fig. 2.3a).

2.7 Conclusions

We used numerical subduction models to investigate how the addition of a plastic rheology component modifies the dependence of subducting plate curvature (R_{min}) on subducting and overriding plate thickness (h_{SP} and h_{OP}), relative to the systematics observed for the isoviscous SPs typically considered. Plasticity is found to reduce the strength of the positive correlation between h_{SP} and R_{min} and increase the strength of the positive trend between R_{min} and h_{OP} . Such a positive trend, between R_{min} and h_{OP} , appears to be present on Earth. We therefore suggest that the structure of the overriding plate, and plastic weakening of the bending region of the subducting slab, are integral components in dictating subducting slab curvature on Earth. In addition to being a key control on the amount of energy dissipated during plate bending, the evolution of slab curvature has possible implications for long term force transmission on subduction zone interfaces.

Chapter 3

The effect of a stress-dependent mantle viscosity on slab rollback rates

3.1 Summary

Subduction on Earth is partitioned between trench-wards plate motion, and lateral slab migration (i.e., trench retreat and advance), in an absolute plate motion (APM) reference frame. We use 3-D, dynamic models of subduction to address the role of a power-law mantle viscosity on rates of trench retreat. For all numerical models tested, we find that a power-law rheology results in reduced rates of trench retreat, and elevated slab dip angles, relative to the equivalent isoviscous mantle model. We analyze the mantle pressure and style of mantle flow, which exhibits only limited variability as a function of mantle rheology, in order to compute estimates of the mantle forces which vary as a function of mantle rheology. We find that the inclusion of a power-law rheology reduces the mantle shear force that resists subducting plate motion to a greater degree than it reduces the dynamic pressure gradient across the slab, which resists trench retreat. Therefore, the inclusion of a power-law mantle rheology favors a shift towards a subduction mode with a reduced trench retreat component, typically a relative reduction of order 25% in our 3-D models. In addition to the inclusion of an overriding plate,

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we suggest that this mechanism may be of importance in reducing the high trench retreat rates often observed in single slab models to levels more in line with the average subduction partitioning observed on Earth at present (i.e., trench velocity $\leq$ plate velocity), for most APM reference frames.

3.2 Introduction

The subduction of negatively buoyant slabs constitutes the primary driving force of plate tectonics (e.g. Jacoby, 1973; Forsyth and Uyeda, 1975). In addition to trench-wards subducting plate motion, subduction can also occur via lateral migration of the subducting plate hinge/trench, in a deep mantle, absolute plate motion (APM) reference frame. While lateral subducting plate migration typically occurs in the subducting plate (SP) direction, i.e. “slab rollback”, advancing trench motion towards the overriding plate (OP) also occurs at a limited number of trenches (e.g. Elsasser, 1971; Carlson and Melia, 1984; Jarrard, 1986). While the choice of absolute reference frame complicates the deduction of trench migration rates and directions (e.g. Heuret and Lallemand, 2005; Lallemand et al., 2008; Schellart et al., 2008; Funicello et al., 2008; Becker et al., 2015), trench migration rates (V_T) are, on average, less than $\sim 50\%$ of total convergence rates (V_C), albeit with large spread (e.g. Schellart et al., 2008; Becker and Faccenna, 2009). SP motion (V_P) accommodates the remainder of the subduction rate, i.e., $V_C = V_P + V_T$. The rates and direction of trench motions have direct consequences for the style of back arc deformation (e.g. Uyeda and Kanamori, 1979; Lallemand and Heuret, 2005), and understanding “typical” V_T / V_C ratios is a prerequisite for using them to constrain APM models (e.g. Williams et al., 2015).

Numerical and analog subduction models, within which the trench migrates self-consistently (i.e., “dynamic models”), have been used extensively to explore the mechanical controls on subducting plate kinematics. The total convergence velocity can, to first order, be predicted by considering the interplay of the slab pull driving force and the resisting forces associated with slab bending and mantle drag (e.g. Conrad and Hager, 1999; Bellahsen et al., 2005; Capitanio et al., 2007). However, what dictates how convergence partitions between subducting plate, V_P , and trench motion, V_T , remains difficult to isolate. Modeling studies have typically focused on isolated subducting slabs, neglecting the effect of an OP. These

studies highlight, in particular, the strong control of slab strength (e.g. Enns et al., 2005; Belahsen et al., 2005; Di Giuseppe et al., 2008; Funiciello et al., 2008; Faccenna et al., 2009) on V_T . In addition to being consistent with observations relating to plate flexure (Billen and Gurnis, 2005), plate-mantle viscosity contrasts of 100-1000 are generally required to produce both the variability in trench migration regimes, and average subduction partitioning (i.e., typically $V_T \leq V_P$) inferred to occur on Earth (e.g. Schellart, 2004; Funiciello et al., 2008; Becker and Faccenna, 2009). However, even for slabs within this intermediate strength range, single slab studies often appear to favor trench motion over plate motion, to a greater degree than that observed on Earth (e.g. Funiciello et al., 2003b; Stegman et al., 2006; Schellart et al., 2007).

For a given slab strength, a significant trench retreat reducing feature of the subduction system appears to be the presence of an overriding plate (OP). Dynamic models that incorporate an overriding plate are now routinely studied in 2-D (e.g., Capitanio et al., 2010a; Čížková and Bina, 2013; Garel et al., 2014; Sharples et al., 2014; Holt et al., 2015a; Petersen et al., 2015), and increasingly so in 3-D (e.g., Yamato et al., 2009; Schellart and Moresi, 2013; Jagoutz et al., 2015; Duarte et al., 2013; Chen et al., 2015). The presence of an OP modifies the force balance acting on the SP, which shifts subduction partitioning toward subducting plate motion, i.e., reducing V_T/V_P , as shown in chapter 1 (Holt et al., 2015a). In addition, in 3-D, the width of the trench has been shown to exert a significant control on rates of trench retreat, with wide trenches typically exhibiting reduced rates in numerical models, and on Earth (e.g. Dvorkin et al., 1993; Stegman et al., 2006; Schellart et al., 2007; Stegman et al., 2010). In addition to these factors, we investigate the role that a stress-dependent mantle rheology, a feature typically neglected in dynamic subduction models, has on the partitioning of subduction between V_T and V_P .

The 3-D modeling studies discussed above typically consider a uniform viscosity mantle, and so neglect the role that lateral viscosity variations in the sub-lithospheric mantle play in slab dynamics. Laboratory experiments, however, predict that deformation in large regions of the upper mantle is dominated by dislocation creep, in which the effective viscosity has a power-law dependence on stress (e.g., Karato and Wu, 1993; Hirth and Kohlstedt, 2003). 2-D, numerical models which fix the trench location show that the inclusion of a stress-dependent rheology decreases the mantle wedge viscosity, which reduces hydrodynamic stresses on the upper surface of the slab (Tovish et al., 1978), and so results in an in-

creased slab dip angle (Billen and Hirth, 2005, 2007). However, uncertainty remains regarding whether variations in trench rollback rate, which exert a first order control on slab dip (e.g. Christensen, 1996), would act to reduce this dip-increasing effect. A limited number of 2-D, numerical models that do allow the trench to move freely have been used to investigate how the slab rheology, particularly the strength-limiting yielding stress, influences slab kinematics and slab-interaction with a more viscous lower mantle (e.g., Čížková and Bina, 2013; Garel et al., 2014), but the role of a stress-dependent mantle viscosity on trench rollback has not been explicitly considered in a dynamic setting. While relative reductions in mantle viscosity triggered by a stress-dependent viscosity intuitively result in elevated slab sinking rates, and greater subduction-induced mantle flow velocities (e.g. Jadamec and Billen, 2012; Billen and Jadamec, 2012), the degree to which such localized weakening shifts the convergence partitioning between V_T and V_P , is less clear. We therefore use 3-D, time-dependent subduction models to probe the effect that a power-law mantle viscosity has on the partitioning of subduction between V_T and V_P .

Coupled to the subduction kinematics is the style of subduction-induced mantle flow. Subduction excites poloidal flow in the vertical plane and, in 3-D and in the presence of lateral viscosity variations, toroidal flow. Return flow associated with a near-stationary trench is dominantly poloidal, whereas horizontal motion of a subducting slab, in the form of slab rollback or advance, promotes toroidal flow around the slab edges (e.g., Dvorkin et al., 1993; Buttles and Olson, 1998; Kincaid and Griffiths, 2003; Schellart, 2004; Funicello et al., 2003b, 2004; Stegman et al., 2006; Piromallo et al., 2006). Understanding the dynamic controls on the style of subduction-induced return flow is important both in aiding the interpretation of shear-wave splitting measurements which can probe horizontal/sub-horizontal mantle flow (e.g. Long and Becker, 2010), constraining the mantle pressure field and the intertwined slab kinematics (e.g. Royden and Husson, 2006), and understanding the locations of off-axis volcanoes associated with subduction-induced upwelling flow (e.g. Faccenna et al., 2010). In this section of the study, we firstly characterize how the presence of an overriding plate affects mantle flow partitioning in dynamically evolving subduction, as previously considered for 3-D numerical models which prescribe slab kinematics (Honda, 2009). Then, by fixing the mechanical properties of the subducting plate and varying the mantle rheology, we assess whether the inclusion of a power-law mantle rheology modifies this style of subduction-

induced mantle return flow.

3.3 Methodology

We use the finite-element code CitcomCU (Moresi and Gurnis, 1996; Zhong, 2006) to construct numerical, time-evolving subduction models in both 2-D and 3-D domains. CitcomCU solves the equations governing convection in an incompressible viscous fluid (Boussinesq approximation) with negligible inertia. 2-D models are less computationally expensive and so allow us to initially probe the parameter space more extensively in order to inform our 3-D model parameters, as well as examine the role of lithospheric visco-plasticity. Subsequently comparing the slabs in 2-D models, which are applicable only to very wide trenches, and 3-D slabs with finite trench widths, allows us to discern the role that mantle flow around the slab edge, which can relieve sub-slab pressures (e.g. Stegman et al., 2006; Royden and Husson, 2006), can play in dictating slab kinematics. 3-D models are then employed to verify the applicability of the systematics observed in the 2-D models, and investigate the dynamic evolution of both the mantle flow field, and the viscosity field. The mechanical and rheological parameters are equivalent for the two suites of models which initiate with a deep initial slab (see Figure 3.1). Numerically, the only differences are that the 2-D models contain only one finite element in the 3^{rd} , y -dimension, similar to those models presented in chapters 1 and 2 (Holt et al., 2015a,b), while 3-D models contain many elements (256) in order to resolve a finite width trench and a passive side plate. That aside, the two models contain the same sized and number of finite elements in the vertical plane. Our explanation of the 3-D model setup in the next section is therefore also applicable to the 2-D models presented in section 3.4.1.

3.3.1 Numerical modeling strategy and setup

In order to target the first order effects of a power-law mantle rheology on slab dynamics, we use a simplified geometrical, mechanical, and rheological setup. We consider two active plates, a subducting plate (SP) and an overriding plate (OP), and a passive side plate which is separated from the subduction system plates by 25 km of mantle material. The side plate, while not participating directly in subduction, prevents both unrealistic mantle flow at the surface, and rapid plate shortening in the trench parallel direction (Yamato et al., 2009).

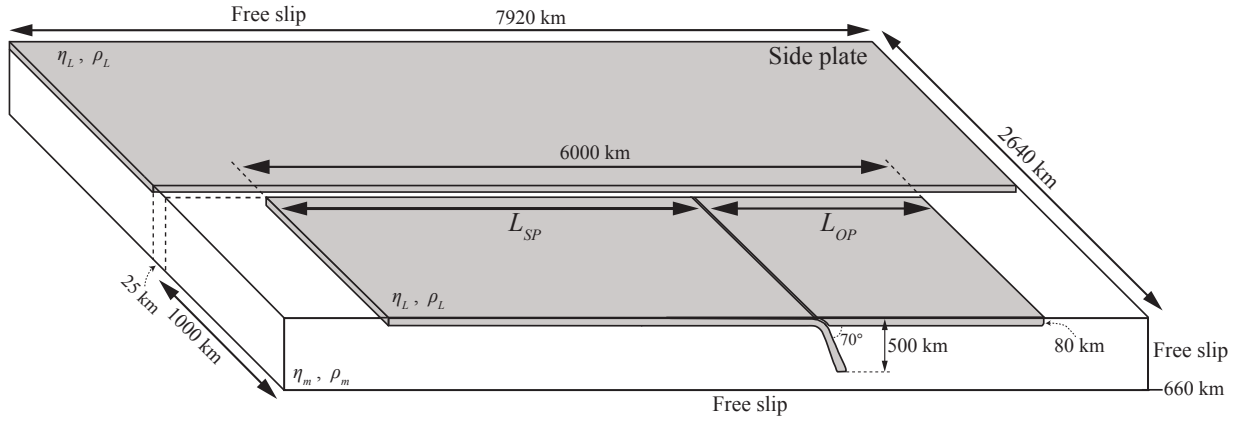


Figure 3.1: Schematic illustration of the initial conditions for the 3-D numerical models. The 2-D numerical models have an equivalent setup but neglect the y -dimension (i.e., dimension with length = 2640 km). Additionally, a subset of the 2-D numerical models has different initial slab geometry, where only a shallow proto-slab is initially prescribed ($z < 150$ km).

The side plate also provides a convenient, stationary reference frame for velocities measured in the subduction systems (e.g. trench migration rates). The trench-perpendicular lengths of the SP and OP exerts a strong control on rollback rates, and so we examine variable plate lengths, relative to the vertical (upper mantle) dimension of 660 km, to assess the effect of a power-law viscosity on models with a wide range of slab kinematics.

In order to investigate the first order physics, we simplify our model setup by ignoring thermal diffusion (e.g. Enns et al., 2005). We therefore consider a density and viscosity field that is dependent purely on composition, where the compositional field is advected using tracer particles (McNamara and Zhong, 2004). The domain size is 7920 km (length, x) $\times$ 660 km (height, z) $\times$ 5280 km (width, y), and all boundaries are mechanically free slip. We initially place 40 tracers within each of the elements, and separate sets of tracers are used to define the lithospheric plates, and the weak “crustal” layer that decouples the lithospheric plates. The finite element dimensions vary from 6 km in regions which are both in the upper 200 km of the domain and proximal to the trench, to 14 km elsewhere. Due to concerns that a free-slip boundary may render the slab tip overly mobile atop the upper-lower mantle boundary, we additionally test the effect of a no-slip lower boundary which approximates a high viscosity lower mantle.

We consider plates with a finite width of 2000 km. For each of the models presented,

we take advantage of the plate symmetry about $y = 0$ (i.e. trench-perpendicular horizontal axis), which enables us to half the computational domain (i.e. $y = 2640$ km). We consider compositional lithospheric plates with a constant density contrast, relative to the sub-lithospheric mantle, of 85 kg/m^3 and a constant thickness of 80 km. The sublithospheric mantle has a reference density of 3300 kg/m^3 .

We model the sub-lithospheric mantle with a composite Newtonian and power-law rheology (e.g. Zhong et al., 1998). We neglect the temperature and pressure dependence of both the Newtonian and power-law viscosity components, which are typically assumed to follow an Arrhenius, exponential dependence. Because the Newtonian (i.e., diffusion creep) and power-law viscosity (i.e., dislocation creep) components are known to have different temperature and pressure dependencies (e.g. Karato and Wu, 1993; Hirth and Kohlstedt, 2003), this simplification likely modifies the spatio-temporal location of the regions within each of the two rheological regimes. However, numerical modeling studies that do include the full laboratory derived rheological formulations tend to observe dislocation creep activation in, qualitatively, the same regions as in our models (Billen and Hirth, 2005, 2007; Garel et al., 2014). Thus, our simplified parameterization serves as a useful starting point to understand the effect of stress-dependent mantle viscosity reductions on slab dynamics in 3-D.

Our effective, asthenospheric viscosity is composed of a isoviscous Newtonian component ($\eta_N = 2.8 \cdot 10^{20} \text{ Pa s}$) and a power-law component, η_P , given by,

$$\eta_P = \eta_N \left(\frac{\dot{\epsilon}_{II}}{\dot{\epsilon}_{II,T}} \right)^{\frac{1-n}{n}}, \quad (3.1)$$

where $\dot{\epsilon}_{II} = \sqrt{\dot{\epsilon}_{ij}\dot{\epsilon}_{ij}/2}$ is the second invariant of the deviatoric strain rate (Einstein summation convention), and n is the power-law exponent ($= 3.5$). $\dot{\epsilon}_{II,T}$ is the transition strain rate, at which the viscosity components are equal (i.e., $\eta_P = \eta_N = 2.8 \cdot 10^{20} \text{ Pa s}$). We therefore focus on the effect of varying $\dot{\epsilon}_{II,T}$ which controls the proportion of the domain within the power-law regime. We use a velocity field convergence criterion of 0.01 for our power-law iterations. For the power-law models which initiate with a notch, we use a slightly reduced Newtonian viscosity ($2.5 \cdot 10^{20} \text{ Pa s}$). An effective mantle viscosity is computed as,

$$\eta_{eff} = \frac{\eta_N + \eta_P}{\eta_N \eta_P} = \left(\frac{1}{\eta_N} + \frac{1}{\eta_P} \right)^{-1} \quad (3.2)$$

While we mainly consider isoviscous lithospheric plates, we additionally test the effect of visco-plastic subducting plates in the 2-D models. In this case, an additional viscosity component, η_{yield} , is computed from a prescribed yield stress, τ_{yield} , as follows (e.g. Schott and Schmeling, 1998),

$$\eta_{yield} = \frac{\tau_{yield}}{2\dot{\epsilon}_{II}}, \quad (3.3)$$

where $\tau_{yield} = (a\sigma_l + b)\lambda$, and a is the friction coefficient (0.6), b is the cohesion (60 MPa), σ_l is the lithostatic pressure, and λ is a constant pre-factor (0.15), comparable to that used in previous studies (e.g. Enns et al. 2005: 0.1; Di Giuseppe et al. 2008: 0.08). η_{yield} is then added as another component in the geometric mean expression for η_{eff} (eq. 3.2). To facilitate decoupling of the SP from the OP, we insert a 15 km thick, weak ($= \eta_N$) crustal layer within SP (e.g. Běhouňková and Čížková, 2008). The crust, defined by compositional tracers, is linearly tapered full thickness at a depth of 200 km (and above), to zero thickness below a depth of 400 km, in order to avoid stress discontinuities. For more details about the implementation of the compositional crust, and resolution tests for similar 2-D numerical models see chapter 1 (Holt et al., 2015a).

Subduction is initiated by allowing the SP to extend below the OP with a center-line radius of curvature (R_{init}) of 150 km. Below the curved portion of the initial lithospheric geometry, we allow the plate to extend to a depth of 500 km with an initial dip angle of 70° (Fig. 3.1). The initial, deep placement of the slab-tip near the model base allows the models to reach a near steady-state for a significant length of time, relative to the more transient model evolution observed in models that are initiated by a shallow proto-slab. More importantly, however, is that such an initial condition results in a suite of models which have a consistent style of evolution, with all slabs bending backward on top of the impermeable base of the domain, as opposed to slabs which bend forward and roll over the lower boundary. While this initial condition reduces the observed variability in slab kinematics, it allows us to zoom into rheological effects on the slab kinematics for the mode which appears most akin to subduction on Earth. Otherwise, much of the kinematic variability after slab interaction with the model base is a direct result of the different styles of slab bending. However, for the 2-D models, we do examine a suite of models in which the initial slab is restricted to shallow depths of < 150 km.

Table 3.1: Numerical model parameters

Symbol	Parameter	Value	Units
Geometry			
h	Domain height (z)	660	km
l	Domain length (x)	7920	km
w	Domain width (y)	5280	km
w_T	Trench width	2000	km
h_l	Lithospheric thickness	80	km
h_{crust}	Crustal thickness	15	km
L_{SP}	Subducting plate length	2000, 3000, 4000	km
L_{OP}	Overriding plate length	2000, 3000, 4000	km
R_{init}	Initial radius of curvature	150	km
z_{notch}	Initial slab depth	150, 500	km
θ_{init}	Initial slab dip angle	70	°
Mechanical and rheological properties			
$\Delta\rho$	Plate-mantle density contrast	85	kg/m ³
ρ_0	Mantle density	3300	kg/m ³
η_{crust}	Crust viscosity	$2.8 \cdot 10^{20}$	Pa s
η_N	Newtonian viscosity	$2.8 \cdot 10^{20}$	Pa s
n	Power-law exponent	3.5	-
$\dot{\epsilon}_{II,T}$	Transition strain rate	$10^{-15}, 5 \cdot 10^{-15}, 10^{-14}, 10^{-13}$	s ⁻¹
Ra_C	Compositional Rayleigh number	$8.55 \cdot 10^5$	-

3.4 Results

3.4.1 2-D Models

We now consider the effect of a power-law sub-lithospheric mantle on 2-D slabs, which are, by definition, equivalent to slabs with infinite width trenches. Numerical models which are initiated by a shallow proto-slab are presented in Figure 3.2. Decreasing the transition state increases the proportion of the asthenosphere deforming dominantly in the power-law viscosity regime (i.e., eq. 3.1). As expected, power-law deformation is focused in the portions of the asthenosphere next to the slab and surface plates (e.g. Billen and Hirth, 2007; Garel et al., 2014). Here, high strain rates result from the shearing of the adjacent asthenosphere by the mobile plates/slab. Additional regions that undergo, more moderate, power-law deformation are the return-flow upwellings that occur below the trailing edge of the subducting plate, and above the slab tip. For a transition strain rate of 10^{-15} s^{-1} , power-law viscosities are in some places reduced by over an order of magnitude, particularly directly adjacent to the subducting slab (Fig. 3.2e). For a transition strain rate of 10^{-14} s^{-1} , the more spatially limited power-law regions have viscosities reduced by a factor of ~ 5 , relative to the Newtonian viscosity (Fig. 3.2c).

Figure 3.2 clearly shows the strong dependence on slab morphology, particularly slab dip angle, on the transition strain rate. As the transition strain rate decreases, i.e. as the non-Newtonian component becomes more dominant, the slab dip angle gradually increases up until the slab bends forward over the model base, for a transition strain rate of 10^{-15} s^{-1} (Fig. 3.2e). As discussed by Ribe (2010), whether the slab leans backwards or forwards at an impermeable model base is a function of the dip angle of the slab, obtained during the prior free-sinking phase, as it impinges on the boundary. For an intermediate transition strain rate of $5 \cdot 10^{-15} \text{ s}^{-1}$, the slab impinges on the lower boundary with a near-vertical dip angle and so, after initially folding forwards, eventually leans backwards. For higher transition strain rates, the slabs lean directly backwards on top of the lower boundary.

Additionally plotted in Figure 3.2 are mantle flow velocities. The mantle flow becomes, on average, faster as the transition strain rate reduces and so, to facilitate comparison of the mantle flow regime, we non-dimensionalize the mantle velocities using a Stokes velocity computed using an average mantle viscosity. A Stokes sinking velocity is computed

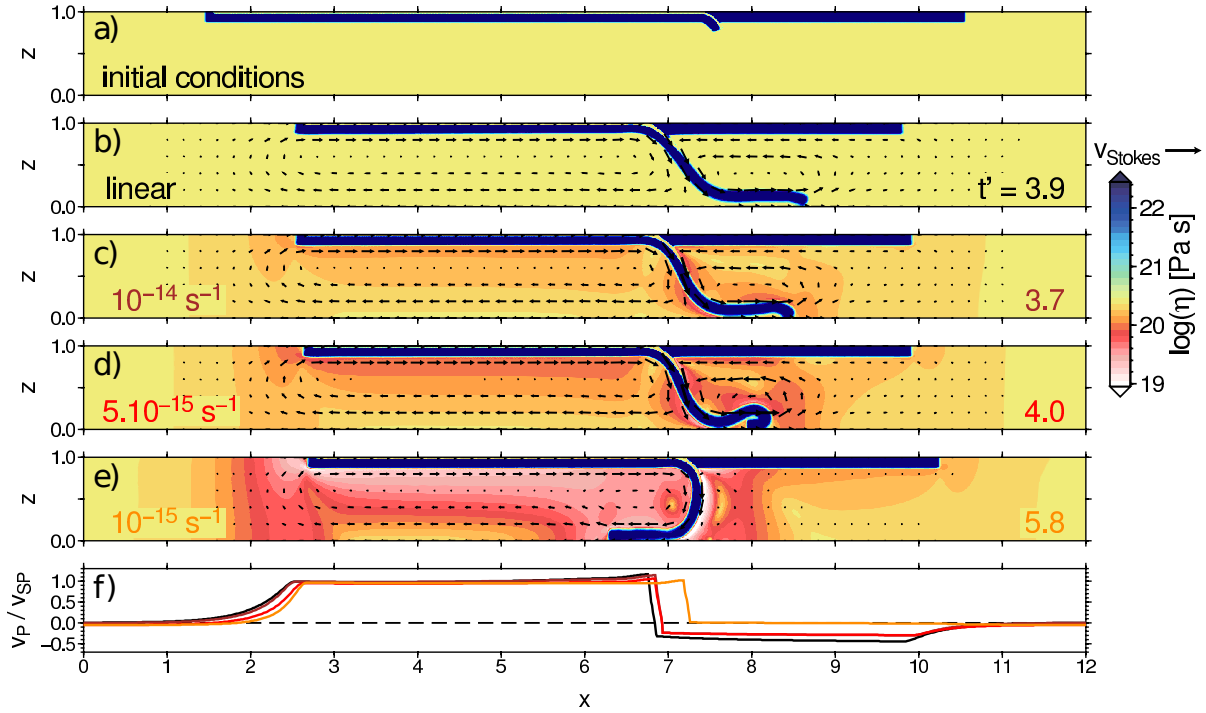


Figure 3.2: Viscosity field snapshots showing the suite of 2-D numerical models initiated with a shallow initial proto-slab, at broadly equivalent time-steps based on slab morphology. In addition to panels showing the models with variable transition strain rates (c-e), the equivalent isoviscous mantle model (b) and initial conditions (a) are shown. Overlain are mantle velocity vectors normalized by a typical Stokes velocity. The model times (bottom right corner) are computed by dividing the dimensional time by a characteristic transient time computed using the Stokes velocity ($t' = t/t_{Stokes} = tv_{Stokes}/660 \text{ km}$). The bottom panel, f), compares the plate velocities, normalized by the average SP velocity, in the four models. Relative to all other models presented, these models have a slightly reduced reference viscosity ($2.5 \cdot 10^{20} \text{ Pa s}$).

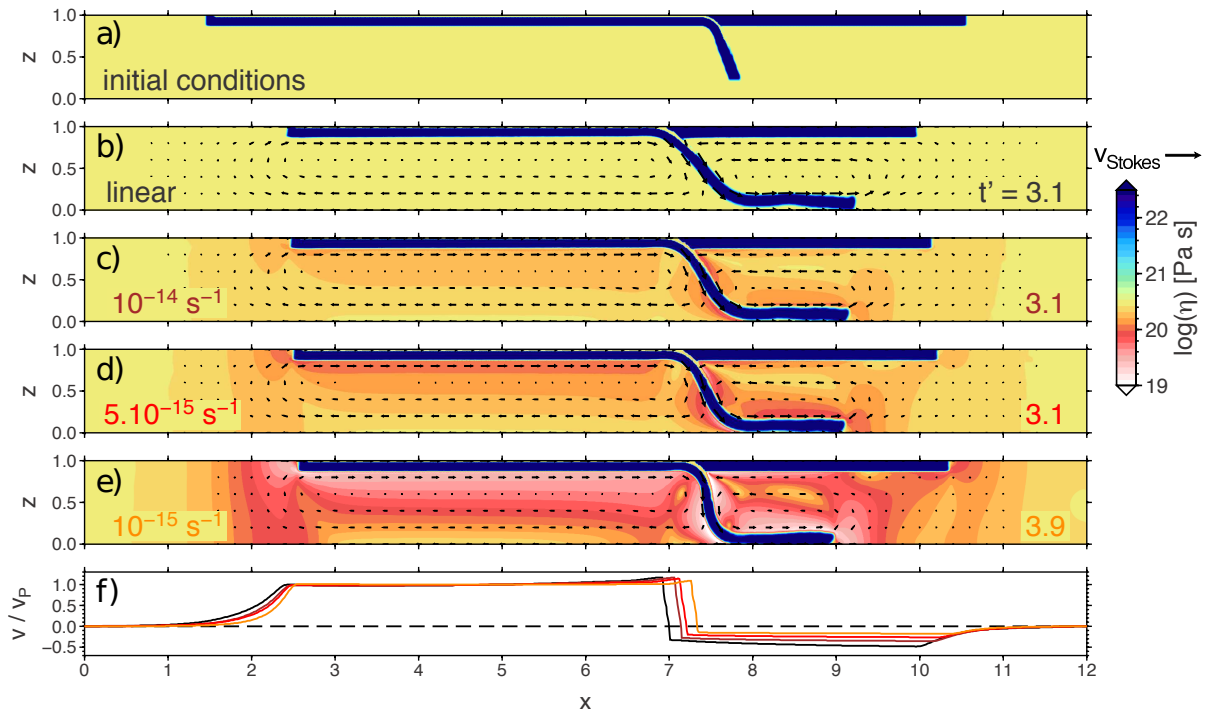


Figure 3.3: As in Figure 3.2, but for the 2-D models with a deeper initial slab geometry.

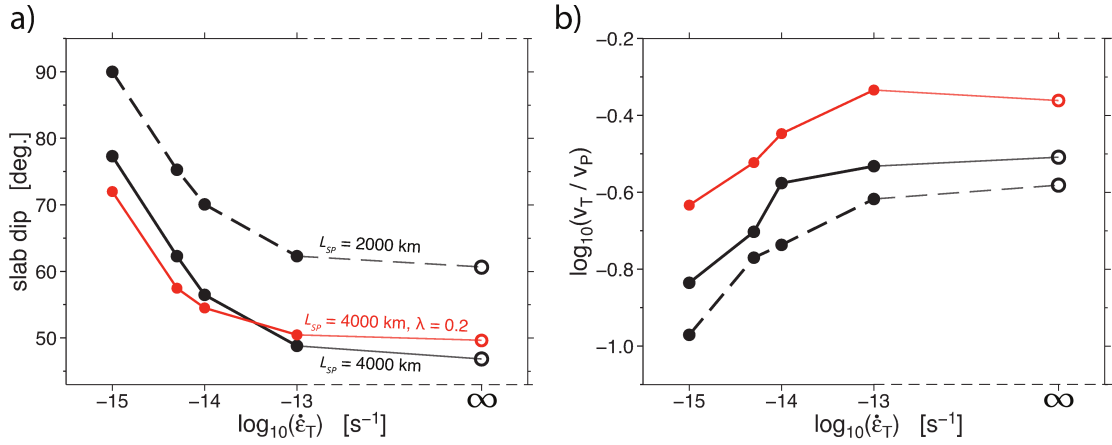


Figure 3.4: Plate kinematics as a function of transition strain rate for the 2-D models with deep initial slab geometry, extracted at time-steps equivalent to those shown in Figure 3.3. In addition to the long SP models shown in Figure 3.3 (i.e., initial $L_{SP} = 4000$ km: black, solid lines), kinematics from short initial SP models (i.e., $L_{SP} = 2000$ km: black, dashed lines), and long SP models with plasticity (i.e., $L_{SP} = 4000$ km and $\lambda = 0.15$: red lines) are plotted. The plasticity suite of models are plotted at an earlier stage of subduction, but still after flattening on the model base, due to later-stage instability (i.e. OP subducts) for the lowest transition strain rate model. Shown is, a), average slab dip angle computed between depths of 200 and 460 km and, b), log of the ratio of trench retreat velocity and SP velocity. The kinematics extracted from the equivalent Newtonian models are placed at an arbitrary, “infinite” transition strain rate, x -location, for illustrative purposes.

as $V_{Stokes} = (2\Delta\rho gh_l^2)/(9\eta_{avg.})$, where $\eta_{avg.}$ is the arithmetic mean viscosity extracted from a vertical, sub-lithospheric profile at a distance of 750 km to the left of the trench. In addition, plate velocities are plotted in Figure 3.2f. To highlight variability in OP velocity, which is approximately equal to the trench velocity, we normalize the plate velocities by the model average SP velocity. In addition to the increasing slab dip angle, the normalized trench velocity decreases as the power-law viscosity component becomes more dominant. For a transition strain rate of 10^{-15} s^{-1} , the trench/OP is near-stationary, although this is likely a result of the forward bending mode of subduction that this slab attains.

As discussed in the previous section, we additionally examine numerical models with initial conditions that result in an equivalent backwards-bending subduction mode for all of the rheologies considered to make comparisons more straight-forward (Figure 3.3). While the magnitude of slab morphology variability reduces, we observe the same systematic increase in slab dip angle for models with an increasingly dominant power-law viscosity

component. In addition, within this single backward-bending subduction mode, the increase in slab dip angle is coupled with a reduction in trench motion rate, relative to that of the SP (Fig. 3.3f). For equivalent models times, and $L_{SP} = 4000$ km, Figure 3.4 shows the variation of the slab dip angle, and the log ratio of the trench retreat rate (V_T) and plate motion rate (V_P), as a function of the transition strain rate. Relative to the reference model with an isoviscous mantle, the dip angle increases by $\sim 30^\circ$ and $\log_{10}(V_T / V_P)$ decreases by ~ 0.35 in the non-Newtonian model with the lowest transition strain rate (10^{-15} s^{-1}). To verify the more general applicability of these general results, we have also examined suites of models with shorter initial SPs (i.e., $L_{SP} = 2000$ km), and a plastic lithospheric viscosity component ($\lambda = 0.15$). For isoviscous mantle models with a similar setup, reducing L_{SP} (i.e. increasing L_{OP}) increases slab dip angle, and reduces V_T / V_P . The addition of plasticity (i.e., viscosity reduction in the bending region) does not change the basic trends observed, but increases the relative rate of trench motion. Interpretation of the models which include plastic yielding is complicated by the feedback between yielding and the degree of power-law flow in the sub-lithospheric mantle. As the power-law transition strain rate is reduced and the mantle moves further into the power-law regime, slab bending stresses increase which, in turn, increases the extent of the SP hinge which yields. We expect that this behavior is responsible for the slight decrease in V_T / V_P observed at high transition strain rates. While further work is required to understand how these non-linear rheological interactions modify slab dynamics, we show, to first order, that the sense of variation in slab dip and trench retreat rate, as a function of mantle transition strain rate, is equivalent for models with plastic yielding, and also for reduced L_{SP} .

3.4.2 3-D Models

We now turn our attention to 3-D numerical models to explore whether the 2-D results are robust in 3-D. Additionally, we probe the effect that such a non-linear rheology has on 3-D mantle flow.

Model evolution

We analyze 3-D models with equivalent properties, and initial conditions, to the suite of 2-D models shown in Figure 3.3, with $L_{SP} = 4000$ km. Figure 3.5 shows the viscosity, and mantle flow, fields for the models both with an isoviscous mantle and a composite power-law rheology with a transition strain rate of $5 \cdot 10^{-15} \text{ s}^{-1}$. As in the 2-D models, the dip angle is elevated in the power-law mantle models, and the power-law regime is active primarily in the portion of the asthenosphere being directly sheared by adjacent plates/slab. In the 3-D models, however, a significant portion of the subduction-induced mantle return flow occurs in the form of toroidal flow around the slab edge (e.g. Buttles and Olson, 1998; Kincaid and Griffiths, 2003; Schellart, 2004; Funiciello et al., 2004), which induces strain rates large enough (i.e. $> 5 \cdot 10^{-15} \text{ s}^{-1}$) for deformation to occur within the power-law regime. We have additionally examined equivalent models within which the OP is absent (Fig. 3.6), and where the base of the model has a no slip boundary condition (Fig. 3.7), as opposed to free slip. In the single slab experiments, the lack of an OP promotes both the development of slab curvature and slab/trench rollback (e.g. Stegman et al., 2006; Schellart et al., 2007; Stegman et al., 2010), relative to the models which include an OP. Therefore, power-law deformation associated with rollback-driven toroidal flow is greater in the single slab model (cf. Figs. 3.5b, 3.6b). In the model with a no slip base, the subducting slab morphology is similar to that observed in the free slip models, but the subduction rates are reduced and so the magnitude and extent of power-law deformation reduces.

As for the 2-D models, we consider models with three different initial SP lengths (Fig. 3.8). While the power-law deformation is generally located preferentially below the SP, as opposed to the OP, a shift towards more wide-spread power-law deformation below the OP occurs for greater L_{SP} . This occurs because as L_{SP} increases, and L_{OP} decreases, the OP moves more rapidly due to slab/trench rollback which results in greater shear strain rates below the OP. Where power-law deformation, and the associated viscosity reduction, occurs within the subduction system dictates the spatial distribution of basal traction on the subduction system which, in turn, dictates the overall effect of a power-law mantle on slab kinematics.

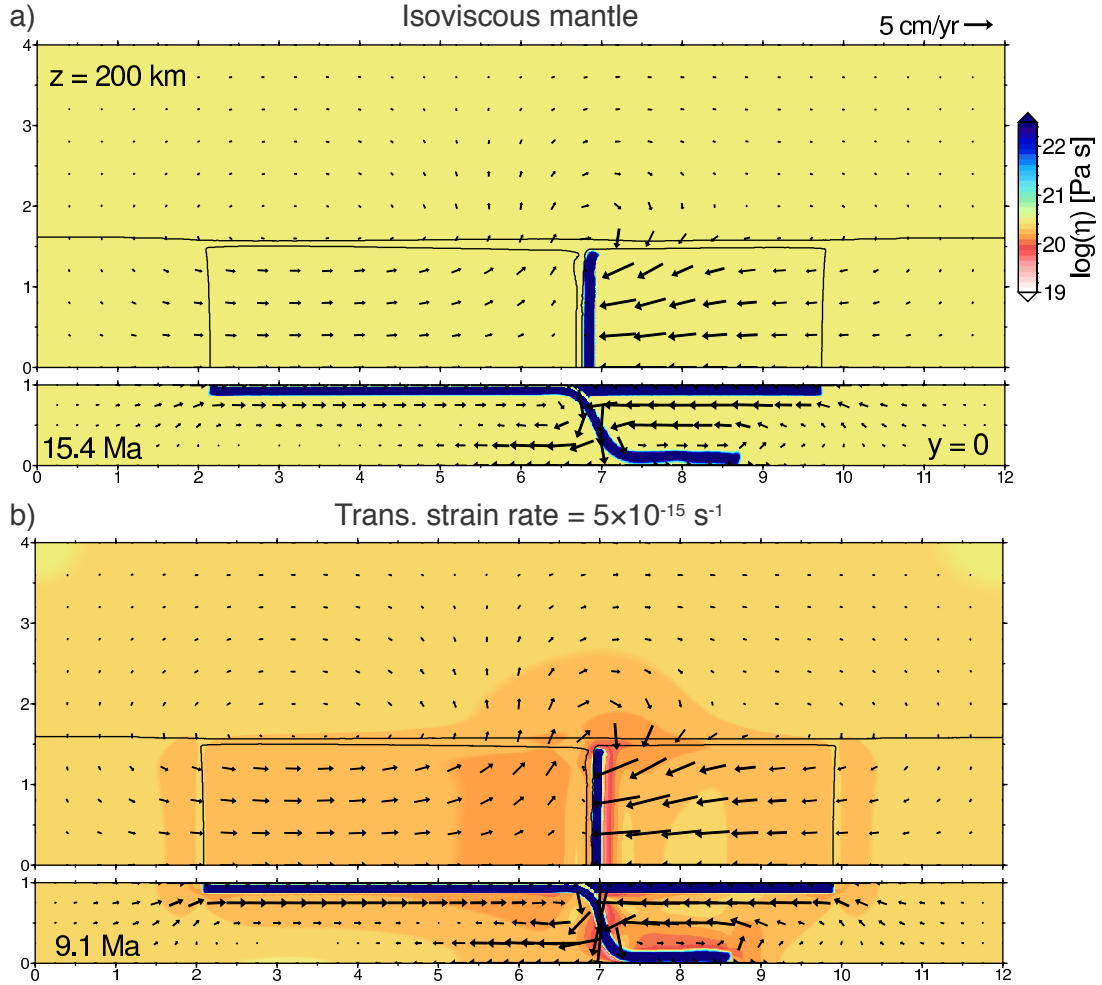


Figure 3.5: Comparison of viscosity and mantle flow fields for 3-D models ($L_{SP} = 4000$ km) with, a), an isoviscous mantle and, b), a composite power-law/Newtonian mantle rheology with a transition strain rate of $5 \cdot 10^{-15} \text{ s}^{-1}$. Both models are plotted at broadly equivalent time steps based on slab morphology. For each model the upper panel shows a horizontal, sub-lithospheric slice ($z = 200$ km), and the lower panel shows a vertical slice through the model/plate mid-plane ($y = 0$). In the horizontal slices, the surface plate geometries are overlain in black.

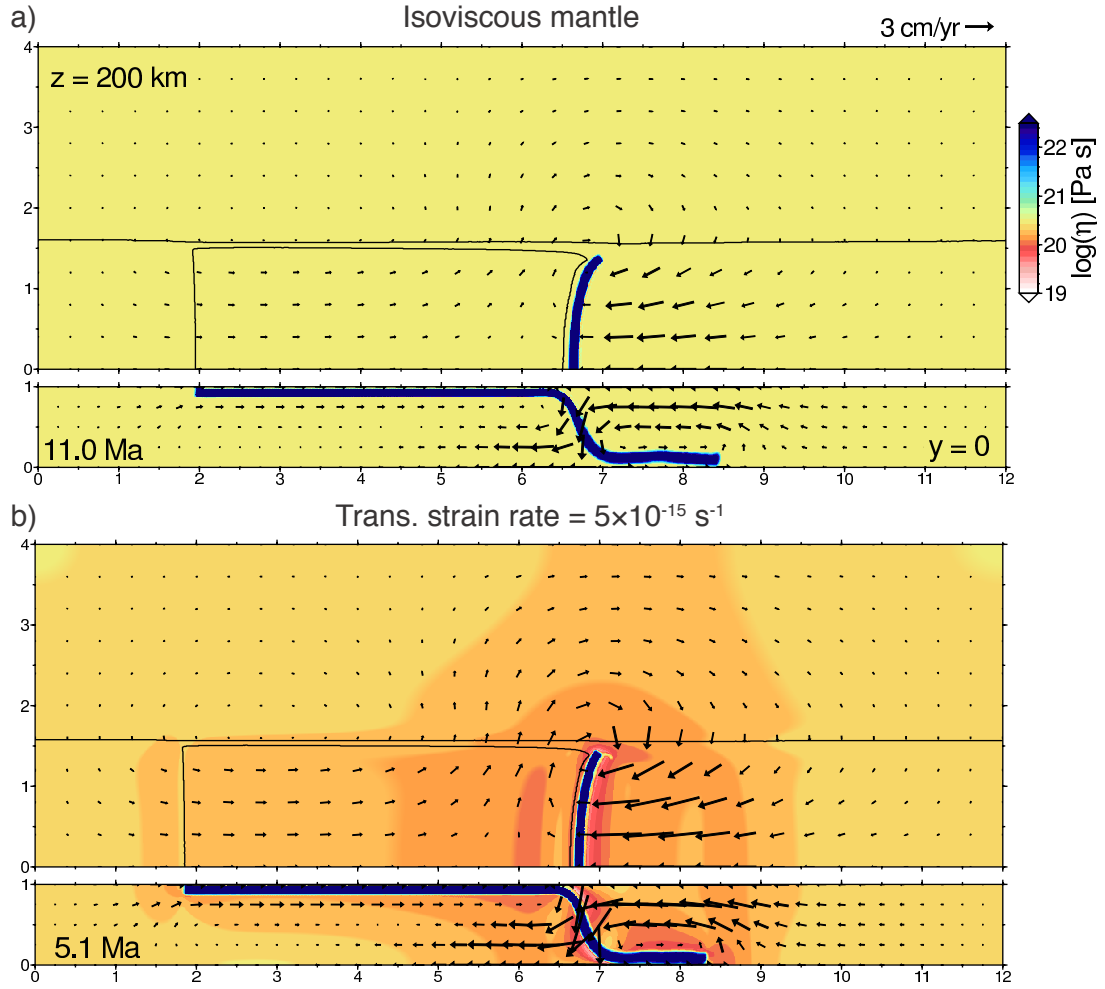


Figure 3.6: As in Figure 3.5, but for the models without an OP.

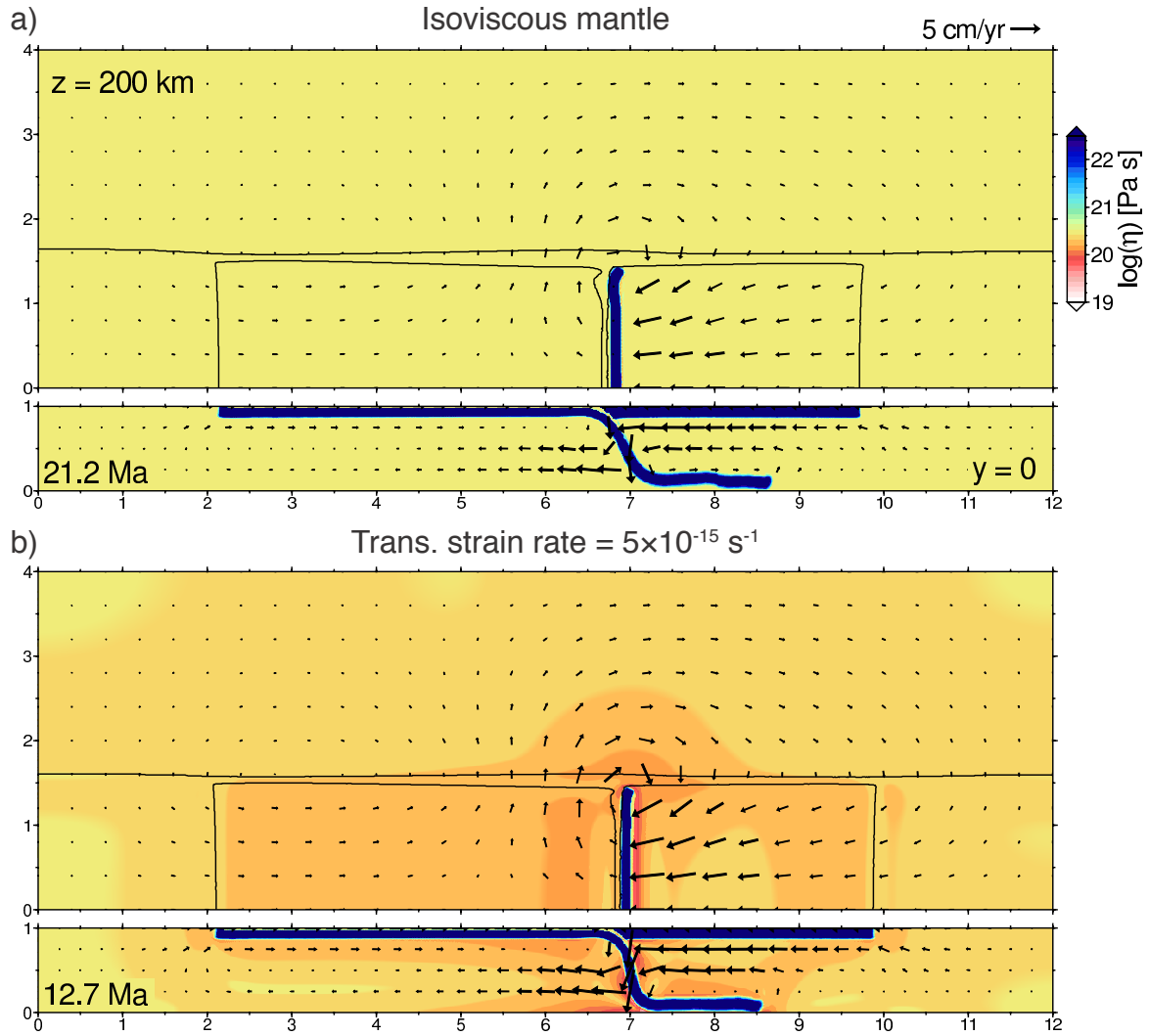


Figure 3.7: [

As in 3.5, but with no slip base]As in Figure 3.5, but for the models with no slip basal boundary conditions.

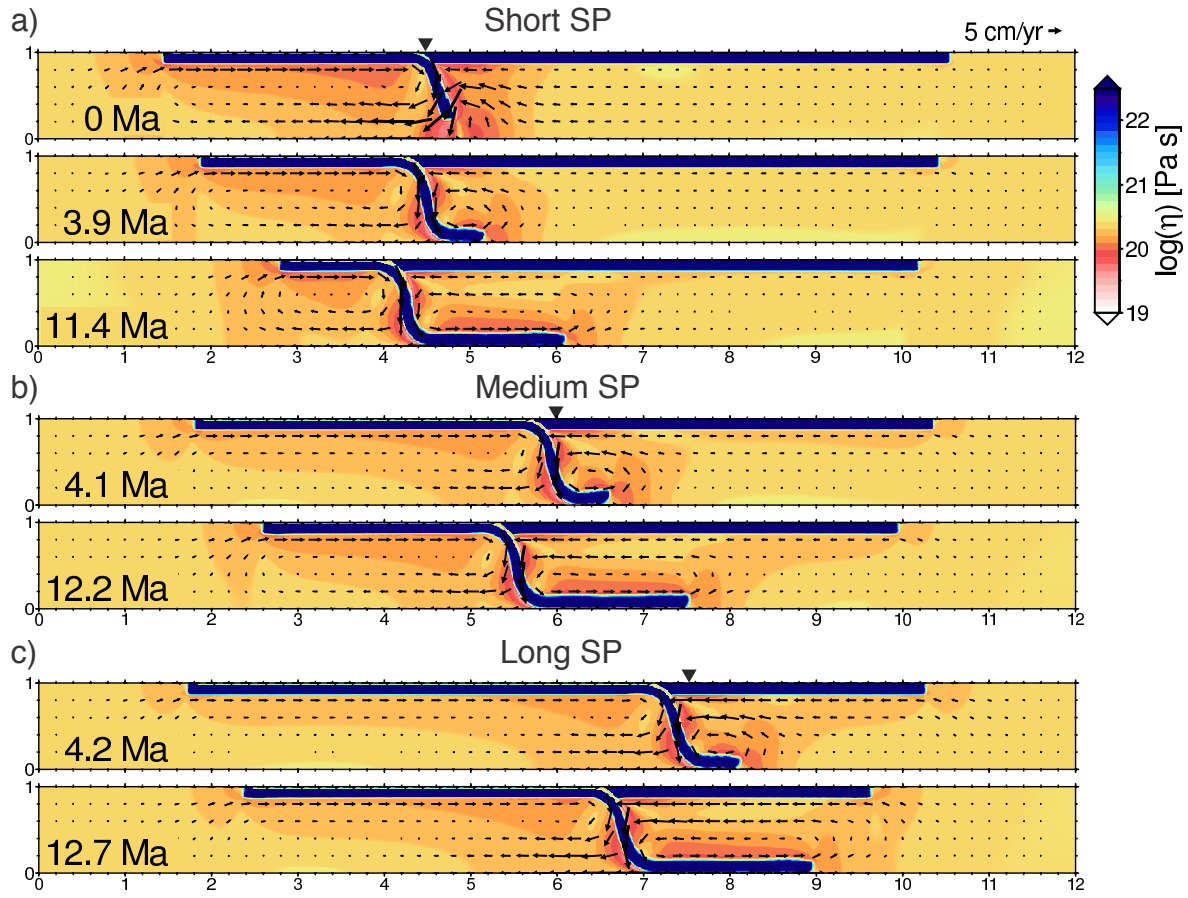


Figure 3.8: Morphological evolution of power-law models with variable initial subducting (L_{SP}) / overriding plate lengths ($L_{OP} = 6000 \text{ km} - L_{SP}$), for a constant transition strain rate of $5 \cdot 10^{-15} \text{ s}^{-1}$. Panels show vertical slices through the model mid-plane ($y = 0$) for, a), the short SP model ($L_{SP} = 2000 \text{ km}$), (b), the medium SP model ($L_{SP} = 3000 \text{ km}$), and, c), the long SP model ($L_{SP} = 4000 \text{ km}$, i.e. Figure 3.5b). The initial solution is additionally shown for the short SP model.

3-D mantle flow

We now compare the 3-D mantle flow field for the long SP ($L_{SP} = 4000$ km) models with, and without, a power-law mantle rheology (transition strain rate of $5 \cdot 10^{-15} \text{ s}^{-1}$). Vertical and horizontal mantle flow velocity, normalized as described in section 3.4.1 for the 2-D models, is shown in Figure 3.9. After normalizing, the flow patterns are qualitatively very similar for the two models. Similarly, for models both without an OP (Fig. 3.10), and with no slip basal boundary conditions (not shown), we find that a power-law mantle rheology also exerts only a minor influence on the broad scale mantle flow patterns. As subduction is partitioned between both slab rollback and subducting plate advance in these models, return flow is partitioned between, respectively, toroidal flow around the slab edges and upwelling in the vertical plane. In all of the models explored, there are multiple regions with significant upwards-directed vertical mantle flow, the locations of which may have implications for the location of decompression melting-induced volcanism (e.g. Faccenna et al., 2010). In addition to upwelling flow at the trailing edges of the two plates, a vigorous mantle upwelling occurs at the lateral edge of the slab-tip, and more moderate vertical flow is associated with flow around the slab edge (i.e. sub-horizontal toroidal flow cell).

While the mantle flow velocities are of greater magnitude in the power-law mantle models due to locally reduced viscosities (e.g. Jadamec and Billen, 2012), the flow patterns, as discussed, appear qualitatively similar. In order to quantify the mantle flow characteristics, we examine the mantle flow partitioning between toroidal and poloidal components (e.g., Bercovici, 1993). Following Tackley (2000), the 2-D velocities on each horizontal layer are decomposed into irrotational (i.e., poloidal) and divergence-free (i.e., toroidal) vector fields. The vertical velocity field is added to the poloidal field, and toroidal/poloidal ratio (TPR) is taken as the toroidal RMS velocity divided by the poloidal RMS velocity. We compute the RMS of the poloidal and toroidal velocity fields for 10 horizontal layers ($\Delta z = 66$ km), and plot vertical profiles of the horizontal RMS velocities in order to characterize both the role of an OP (Fig. 3.11), and a power-law mantle rheology (Fig. 3.12), on the partitioning of mantle flow.

We examine the flow field for the our two plate length end-members, $L_{SP} = 2000$ and 4000 km. As discussed, slab rollback is dominant in the $L_{SP} = 4000$ km model, which

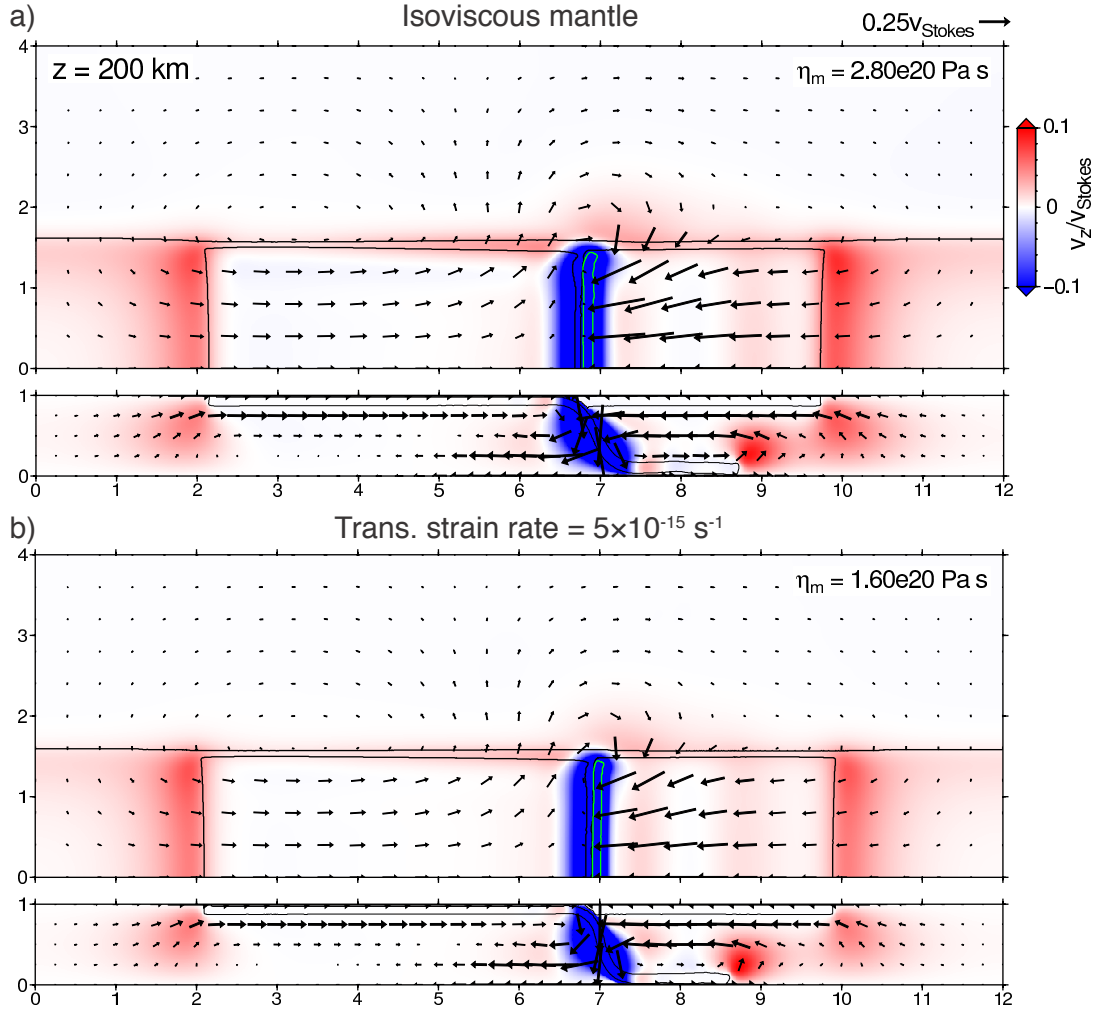


Figure 3.9: Non-dimensional mantle velocity field for the models shown in Figure 3.5, at equivalent time-steps. Velocities are normalized by a Stokes velocity computed using an average sub-lithospheric viscosity (shown in top-right corner), taken from a sub-lithospheric profile at 500 km to the left of the trench. For each model, the top panels show the vertical velocity extracted at a depth of 200 km, overlain by the mantle velocity vectors. The lower panels show the vertical velocity field plotted in the vertical plane, at $y = 0$.

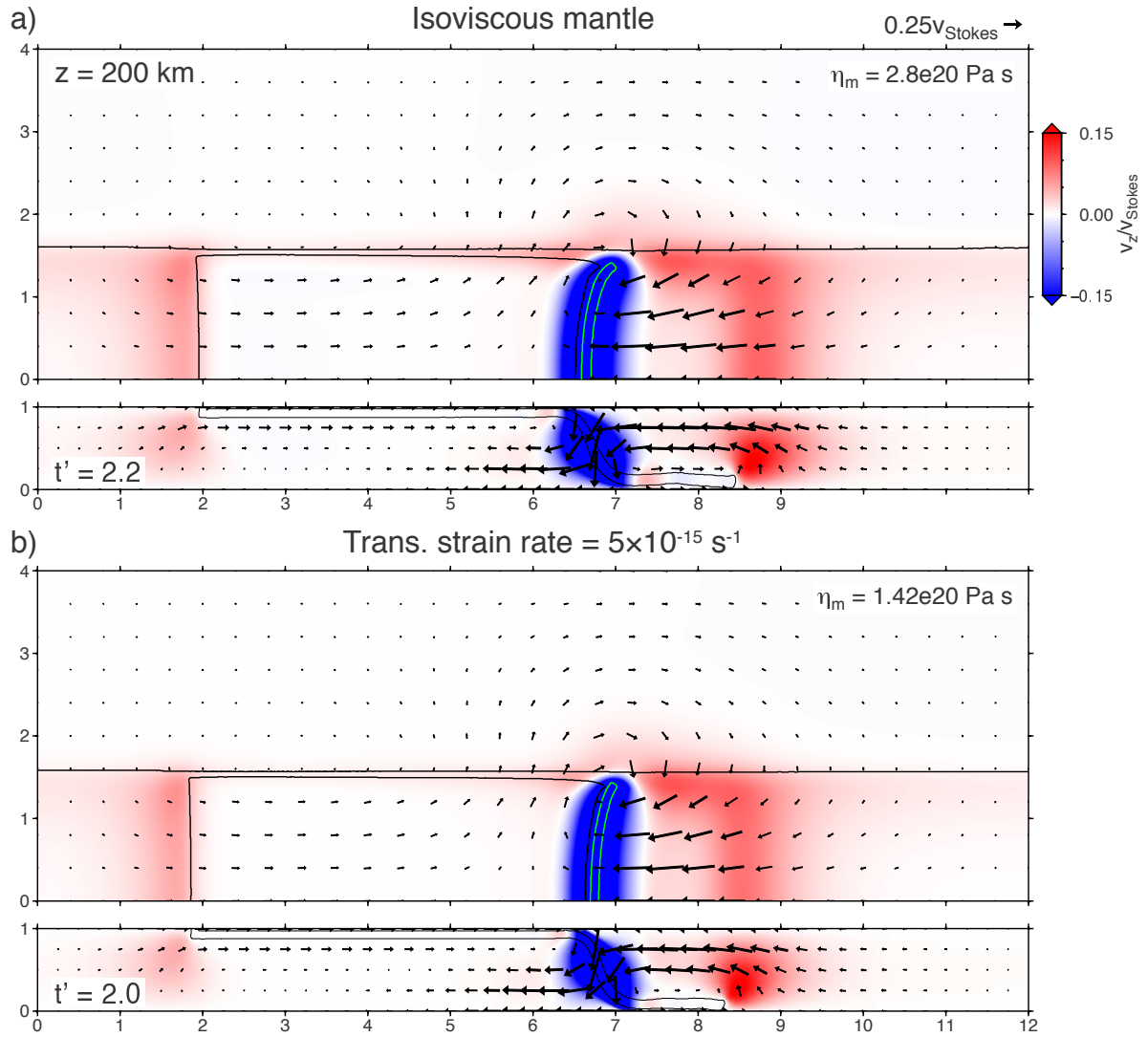


Figure 3.10: As in Figure 3.9, but for the models without overriding plates.

results in the dominance of rollback-induced toroidal flow (i.e. $\log_{10}(\text{TPR}) > 0$) over a substantial depth range ($200 \text{ km} < z < 460 \text{ km}$), for both mantle rheologies (Fig. 3.12). Toroidal flow around the slab edge is focused in the mid-upper mantle, while poloidal mantle flow is dominant only near both the surface and the model base. In contrast, for both rheologies, the shorter SP model exhibits dominantly poloidal flow at all depths. The absence of an OP, by enhancing the relative importance of slab rollback, increases TPR at mid-mantle depths (Fig. 3.11). This effect is particularly prominent for the short SP models, in which the absence of a large OP exerts a stronger effect on slab dynamics than in the long SP model (cf. Figs. 3.11d, e). In line with the qualitative mantle flow similarity observed in Figure 3.9, the inclusion of a power-law mantle has only a moderate effect on the TPR (Fig. 3.12). For the short SP model, a power-law mantle rheology reduces $\log_{10}(\text{TPR})$ by 0.05-0.1, and for the long SP model, the TPR is approximately equivalent (slightly elevated). As observed in the 2-D models, a power-law mantle rheology decreases the importance of slab rollback, and so we propose that the reduction in TPR, for the short SP model, stems from the reduction in relative rollback rate. For the long SP model, however, the relative increase in rollback does not have such an expected effect. We propose that this is because, in addition to the effect that reducing slab rollback has on reducing toroidal flow, the lateral viscosity variations induced by a power-law viscosity likely promote toroidal flow. As discussed in the next section, a power-law viscosity has a more minor effect on slab kinematics relative to the models with shorter SPs (Fig. 3.13c), and so one may expect the lateral viscosity variation effect to dominate here.

Slab kinematics

As for the 2-D models presented in section 3.4.1, we find that a power-law mantle rheology reduces the relative magnitude of trench retreat, V_T , relative to trench-directed subducting plate velocity, V_P . This is true for each of the subducting plate lengths examined, as shown in Figure 3.13 where we plot the temporal evolution of the non-dimensional plate velocities, in the side plate (/box sides) reference frame.

Generally, V_P is greater than V_T (i.e., negative $\log(V_T/V_P)$; Fig. 3.13c), with the only exception being the isoviscous mantle, long SP model. After the initial transient, bending phase (i.e., $t < t_{\text{transient}} \approx t_{\text{Stokes}}$), both plate velocities generally attain a near-constant, steady-state value. As discussed previously, with increasing L_{SP} , and decreasing L_{OP} , the

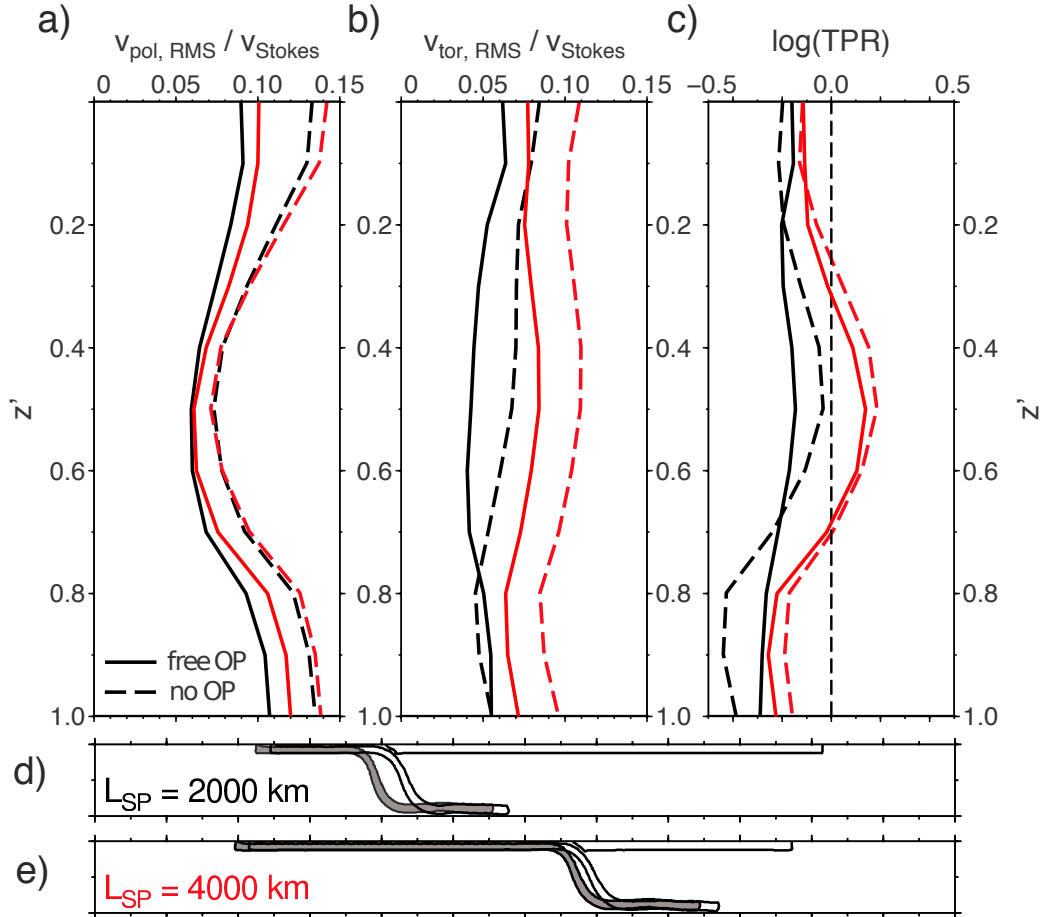


Figure 3.11: Partitioning of mantle flow between poloidal ($\nabla_h \times \vec{v}_{pol} = 0$) and toroidal ($\nabla_h \cdot \vec{v}_{tor} = 0$) components for models with (solid lines) and without OPs (dashed lines), for the initial SP length end-members (black: $L_{SP} = 2000$ km, red: 4000 km). Shown are vertical profiles, which are composed of velocities averaged over 10 horizontal slices ($\Delta z = 66$ km). Plotted is, a), RMS of poloidal velocity, b), RMS of toroidal velocity and, c), the log ratio of the toroidal to total poloidal RMS velocities (e.g., Piromallo et al., 2006). Velocities are normalized by a Stokes velocity (equivalent for all isoviscous mantle models), as discussed in the main text. Lower panels show corresponding slab morphologies for, d), $L_{SP} = 2000$ km models, and, e), $L_{SP} = 4000$ km models, where filled grey slabs correspond to the single slab models.

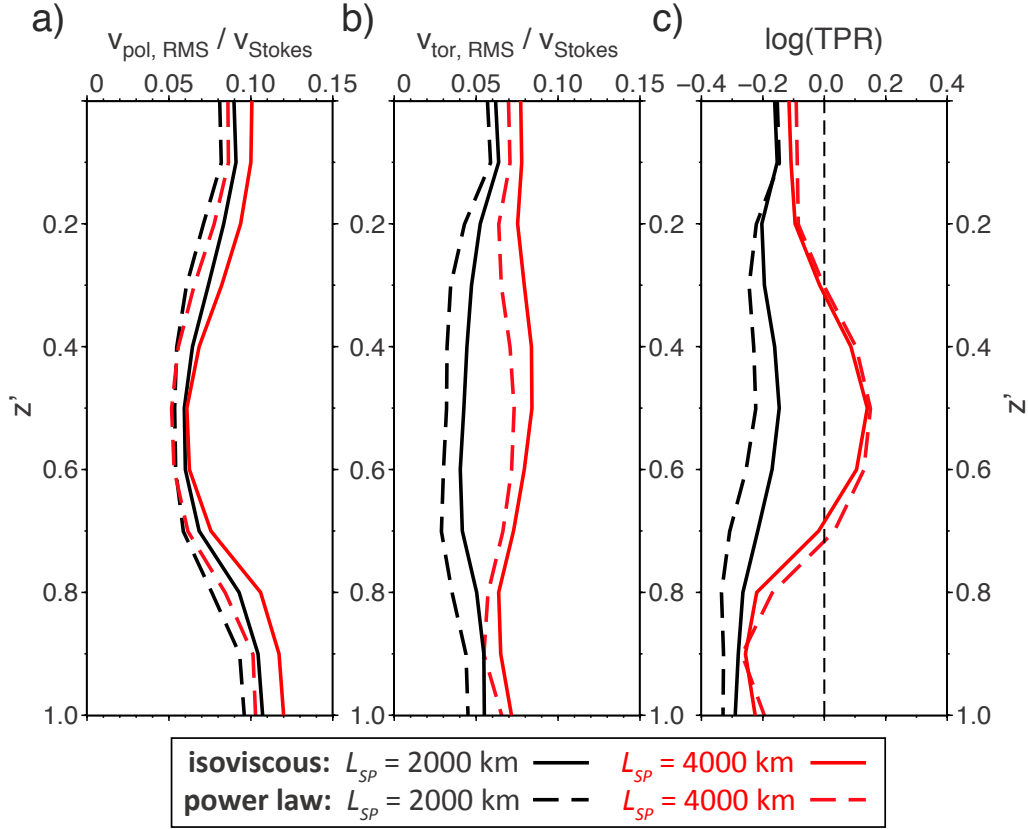


Figure 3.12: Partitioning of mantle flow between poloidal ($\nabla_h \times \vec{v}_{\text{pol}} = 0$) and toroidal ($\nabla_h \cdot \vec{v}_{\text{tor}} = 0$) components for models with an isoviscous mantle rheology (solid lines) and with a power-law rheology with a transition strain rate of $5 \cdot 10^{-15} \text{ s}^{-1}$ (dashed lines), for the initial SP length end-members (black: $L_{SP} = 2000$ km, red: 4000 km). Shown are vertical profiles, which are composed of velocities averaged over 10 horizontal slices ($\Delta z = 66$ km). Plotted is, a), RMS of poloidal velocity, b), RMS of toroidal velocity and, c), the log ratio of the toroidal to total poloidal RMS velocities (e.g., Piromallo et al., 2006). Velocities are extracted at the time steps plotted in Figures 3.5 and 3.9.

plate convergence becomes increasingly accommodated by trench retreat (i.e. V_T/V_P increases). For all plate lengths, the non-dimensional V_P of the power-law rheology models (dashed lines) is generally comparable, but slightly reduced relative to the isoviscous mantle models (solid lines). This similarity occurs because the average viscosity used to compute V_{Stokes} is extracted from SP side of the trench (i.e. the portion of the mantle which dictates V_P). In contrast, the power-law mantle viscosity models exhibit a much stronger reduction in V_T . This can be seen in Fig. 3.13c, where the temporal evolution of the log of the ratio of V_T and V_P is plotted. For every plate length, $\log(V_T/V_P)$ is consistently reduced in the power-law mantle models. The relative reduction in V_T reduces as the SP length increases (for $L_{SP} = 2000$ km, $\Delta \log(V_T/V_P) \approx 0.15$; for $L_{SP} = 4000$ km, $\Delta \log(V_T/V_P) \approx 0.07$). We additionally examine isoviscous mantle models with average viscosities equivalent to those observed in the power-law models (as extracted from viscosity profiles taken below the SP). This is done to examine whether the localized viscosity reductions, or the bulk mantle viscosity reduction (which reduces the plate-mantle viscosity contrast), are responsible for the relative V_T reduction. Reducing the mantle viscosity in the Newtonian models does not produce the reduced V_T/V_P observed in the power-law mantle models (Fig. 3.13c: lighter colored solid lines), and so we attribute the reduction in relative trench motions to the localized viscosity reductions that occur directly adjacent to the lithospheric plates.

As in the 2-D models (Fig. 3.4a), the dip angle increases as power-law deformation becomes more dominant (i.e., transition strain rate decreases). For the long SP models, Figure 3.14 illustrates the dependence of V_T/V_P and the, inversely correlated, slab dip on the transition strain rate. For transition strain rates of $5 \cdot 10^{-13} \text{ s}^{-1}$, and above, the mantle is almost entirely in the Newtonian regime, and so the slab kinematics are nearly equivalent to those observed in the isoviscous mantle model. While the sense of kinematic variability, as a function of transition strain rate, is equivalent for the 2-D and 3-D models, the magnitude of variation is not. The $\log_{10}(V_T/V_P)$ ratio is reduced by 0.09 going from an isoviscous mantle to a transition strain rate of $5 \cdot 10^{-15} \text{ s}^{-1}$ for the 3-D models (Fig. 3.14), but the equivalent 2-D models produce a much greater variation of 0.19 (Fig. 3.4b).

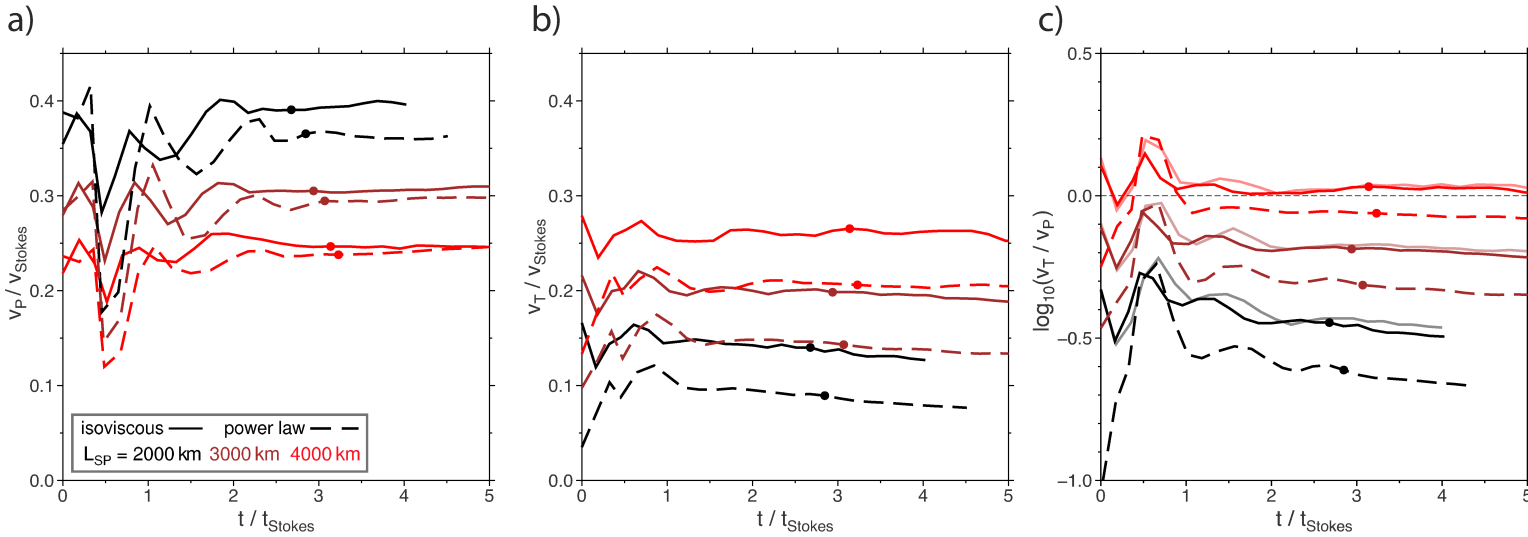


Figure 3.13: Temporal evolution of plate kinematics for isoviscous mantle (solid lines), and power-law mantle (dashed lines: $5 \cdot 10^{-15} \text{ s}^{-1}$), models with variable initial SP lengths (black: $L_{\text{SP}} = 2000$ km, brown: 3000 km, red: 4000 km). Panels show, a), the trench-wards SP velocity (side plate / box reference frame), b), the trench retreat velocity and, c), the log of the trench retreat velocity divided by the SP velocity. The lighter, solid lines, on panel c), display the log ratio for additional Newtonian models which have a plate-mantle viscosity contrast that is set to be equivalent to that of the power-law models with the same L_{SP} . Plate velocities are normalized by Stokes velocities computed at times indicated by points.

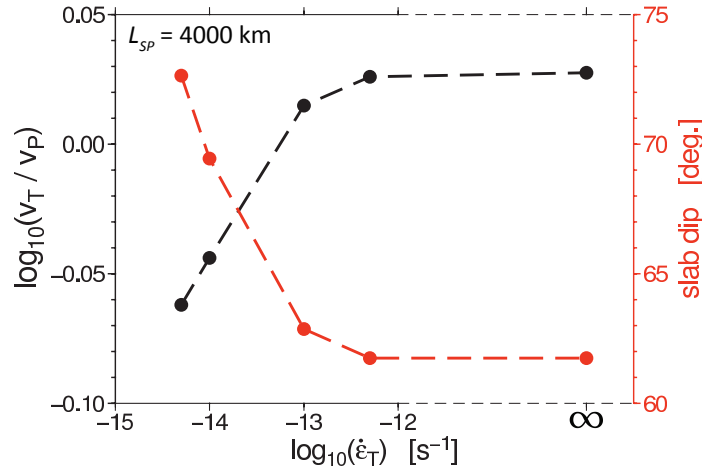


Figure 3.14: Plate kinematics as a function of transition strain rate for the 3-D models with initial $L_{SP} = 4000$ km, extracted at time steps equivalent to those shown in Figures 3.5 and 3.9. Shown is average slab dip angle computed between depths of 200 and 460 km (red), and, log ratio of trench retreat velocity and SP velocity (black). The kinematics extracted from the equivalent Newtonian models are placed at an arbitrary, “infinite” transition strain rate, x -location, for illustrative purposes.

3.5 Discussion

3.5.1 Slab dynamics

While we employ a simplified parameterization of dislocation creep, time-dependent numerical modeling studies that do include the full laboratory-derived rheological formulations tend to observe dislocation creep activation in, approximately, the same regions as in our models, particularly directly below the lithospheric plates and adjacent to the sinking slab (e.g. Billen and Hirth, 2005; Čížková et al., 2007; Billen and Hirth, 2007; Androvičová et al., 2013; Garel et al., 2014). While further work is required to constrain the dynamic consequences of a more realistic mantle rheology (e.g. Karato and Wu, 1993; Hirth and Kohlstedt, 2003), this similarity gives us confidence that our simplified parameterization is able to capture the first order dynamic effects associated with dislocation creep in the sub-lithospheric mantle. As observed in previous numerical models, a primary effect of reduced viscosities adjacent to the subducting slab include elevated slab sinking rates (Billen and Hirth, 2005, 2007; Garel et al., 2014; Holt et al., 2015a), and rapid mantle flow velocities (Jadamec and

Billen, 2012; Billen and Jadamec, 2012).

In all of our models the slab dip angle is reduced by a power-law mantle rheology. As proposed by Billen and Hirth (2005, 2007), this occurs in part because relative viscosity reductions in the mantle wedge reduce the magnitude of negative pressure in the mantle wedge, and so the associated lifting torque (Stevenson and Turner, 1977; Tovish et al., 1978). This reduction in across-slab dynamic pressure gradient (i.e., $\Delta P = P_{sub-slab} - P_{wedge}$) is, averaged over the various plate lengths, ~ 5 MPa (Fig. 3.15). For an average dip angle of $\approx 70^\circ$, such a reduction in ΔP produces a reduction in the lifting force, $F_{\Delta P}$, of ≈ 3.5 TN/m (for our models, $F_{slab\ pull} \approx 44$ TN/m). Billen and Hirth (2005, 2007) hold the trench position fixed and so only consider the role of a reduced ΔP angle, assuming trench migration is minimal. However, dip angle is strongly affected by the magnitude and direction of trench retreat, V_T (e.g. Christensen, 1996; Ribe, 2010). In our numerical models, we find that a power-law mantle viscosity reduces the relative magnitude of trench retreat which, additionally, promotes greater slab dip angles. We, therefore, suggest that allowing the trench to migrate freely will enhance the slab dip increasing effect of a dislocation creep observed by Billen and Hirth (2005, 2007). However, a reduction in $F_{\Delta P}$, which has a horizontal component directed towards the mantle wedge, constitutes a reduction in one of the major forces resisting trench/slab rollback (e.g. Royden and Husson, 2006; Stegman et al., 2006), and so understanding why slabs subducting into a power-law mantle exhibit reduced V_T/V_P requires an analysis of the additional forces associated with mantle flow (sec. 3.5.3).

3.5.2 Mantle flow and 3-D effects

In addition to slab kinematics, we examine the impact of a power-law mantle viscosity on 3-D mantle flow. In our numerical models, we find that a power-law rheology does not significantly impact the mantle flow patterns (cf. Christensen, 1984), nor the partitioning of flow between poloidal and toroidal components (cf. Čadek et al., 1993). When assessing the magnitude of the subduction-resisting forces associated with mantle flow (Section 3.5.3: Fig. 3.17), such mantle flow similarity is convenient in that it allows similar expressions to be used to compute mantle shear stresses for both rheologies. We suggest that the lack of variation in mantle flow partitioning is due to competing power-law effects, namely reduced rollback rates and increased lateral viscosity variations. However, it could also be

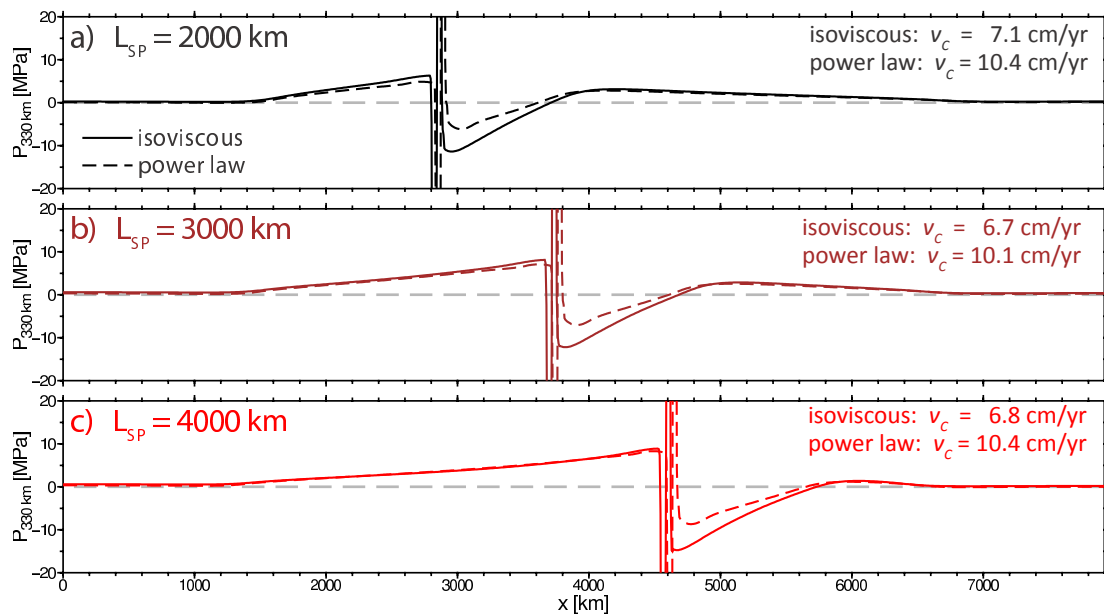


Figure 3.15: Dynamic pressure profiles taken at a constant depth of 330 km (i.e. mid-box) for isoviscous mantle (solid lines) and power-law mantle models (transition strain rate = $5 \cdot 10^{-15} \text{ s}^{-1}$; dashed lines), for models with various initial plate lengths (black: $L_{SP} = 2000$ km, brown: 3000 km, red: 4000 km). The curves are shifted to give equivalent (zero) far-field pressures, in order to focus on the across-slab pressure gradients (ΔP). Pressures are extracted at the time-steps shown in Figures 3.5 and 3.9.

the case that the transition strain rate imposed in our models, which controls the degree of power-law deformation, is still too high to generate the full extent of dislocation creep active on Earth. Indeed, previous numerical modeling studies, which utilize laboratory derived rheologies, appear to exhibit stronger mantle flow localization for models with power-law mantle viscosities (Jadamec and Billen, 2012; Billen and Jadamec, 2012). However, these instantaneous computations do not consider the dynamic evolution of the subducting slab, and so it is not immediately obvious that stable subduction would occur for these setups. A combination of these two modeling strategies, i.e., dynamically evolving 3-D subduction and a laboratory-derived rheological parameterization, would enable the effect of dislocation creep on the large-scale mantle flow field to be more rigorously assessed.

In all of the models explored, there are multiple regions with significant upward-directed vertical mantle flow. In addition to upwelling flow at the trailing edges of the two plates, a vigorous mantle upwelling occurs at the lateral edge of the slab-tip, and more moderate vertical flow is associated with flow around the slab edge (i.e. sub-horizontal toroidal flow cell). These flow features are consistent with the study of Faccenna et al. (2010), where these types of upwellings are observed in single slab numerical models, and invoked to explain the anomalous off-arc volcanism observed in locations such as the western US and the Mediterranean, and the analog modeling studies of Schellart (2010) and Strak and Schellart (2014). Our models verify the presence of such upwellings in dynamic subduction models that incorporate both an OP, and a stress-dependent mantle rheology. Additionally, beneath the OP and ~ 500 km away from the trench, we observe moderate power-law deformation which could have potential implications for melt/volatile transportation in the mantle wedge/fore-arc region.

Relative to the 2-D models, we find that the magnitude of variation of both slab dip and trench retreat rates induced by variable mantle rheology is significantly reduced in 3-D. In 3-D, the presence of a finite width trench enables toroidal mantle flow around the slab edge, which can act to relieve sub-slab pressures, thereby enabling rapid slab rollback (e.g. Funicello et al., 2003b; Schellart, 2004; Stegman et al., 2006; Royden and Husson, 2006; Stegman et al., 2010). However, in 2-D, such a mechanism for relieving elevated subduction pressures is absent. Therefore, variations in across-slab pressure gradients induced by a variable mantle rheology (Tovish et al., 1978; Billen and Hirth, 2005, 2007) are likely to be greater

in 2-D, and so require greater variation in the slab kinematics to accommodate such large mantle pressure variations (i.e., satisfy the appropriate force balances). The presence of a viscous lower mantle, neglected here for simplicity, may reduce the magnitude of this 2-D effect by enabling sub-slab pressure build-up to be relieved by deep, poloidal flow (e.g., Faccenna et al., *submitted*).

3.5.3 A mechanism for reduced trench retreat

A simplified analysis of the magnitudes of the forces resisting subduction may provide further insights for why a power-law asthenospheric viscosity promotes subduction to occur preferentially via SP motion, V_P , as opposed to trench retreat, V_T . Previous studies have utilized adaptations of the expression derived by Conrad and Hager (1999), which consists of a balance between slab pull and the subduction-resisting forces (i.e., slab bending, mantle drag, and interfacial coupling), to predict convergence rates (V_C) for a range of lithospheric properties (e.g. Bellahsen et al., 2005; Funiciello et al., 2003b). Here, we instead compute approximations of the forces resisting both trench retreat and SP motion, for our observed V_C and across-slab pressure gradients (ΔP). We focus on the forces associated with mantle flow, as these are the force components directly modified by the presence of a power-law rheology. Therefore, we do not explicitly consider slab pull and slab bending which, respectively, drive and resist subduction (e.g. Conrad and Hager, 1999; Buffett, 2006).

While the inclusion of a power-law viscosity component significantly modifies the mantle viscosity (e.g. Fig. 3.5), the style of subduction-induced mantle return flow is comparable for models with and without such a mantle rheology (e.g. Fig. 3.9). We therefore assume that the ratios between the plate/slab velocities and the adjacent mantle velocities are equivalent for both mantle rheologies. We compute the plate/mantle velocity ratios from the medium SP model, with $L_{SP} = 3000$ km, and then simply use the value averaged over both rheologies for the subsequent shear force calculations. Figure 3.16 shows the mantle velocity profiles used for this calculation and verifies the similarity between the flow profiles for the two models. Beneath the SP, the asthenosphere is flowing slower than the plate, and so exerts basal tractions which resist subducting plate motion, V_P (Fig. 3.16a). Conversely, the subduction-induced flow beneath the OP is faster than the plate motion which results in basal tractions that are driving trench-wards OP motion (Fig. 3.16c). The subduction-resisting shear

force per unit length on the SP is given by the product of an average sub-lithospheric shear stress and the SP length,

$$F_{SP} = (2\eta_{m,SP} \dot{\epsilon}_{shear})X_{SP} \approx \eta_{m,SP} \left(\frac{V_P - V_{m,P}}{\Delta z} \right) X_{SP}, \quad (3.4)$$

where V_P and $V_{m,P}$ are, respectively, the SP velocity and the velocity of sub-lithospheric mantle, and Δz is the depth that $V_{m,SP}$ is extracted at, relative to the base of the lithospheric plate (e.g. Fig. 3.16). $\eta_{m,SP}$ is the average sub-lithospheric viscosity beneath the SP, and X_{SP} is the SP length over which this average shear stress acts ($= L_{SP} - 1000$ km). Similarly, the OP motion-driving force is given by,

$$F_{OP} \approx \eta_{m,OP} \left(\frac{V_C - V_P - V_{m,OP}}{\Delta z} \right) X_{OP}, \quad (3.5)$$

where V_C is the convergence rate ($= V_T + V_P$), and X_{OP} is the approximate OP length over which this average shear stress acts. X_{OP} does not appear to have a dependence on plate length and so, based on inspection of Figure 3.16 and similar figures with flow profiles plotted at different distances from the slab, we use an average X_{OP} of 1200 km. In addition we compute the approximate, up-dip directed shear force, F_{slab} , acting on the slab,

$$F_{slab} \approx 2\eta_{m,slab} \left(\frac{V_C - V_{m,slab}}{\Delta n} \right) \frac{h}{\sin(\theta)}, \quad (3.6)$$

where $V_{m,slab}$ is the slab-parallel mantle flow velocity at a slab-normal distance of Δn away from the slab edge, h is the depth-extent of the slab ($= 660 \text{ km} - h_l = 580 \text{ km}$), and θ is the dip angle. The additional factor 2 occurs due to mantle shear occurring on both sides of the slab. In the power-law numerical models, mantle viscosity is a function of the deviatoric strain rate, $\dot{\epsilon}_{II}$. Under the assumption that the only non-zero strain rate component is the shear strain rate (in the vertical plane), $\dot{\epsilon}_{II} = \dot{\epsilon}_{shear}$, and so, for example, the power-law component of the sub-lithospheric viscosity (i.e., eq. 3.1) below the SP can be written as,

$$\eta_{P,SP} = \eta_N \left(\frac{V_P - V_{m,P}}{2 \Delta z \dot{\epsilon}_{II,T}} \right)^{\frac{1-n}{n}}, \quad (3.7)$$

which is then combined with η_N to get the effective viscosity (eq. 3.2), and inserted into the expression for F_{SP} (eq. 3.4). An approximation for the total, mantle force resisting subducting plate motion, V_P , is taken to be the sum of the subduction-resisting shear forces

associated with the flat-lying portion of the SP and the slab, $F_P = F_{SP} + F_{slab}$. Similarly, the approximate, mantle-derived, force resisting trench retreat is composed of a component associated with an along-slab mantle pressure gradient, ΔP , and the force driving OP motion, i.e. $F_T = F_{\Delta P} - F_{OP}$. Within this F_T is the assumption that F_{OP} – the horizontal mantle shear force on the OP – is transmitted to the SP in the form of a horizontal force promoting rollback (i.e., by “pushing” the slab hinge, e.g. van Dinther et al., 2010). While a proportion of F_{OP} is likely dissipated in the weak plate boundary, we neglect this in order to simplify our calculations. The other component of F_T , the horizontal $F_{\Delta P}$, is computed using the average pressure gradient extracted from our numerical models ($F_{\Delta P} = \Delta P(h/\sin(\theta))$), computed between two points 330 km either side of the slab, at mid-mantle depths ($z = 330$ km).

We consider a power-law rheology equivalent to that implemented in the numerical models, with a transition strain rate of $5 \cdot 10^{-15} \text{ s}^{-1}$. As input into our simplified model we use the average θ , V_C , and ΔP observed in the isoviscous (65° , 6.9 cm/yr, 18 MPa) and power-law mantle viscosity models (75° , 10.3 cm/yr, 13 MPa), but ignore variation associated with the variable plate lengths (Fig. 3.15) in order to keep the calculations as general as possible. Despite this, we do include the average variability in ΔP and V_C , between the power-law and isoviscous models. This is included because the power-law reduction in ΔP and increase in V_C both promote trench retreat in our calculation and so we wanted to determine whether, despite this, the relative force resisting trench motion, i.e. F_T/F_P , would increase.

For our simplified expressions, Figure 3.17 shows the magnitude of the trench resisting (F_T) and plate motion resisting forces (F_P), normalized by the slab pull force ($\approx 44 \text{ TN/m}$), associated with the viscous mantle, as a function of relative trench motion, V_T/V_C (i.e., 0 = pure plate motion, 1 = pure trench motion). For a fixed trench, F_T is composed solely of the resisting force associated with mantle pressure (Fig. 3.17a). With increasing trench motion, the trench retreat resisting force reduces, as a result of an increasing F_{OP} which acts to promote retreat (eq. 3.5). The magnitude of this reduction in F_T is lower when a power-law rheology is considered, as F_{OP} reduces due to the relative reduction in asthenospheric viscosity, despite elevated V_C . The force resisting plate motion, F_P , is shown in Figure 3.17b. Generally, F_P is reduced for the power-law models. This occurs because the magnitude of the subduction-resisting shear tractions associated with both the flat-lying portion of the SP (F_{SP}), and the slab (F_{slab}), are reduced when a power-law viscosity is considered. Additionally, models with

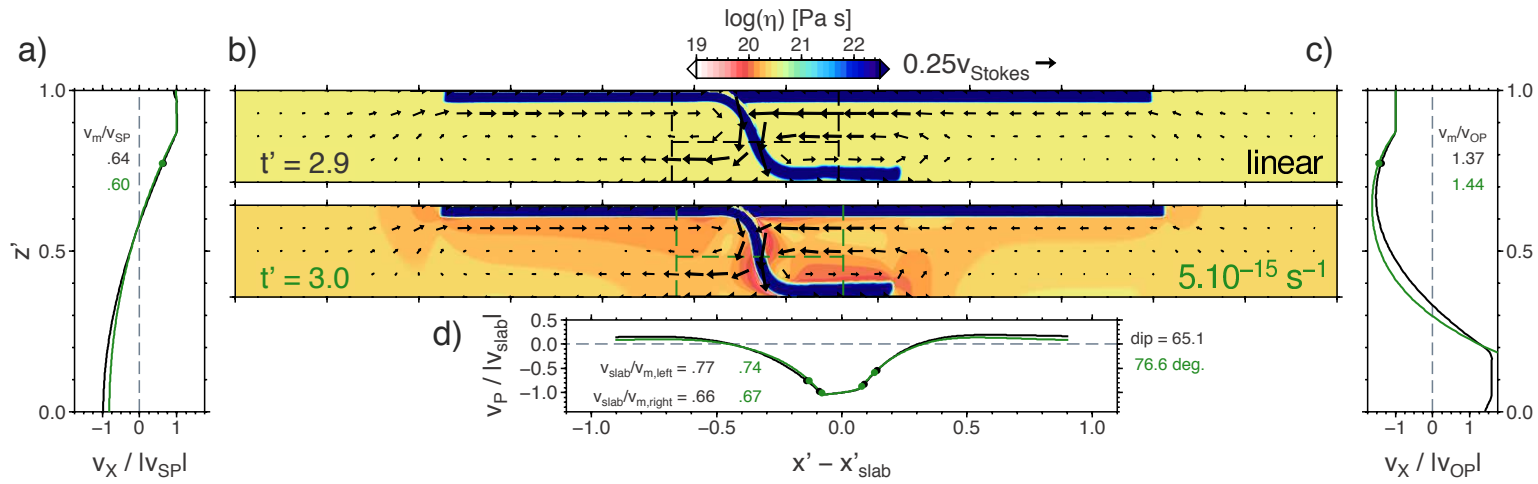


Figure 3.16: Mantle flow profiles for the medium SP models ($L_{SP} = 3000$ km), with both an isoviscous mantle and a transition strain rate of $5 \cdot 10^{-15} \text{ s}^{-1}$. Shown are normalized vertical profiles of horizontal velocity (v_x) extracted, a), 600 km from the left of the slab (i.e., sub-slab mantle), and, (c) 600 km to the right of the slab (i.e., mantle wedge side). Also shown are, b), model viscosity fields and, d), slab parallel mantle velocity extracted along a horizontal profile at mid-mantle depth ($z = 370$ km), normalized by the slab sinking velocity. Indicated by text are absolute plate velocities, mantle over plate velocity ratios ($\Delta z = 70$ km, $\Delta x = 50$ km), and slab dip angles.

longer SPs have elevated F_P due to effect of increasing X_{SP} on F_{SP} , and the curves have negative trends due to the dependence of F_{SP} on V_P (eq. 3.4).

We then plot the ratio of the trench retreat driving force and the plate motion resisting force, F_T/F_P (Fig. 3.17c). Also included on Figure 3.17c are points indicating the V_T/V_C values observed in our numerical models, and the present day, median V_T/V_C values observed on Earth (Heuret and Lallemand, 2005), for a range of APM reference frames. As the trenches in our numerical models are, by design, retreating, we consider only retreating trenches for the plotted Earth V_T/V_C averages. However, we note that the V_T/V_C range is comparable when advancing trenches are included (i.e., with advancing trenches: $V_T/V_C = 0.32\text{--}0.55$; without advancing trenches: $0.33\text{--}0.58$). Our models generally lie within these kinematic ranges, with the exception of the short SP models with $L_{SP} = 2000$ km, in which V_P is overly dominant (Fig. 3.17c). We consider F_T/F_P to be an approximate measure of the degree to which the mantle forces discussed above, favor trench retreat (i.e., low values) or plate motion. This measure does not consider other important forces, both driving (i.e. slab pull) and resisting (e.g., slab bending), in the system, and so only constitutes an approximate metric for rollback vs. trench motion. However, it does mirror the first-order features observed in the slab kinematics (Fig. 3.13), suggesting the primary importance of these mantle related effects. Firstly, plate motion is increasingly favored as L_{SP} decreases, as observed in the numerical models. For the simplified force balances, this occurs primarily because the plate motion resisting force, F_{SP} , reduces as L_{SP} decreases. More importantly, F_T/F_P is elevated for the power-law models, for each L_{SP} . This occurs because the relative viscosity reductions induced by the power-law rheology reduce the mantle shear resistance to both horizontal plate sliding (F_{SP}) and slab sinking (F_{slab}), to a greater degree than a reduction in mantle wedge pressure (Billen and Hirth, 2005; Tovish et al., 1978) reduces the mantle resistance to trench retreat. In addition to this, the power-law rheology reduces the basal tractions on the OP, F_{OP} , which also acts to reduce the force driving trench retreat. We therefore suggest that it is these effects which preferentially promote SP motion, over trench retreat, for the models with a power-law mantle. A more subtle feature of our analysis is that the difference in the force partitioning (i.e., $\Delta(F_T/F_P)$), between the isoviscous and power-law mantle models, is greater for the models with low L_{SP} , particularly at the V_P/V_C values observed in the numerical models (points on Fig. 3.17c), consistent with the observed increase in $\Delta V_T/V_P$

for long SP models (cf. Figs. 3.13c, 3.17c).

3.5.4 Model and analysis limitations

While this force analysis has enabled us to probe the mantle forces that control the partitioning of subduction between rollback and plate advance for our kinematic scenarios, it does not allow us to directly predict V_T/V_P . The computation of full force balances, which include the additional important forces (i.e., slab pull, slab bending, plate interface coupling) (Royden and Husson, 2006; Billen, 2008), are required to dynamically compute V_C and ΔP , and so in turn assess the magnitude of this effect for various subduction setups (e.g., lithospheric and mantle strengths). However, simply extracting these components from the numerical models is sufficient to illustrate our proposed mechanism for trench retreat reduction.

In addition, our force assessment focuses on the variation in mantle forcing as a function of mantle rheology, thereby neglecting forces directly associated with the lithospheric plates. This is reasonable as slab density, viscosity, and thickness are fixed throughout this study. An exception, and potential source of lithospheric-induced force variation, is the variation of slab radius of curvature, R , which exerts a strong control on the slab bending force for a viscous slab ($F_B \propto \eta_l v_C (h_l/R)^3$) (Conrad and Hager, 1999), as a function of both mantle rheology and time (e.g. Becker et al., 1999; Stegman et al., 2006; Capitanio et al., 2009). A more rigorous analysis is required to assess potential feedbacks between slab curvature and a stress dependent mantle rheology.

While a simplified mechanical and geometric setup has allowed us to extract the first order effects of a power-law mantle rheology on slab kinematics, we suggest that both the lithospheric rheology and the properties of the lower mantle may play important roles. The rate of trench retreat and the degree of subducting slab weakening, in the form of plastic yielding or a maximum viscosity cut-off, have been demonstrated to exert first order controls on the propensity of a slab to penetrate into a strong lower mantle (Zhong and Gurnis, 1995; Christensen, 1996; Čížková et al., 2002; Enns et al., 2005; Čížková et al., 2007; Billen, 2008; Garel et al., 2014). Future modeling work focused on understanding how slab impingement upon a strong lower mantle influences the rollback systematics observed here is required to refine our understanding of this mechanism of trench retreat rate reduction.

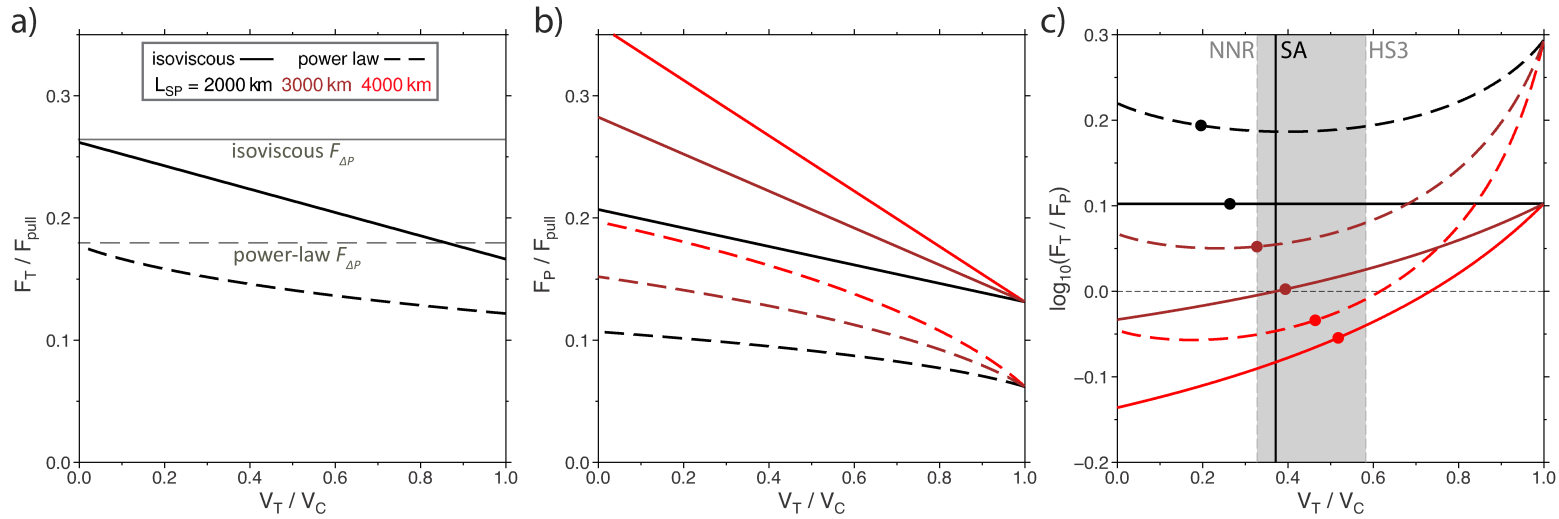


Figure 3.17: Simplified analytical calculations of the trench retreat, and SP motion, resisting forces for isoviscous and power-law mantle ($\dot{\epsilon}_{II,T} = 5 \cdot 10^{-15} \text{ s}^{-1}$) rheologies. Plotted is, a), the approximate force resisting trench retreat, b), the approximate force resisting SP motion, and, c), the log of the ratio of the two forces, all as a function of the ratio of the trench retreat velocity (V_T) and the total convergence rate (V_C). Curves are shown which correspond to each of the subducting plate lengths modeled (in our approximation, F_T is not a function of L_{SP}), and forces are normalized by the slab pull force ($F_{pull} = 44 \text{ TN/m}$). The filled circles indicate the V_T / V_C observed in the numerical models for the corresponding rheology and L_{SP} . Also included on panel c) are vertical bars indicating the median V_T / V_C in three different reference frames: No-net rotation (0.33), spreading-aligned (0.37), and hotspot (0.58) (Kreemer et al., 2003; Becker et al., 2015; Gripp and Gordon, 2002). Average kinematic properties used for the calculation (extracted from the numerical models) are listed in the main text.

3.6 Conclusions

We used 3-D, dynamic models of subduction to address the role of a power-law mantle viscosity on rates of trench retreat. For a suite of models that exhibit a broad range of slab kinematics, we find that a power-law rheology results in reduced rates of trench retreat, V_T and elevated slab dip angles, relative to the equivalent isoviscous mantle model. We analyze the style of mantle flow, which exhibits only limited variability as a function of mantle rheology, and mantle pressure, in order to compute estimates of the forces resisting both V_T , and subducting plate advance, V_P . We find that the inclusion of a power-law rheology reduces the mantle shear force, which resists V_P , to a greater extent than it reduces the dynamic pressure gradient across the slab, which resists V_T . Thus, the inclusion of a power-law mantle rheology favors a shift towards a subduction mode with a reduced trench retreat component. In addition to the inclusion of an overriding plate, we suggest that this mechanism may be of key importance in reducing the high relative V_T rates often observed in single slab models to levels more in line with the average subduction partitioning observed on Earth (i.e., $V_T \leq V_P$). More sophisticated numerical models, and analytical force balances, are required to more rigorously map out the “Earth-like” parameter space where this mechanism is viable.

Chapter 4

The dynamics of double slab subduction

4.1 Summary

We use dynamically consistent, 3-D numerical and analytical models to investigate how the dynamics of a subducting slab can be affected by the presence of an additional subducting slab. Such slab-slab interactions have been recently proposed as a mechanism for inducing a range of interesting plate kinematics, including rapid plate convergence and trench advance. In a consistent geodynamic framework, we probe the dynamics of double slab systems with all of the possible dip directions, for parallel trenches. We find that each of the double subduction configurations exhibit strikingly different plate kinematics, owing to the variable mobilities of the subducting/upper plates which are in some cases free to move, and in other cases attached to subducting/upper plates of the other subduction system. By isolating compartments of the mantle, double slab subduction systems promote enhanced toroidal flow in the mantle, and induce a wide range of lateral pressure gradients across the subducting slabs, relative to the single slab models. We find that much of the observed kinematic variabil-

The work presented is in collaboration with Leigh H. Royden (MIT), who conducted the analytical modeling component, and is in preparation for submission to **Journal of Geophysical Research** (Holt, Royden, Becker). Additionally, this extensive modeling study was motivated and inspired by an earlier study published in **Nature Geoscience** (Jagoutz, Royden, Holt, Becker).

ity for the slabs with free upper plates can be attributed to variable pressure gradients across the slab. This is also the case for the single slab models and, despite widely variable slab pressure gradients, such double slab systematics can be captured by our single slab analytical force balances (Royden et al., *in prep.*). However, the slabs that have upper plates that are either fixed in place, or attached to another subducting plate, behave less predictably. A pertinent example being the rear slab in the model where the slabs dip in the same direction. Here, as previously proposed for the Izu-Bonin-Marianas slabs (Čížková and Bina, 2015), slab pull from the slab in front exerts a horizontal force on the rear slab, which induces trench advance and an over-vertical slab. For our reference dynamic model, a horizontal force that is $\sim 10\%$ of the slab pull force is transmitted between the subduction zones. Instantaneous computations, using the slab geometry extracted from the dynamic model, suggest a weak lower mantle, narrow subduction systems, strong lithospheric plates, and strong crustal channels promote strong horizontal coupling between two subduction zones dipping in the same direction.

4.2 Introduction

Regional interactions between multiple subducting slabs dipping in the same direction have recently been proposed to explain enigmatic slab kinematics in a number of subduction zones, pertinent examples being the rapid pre-collisional plate convergence of India and Eurasia (White and Lister, 2012; Jagoutz et al., 2015), and the advancing motion of the Izu-Bonin-Marianas trench (Carlson and Melia, 1984; Čížková and Bina, 2015). During the Late Cretaceous, convergence between the Indian and Eurasian plates was 50–100% faster than that observed at any present-day convergent margins, and an advancing style of trench motion is observed only for a select few subduction zones (e.g., Heuret and Lallemand, 2005; Faccenna et al., 2007; Lallemand et al., 2008; Schellart, 2008b; Jagoutz et al., 2015; Becker et al., 2015). Such non-standard slab kinematics, particularly the occurrence of trench advance, are, however, not typically observed in dynamic, numerical subduction models that consider an isolated subducting slab (e.g., Funiciello et al., 2003b; Enns et al., 2005; Stegman et al., 2006; Capitanio et al., 2010a; Garel et al., 2014; Holt et al., 2015a). An additional subducting slab, in close proximity to the another slab, is therefore an appealing mechanism to widen the range of viable slab kinematics beyond those inferred from single slab numerical/analog experi-

ments (e.g., Becker et al., 1999; Kincaid and Griffiths, 2003; Funiciello et al., 2003a; Enns et al., 2005; Stegman et al., 2006). Here, we use 3-D numerical models, augmented by force-balance based analytic models, to explore how an additional subducting slab can modify subducting kinematics relative to that observed in single slab subduction models.

Such slab-slab interactions present an extreme case of force transmission between plate margins. Subduction-related forces can conceivably be transmitted between subduction zones through subduction-induced basal tractions and/or direct force transmission across a plate boundary. While there are a limited amount of such double-slab subduction systems active today, developing a dynamic understanding of how slabs can “talk” to each other in these systems is of first order importance for its potential applicability to assessing the role that longer-distance plate margin interactions can have in to the temporal evolution of the global system of subduction zones.

An important aspect of this study is the modeling of the various double slab configurations in a consistent geodynamic framework, i.e., with both a constant numerical setup and constant model parameters, and the relation of the modeled double slab dynamics to the single slab reference models, both numerical and analytical. Understanding the dynamics of slabs in double subduction systems relative to the more simplistic isolated slab is an important step towards a physics-based parameterization of double slab kinematics in situations with limited observational constraints, for example in plate tectonic reconstructions.

However, numerical models of dynamically consistent double slab subduction are computationally challenging. Previous numerical models of double subduction have either omitted the upper plates (e.g., Di Leo et al., 2014), neglected the time-evolving component of the double subduction system (e.g., Lin and Kuo, 2016), or restricted the model domain to 2-D (e.g., Mishin et al., 2008; Čížková and Bina, 2015). Particularly, the inclusion of the 3rd, trench parallel, dimension is of great importance in such double slab systems because, in the absence of significant vertical flow through the strong lower mantle, toroidal mantle flow around slab edges (e.g., Kincaid and Griffiths, 2003; Schellart, 2004; Piromallo et al., 2006; Stegman et al., 2006) is required for the two subduction systems to move relative to one another. Therefore, it is likely that trench migrations rates in double slab systems are even more intimately coupled to toroidal flow/plate width than the equivalent single slab systems (Bellahsen et al., 2005; Stegman et al., 2006; Schellart et al., 2007). We therefore consider subduction systems with

finite plate widths, embedded within a 3-D model domain.

We explore double slab subduction for the three slab dip permutations: (i) Slabs that are attached to the central plate and dip away from the center (“Outward” dipping, e.g., Molucca Sea, Adriatic plate), (ii) Slabs that are attached to the two side plates and dip towards the center plate (“Inward” dipping, e.g., Philippines, Taiwan), (ii) Slabs that dip in the same direction (“Same dip polarity”, e.g., Cretaceous India-Eurasia, Izu-Bonin-Marianas and Ryuku system). Figure 4.1 is from Jagoutz et al. (2015), and shows a schematic illustration of the slab and plate geometries in the Tethyan region during the Cretaceous, prior to the collision of India and Eurasia when two north-dipping subduction zones co-existed in such a same dip polarity configuration. This simplified tectonic evolution is equivalent to that assumed in the analytical models of Jagoutz et al. (2015) that were used to investigate double subduction as a driver of rapid plate convergence rates.

Here, we build on the comparison of fully numerical finite element models and analytic subduction models in Jagoutz et al. (2015), and presented for single subduction systems in Royden et al. (*in prep.*), to explore how subducting slab kinematics, particularly trench and plate motions, can be affected by the presence of an additional slab, for trenches that are initially parallel. While the trenches within double slab subduction zones are generally significantly non-linear, e.g., Izu-Bonin-Marianas, this parallel-trench assumption allows us to initially target the first-order physics of these systems in a generalized setup. An obvious next step is the adaptation of these simplified, and restricted, initial conditions to target individual subduction systems, as routinely done for instantaneous computations (e.g., Alpert et al., 2013; Lin and Kuo, 2016).

Firstly, we characterize double slab plate kinematics for each of the slab configurations. Relative to a single slab reference, we then analyze the temporal evolution of the asthenospheric pressure and the coupled mantle flow. We then explore the degree to which the computed asthenospheric pressure gradients can explain the observed slab kinematics, and use the single slab analytic model to determine which, and to what extent, double slab models obey the single slab systematics described in Royden et al. (*in prep.*). We then focus on the setup within which the slabs dip in the same direction. Unlike the other double slab configurations, there is no plane of symmetry between the two subduction systems, and so this system becomes more complicated, as both slabs behave distinctly. Following the pro-

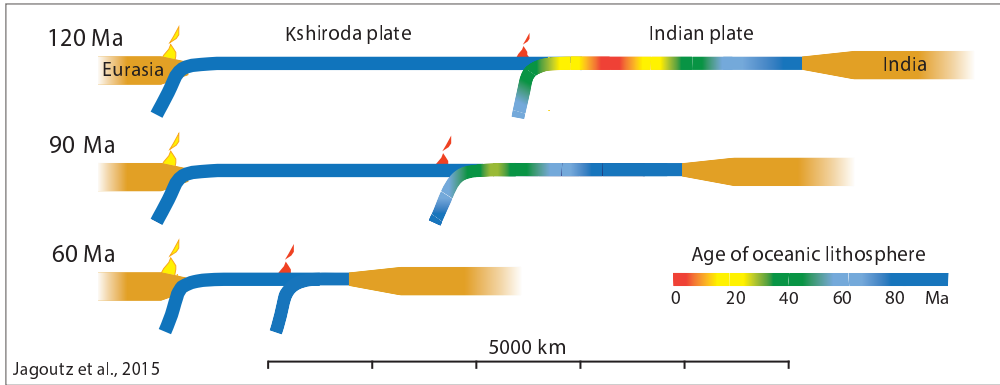


Figure 4.1: Simplified cross-sections of double slab subduction geometry in the Tethys region during the Cretaceous (Figure 2 of Jagoutz et al., 2015).

posal of Čížková and Bina (2015) that slab pull transmitted from one slab to another can induce a upper plate-directed horizontal force great enough to induce trench advance, we directly compute the interface stresses/forces in our 3-D, dynamic models in order to determine the degree to which these coupling stresses differ between the two subduction zones in the double slab system. To develop an understanding of the geometrical and mechanical constraints on this horizontal force transmission, we then use instantaneous models to test how the first order properties of the subduction system, such as lateral plate width and lower mantle strength, modify slab coupling relative to that observed in the reference setup.

4.3 Methodology

4.3.1 Numerical models

We use the finite-element code CitcomCU (Moresi and Gurnis, 1996; Zhong, 2006) to model subduction in a 3-D, Cartesian domain. CitcomCU solves the equations governing convection in an incompressible viscous fluid (Boussinesq approximation) with negligible inertia. In order to investigate the first order physics, we simplify our model setup by ignoring thermal diffusion. We therefore consider a density and viscosity field that is dependent purely on composition, where the compositional field is advected using tracer particles (McNamara and Zhong, 2004). We initially place 40 tracers within each of the elements, and separate sets of tracers are used to define the lithospheric plates, and the weak “crustal” layer that

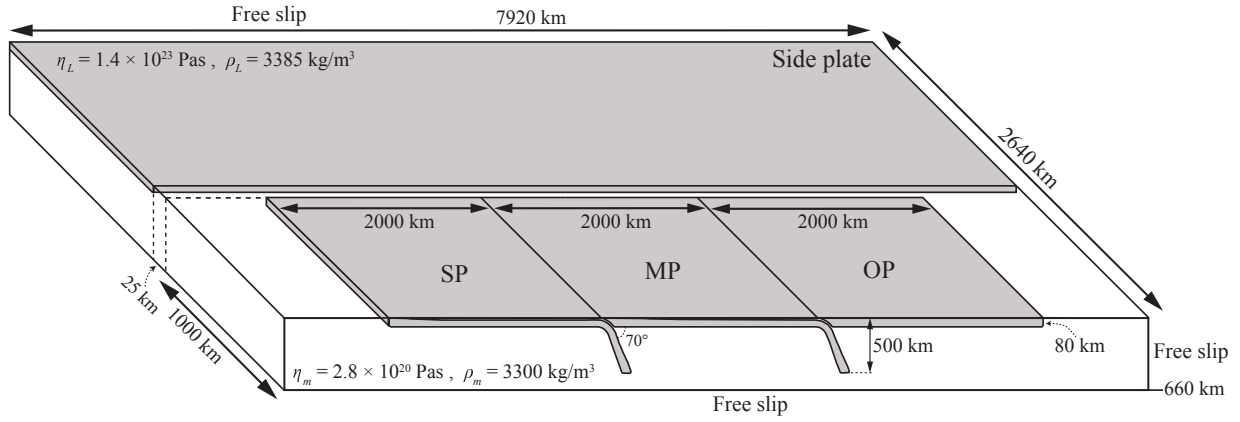


Figure 4.2: Schematic illustration of the initial conditions for the “same polarity” numerical model. Additional, opposite polarity, models have equivalent setups but with variable initial slab dip polarities, and an additional symmetry plane at the center of the middle plate.

decouples the lithospheric plates. The domain size is 7920 km (length, x) $\times$ 660 km (height, z) $\times$ 5280 km (width, y), and all boundaries are mechanically free slip. We consider 3-plate systems, within which each of the plate lengths are 2000 km (Fig. 4.2). We consider plates with a finite width of 2000 km for the reference suite of models. For each of the models presented, we take advantage of the system symmetry about $y = 0$ (i.e., trench-perpendicular horizontal axis), which enables us to half the computational domain (i.e., $y = 2640$ km). In addition, the double slab models within which the two slabs have opposite dip polarities (“outward”, “inward” dipping) have an additional symmetry plane through the center of the middle plate, which further reduces computational expense. The finite element dimensions vary from 6 km in regions which are both in the upper 200 km of the domain and proximal to the trench, to 14 km elsewhere.

In order to target first order dynamics, we use a simple mechanical and rheological subduction setup. We consider compositional lithospheric plates with a constant density contrast, relative to the sub-lithospheric mantle, of 85 kg/m³ and a constant thickness of 80 km. The sublithospheric mantle has a reference viscosity of 2.8×10^{20} Pa s and density of 3300 kg/m³. These parameters correspond to a compositional Rayleigh number of 8.55×10^7 . We use a linear viscous rheology, where the lithospheric plates are a factor of 500 more viscous than the sub-lithospheric mantle. We also include a stiff core in the center of the plates

Table 4.1: Reference numerical model parameters

Symbol	Parameter	Value	Units
Geometry			
h_l	Lithospheric thickness	80	km
h_{core}	Core thickness	20	km
h_{crust}	Crustal thickness	15	km
h	Domain height (z)	660	km
l	Domain length (x)	7920	km
w	Domain width (y)	5280	km
R_{init}	Initial radius of curvature	150	km
z_{notch}	Initial slab depth	500	km
θ_{init}	Initial slab dip angle	70	deg.
Mechanical properties			
$\Delta\rho$	Plate-mantle density contrast	85	kg/m ³
ρ_0	Mantle density	3300	kg/m ³
η'	Plate/mantle viscosity contrast	500	-
η_0	Mantle viscosity	2.8×10^{20}	Pa s
η_{crust}	Crust viscosity	2.8×10^{20}	Pa s
η_{core}	Core viscosity	1.4×10^{24}	Pa s
Ra_C	Compositional Rayleigh number	8.55×10^5	-

with a thickness of 20 km and a viscosity that is $10\times$ greater than the regular lithosphere ($= 5000\eta_{mant}$). Such a thin core acts to significantly increase the uni-axial/stretching strength of a viscous sheet ($\sim \eta H$), relative to the bending strength ($\sim \eta H^3$). To facilitate decoupling of the down-going plates from the upper plates, we insert 15 km thick, weak ($= \eta_{mant}$) crustal layers within the subducting plates (SP) (e.g., Běhounková and Čížková, 2008; Stegman et al., 2010). The crust, defined by compositional tracers, is linearly tapered full thickness at a depth of 200 km (and above), to zero thickness below a depth of 400 km, in order to avoid stress discontinuities (for more details, see chapter 1: Holt et al. 2015a)

To prevent unrealistic mantle flow at the surface, and rapid plate shortening in the trench parallel direction (y), we also include a side plate which is separated from the subduction system plates by 25 km of mantle material (e.g., Yamato et al., 2009). The side plate also provides a stationary reference frame for velocities measured in the subduction systems, e.g., trench migration rates. Subduction is initiated by allowing the SP to extend below the OP with a center-line radius of curvature (R_{init}) of 150 km. Below the curved portion of the initial lithospheric geometry, we allow the plate to extend to a depth of 500 km with an ini-

tial dip angle of 70° . This dip angle, and the subducting slab evolution, is broadly consistent with the dip observed in models which start with a shallower initial subducting plate notch, during the free-sinking phases. We choose to initially place the tip of the subducting plate near to the free-slip lower boundary in order to focus on the latter stages of slab evolution, as the complementary analytical models are most applicable to this phase where the slab is supported by the model base (Royden et al., *in prep.*). Initially placing the slab-tip near the model base allows the models to reach a near-steady state for a significant length of time, relative to the more transient model evolution observed in model which are initiated by a shallow proto-slab.

4.3.2 Analytical models

We augment our numerical models with analytical models, as recently presented for single slab setups (Royden et al. *in prep.*). *FASTER* is a simplified version of the the time-dependent semi-analytic models presented in Royden and Husson (2006), and initially tested against numerical models in Jagoutz et al. (2015). For a prescribed set of mechanical parameters (e.g., slab strength, density, and thickness), *FASTER* predicts the appropriate slab kinematics (i.e., plate velocities, dip angle) by requiring the viscous slab to adhere to a set of force balances (slab-normal, horizontal, and vertical directions). The force balances include forces related to slab buoyancy, plate coupling, and viscous stresses associated with mantle flow. These analytic models allow us to isolate the key physics of the numerical double slab systems, and assess which aspects can be captured by this simplified modeling strategy.

4.4 Results and Discussion

4.4.1 Slab kinematics

The morphological evolution of the various double slab systems is highly variable, depending on relative dip direction of the two slabs. As described above, the “inward dipping” and “outward dipping” models contain a plane of symmetry at $x = 6$, which simplifies the interpretation of slab evolution by allowing us to consider just one slab in these models. As a result of this, the inward dipping case corresponds to a single slab model within which

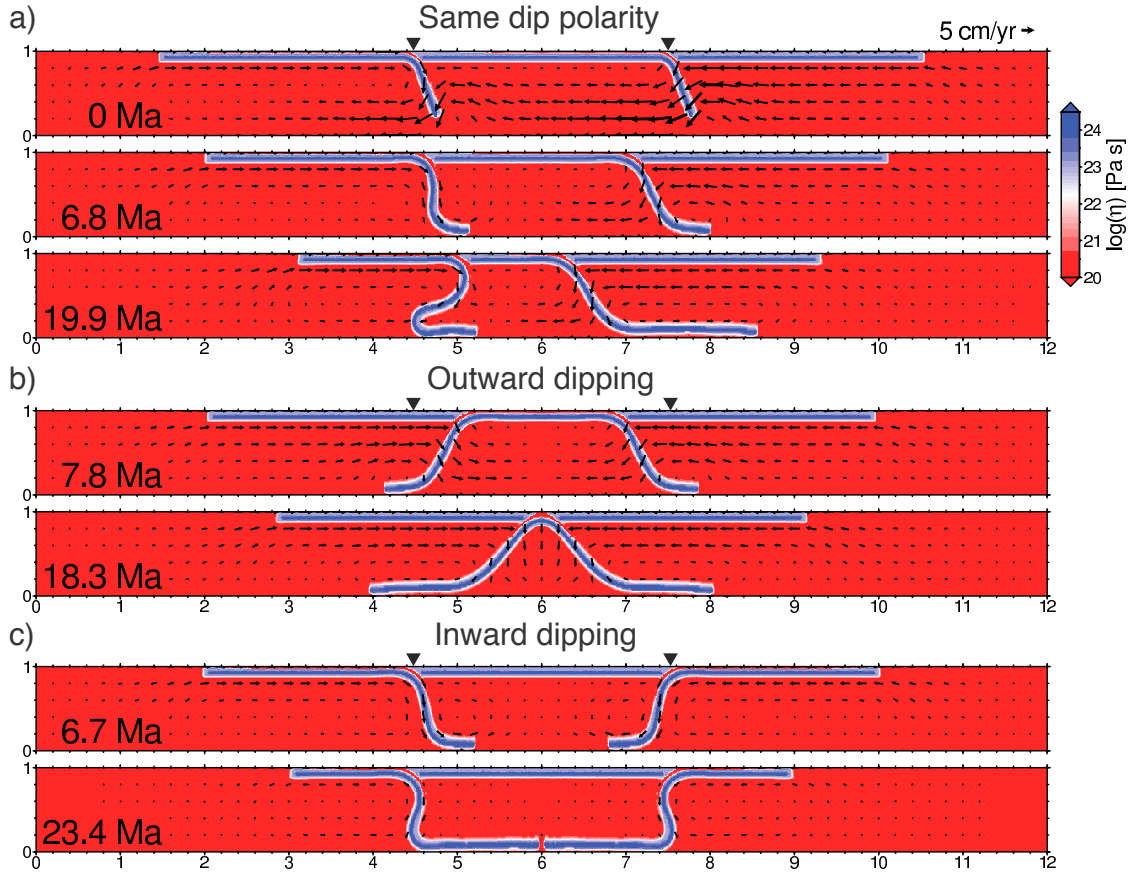


Figure 4.3: Viscosity field snapshots showing the model evolution for the three double slab subduction models. In addition to two panels at equivalent numerical timesteps ($t' = 400, 1100$), the initial geometry/velocity solution for the “same dip polarity” model is shown. The 2-D slices shown are taken through the model mid-plane, $y = 0$. Overlain are velocity vectors.

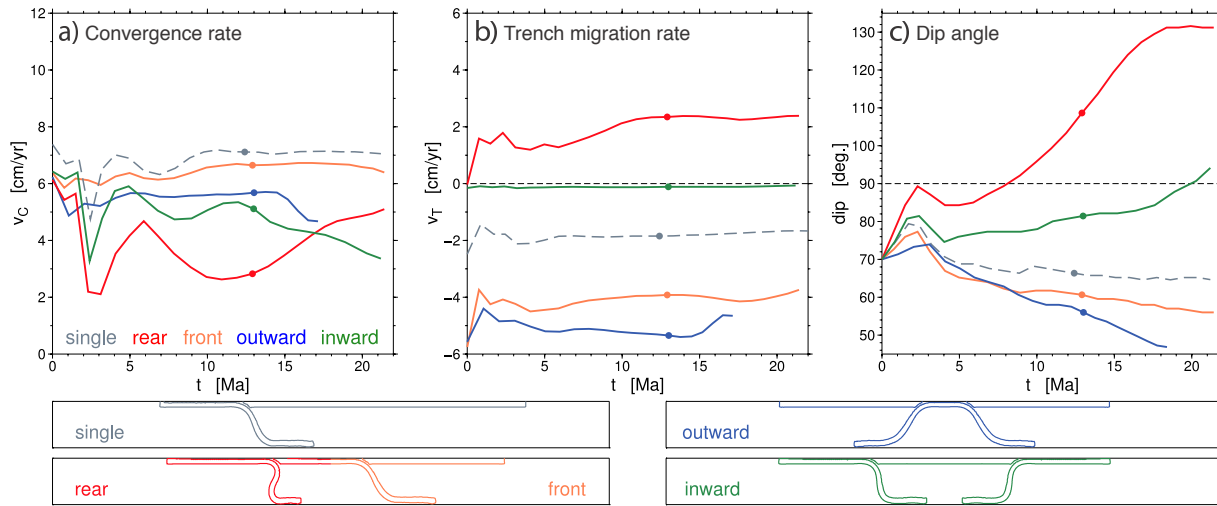


Figure 4.4: Slab kinematics for the double slab models. For the “outward” (blue) and “inward” dipping (green) models, only one curve is shown in each plot due to the model symmetry (i.e., the two slabs are equivalent in these models). In the “same polarity” model, curves corresponding to the two distinct slabs are shown (red = rear, orange = frontal). In addition, a single slab model (gray) is shown for reference. The panels show, a), the convergence rate between the two plates either side of the corresponding trench, b), the trench/subduction hinge migration rate, relative to the passive side plate (negative = trench retreat, positive = trench advance) and, (c), the slab dip angle computed between depths of 200 and 460 km.). Below the kinematics panels, slab outlines illustrate the models shown (time-steps plotted are shown as points on the kinematics plots).

the upper plate is attached (or, “fixed”) to a free-slip side boundary at $x = 6$ and the outward dipping case is, in contrast, equivalent to a single slab model within which the subducting plate is fixed at this boundary. The impact of restricting the mobility of either of the two plates has been studied extensively in 2-D numerical models (e.g., Capitanio et al., 2010a), and the basic kinematic evolution of the inward and outward dipping models can be understood in the context of this previous work. In addition to the morphological evolution of the subduction systems shown in Figure 4.3, we present the associated temporal evolution of the slab kinematics in Figure 4.4. Here, we plot the convergence rates of the individual trenches (V_C), the trench migration rate relative to the stationary side plate (V_T), and the average slab dip angle. Because the lithospheric plates contain strong cores, upper plate stretching is negligible and so the trench migration rate is approximately equivalent to the upper plate velocity. Thus, convergence/subduction is partitioned between subducting plate motion (not explicitly shown) and trench/upper plate motion (i.e., $V_C = V_{SP} - V_T$). For the symmetric models (outward and inward dipping), the kinematics are shown for the slabs which dip towards the right, and thus negative velocities correspond to left-directed motion (e.g., trench retreat), and positive velocities are directed to the right (i.e., subducting plate motion, trench advance). For all of the models presented, the kinematics (and pressure/mantle flow) undergo a transient at the beginning of the model run ($t < 5$ Myr) as the slab bends backwards onto the lower, free-slip boundary. Subsequently, the slab kinematics (e.g., V_T) are less time-dependent, and subduction reaches a pseudo-steady state for most of the slab configurations. Therefore, we focus our analysis/discussion on this latter phase of subduction.

In the outward dipping model, subduction occurs almost entirely by trench migration (i.e., $V_C = -V_T$). Effectively fixing the subducting plate in this model limits subducting plate motion as the plate can only move towards the trench via tensile stretching, which is minor due to the high tensile strength of our lithospheric plates. The observed individual slab convergence rate, of 5–5.5 cm/yr ($= -V_T$), is reduced relative to that observed for a single slab model with equivalent mechanical/rheological properties and an equivalent subducting plate length ($V_{C, \text{single}} = 7.0$ cm/yr). In addition, slabs in the outward dipping model are shallowly dipping, with dip angles that are strongly time-dependent, but always reduced relative to the single slab reference model (single slab dip $\approx 65^\circ$). As slab rollback progresses, the forearc regions of the upper plates eventually collide at ~ 15 Myr (Fig. 4.3b), and convergence ceases.

In contrast, fixing the upper plate, as in the inward dipping model, causes subduction to occur almost entirely by subducting plate advance, with negligible trench motion (i.e., $V_C = V_{SP}$, $V_T \approx 0$). The individual slab convergence rate ($V_{C, inward} = 3.5\text{--}5.5$ cm/yr) is again reduced relative to the single slab case, but the slab dip is significantly elevated with respect to that observed for the reference single slab model. As observed for the outward dipping slab, the inward dipping slab dip angle is highly time-dependent, and gradually increases from the 70° dip angle initially imposed to an angle greater than vertical.

The “same dip polarity” model does not contain such a plane of symmetry and so here the two slabs have distinct evolutions (Fig. 4.3a). After the slab-tip bends atop the free-slip lower boundary, the frontal (i.e., right) slab, which dips away from the central, or intra-arc, region, evolves in a manner similar to the outward dipping slabs with a dip angle ($55\text{--}70^\circ$) reduced relative to the single slab model. In contrast, after resting on the lower boundary, the rear (i.e., left) slab gradually leans forward to obtain a slab dip angle greater than 90° for the vast majority of the model run (Fig. 4.4c). In comparison to the inward dipping model, the rear slab here also has an upper plate which is not mobile, but instead of being fixed in place it is attached to the subducting plate of the other subduction zone. The frontal subduction zone is not dissimilar to the outward dipping scenario in that the motion of subducting plate is restricted, however here the subducting plate is not fixed but is coupled to the rear subduction zone.

Within this same dip polarity model, the two trenches migrate in opposite directions (Fig. 4.4b). In the side plate/box reference frame, the frontal trench retreats towards the subducting plate, as is the typical mode of trench motion in such numerical/analog subduction experiments (e.g., Funiciello et al., 2003a; Schellart, 2004; Bellahsen et al., 2005; Enns et al., 2005; Stegman et al., 2006). However, the rear trench advances towards its upper plate (the middle plate), a mode not observed in any of the other double slab subduction systems, or for any of our single slab numerical models (Royden et al., *in prep.*). Because the two trenches move in opposite directions, the central asthenospheric compartment gradually reduces in size throughout the model evolution. As in the other double slab configurations, the convergence rate of each of the individual trenches is reduced relative to the single slab reference, and single slab models with variable initial plate lengths (Fig 4.4c), but the total convergence (between the left- and right-most plates) is not equally partitioned between the two subduc-

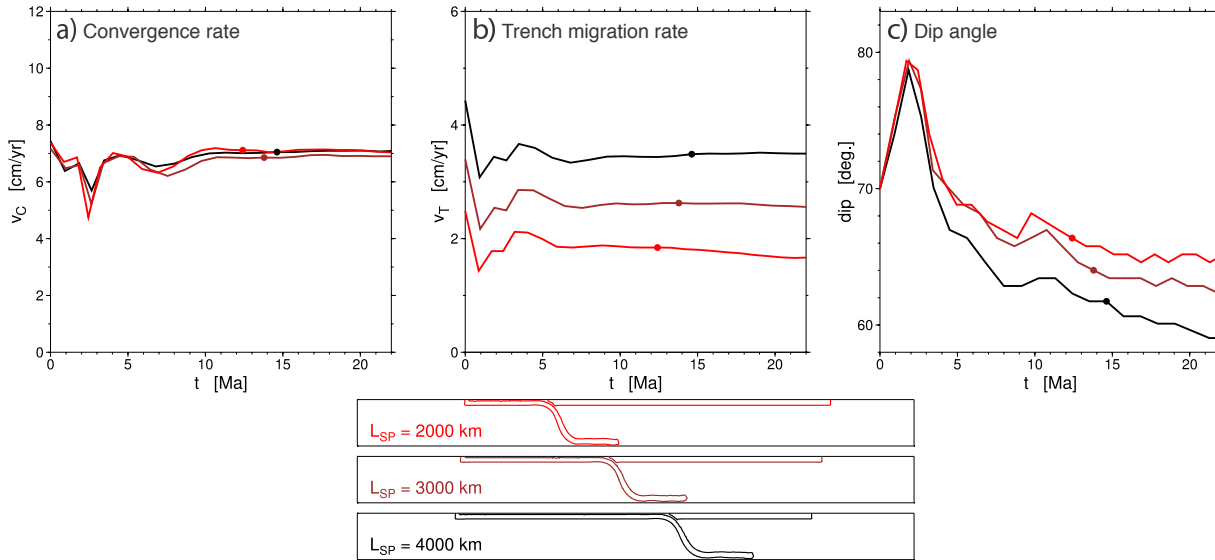


Figure 4.5: Slab kinematics for the single slab models with various initial subducting plate lengths ($L_{SP} = 2000$ km, 3000 km, 4000 km). The kinematic quantities extracted are equivalent to those presented in Figure 4.4.

tion systems. The frontal subduction zone exhibits convergence at a reasonably constant rate of 6.5 cm/yr, while the convergence rate is strongly variable (3-5 cm/yr) but always lower than the frontal subduction zone. While each of the individual trenches in double slab systems have convergence rates reduced relative to the equivalent single slab model, we find that each of the double slab systems have elevated total convergence rates between the left- and right-most plates for the entirety of the model runs.

4.4.2 Mantle pressure and flow

Single, isolated subducting slabs with equivalent mechanical properties subduct via a combination of trench retreat and subducting plate advance, which results in a mantle wedge with strongly negative dynamic pressure and a sub-slab region (i.e., left of right dipping slab) with positive dynamic pressure (Fig. 4.7). In a 3-D setting, such a pressure field induces asthenospheric flow both in the vertical plane (poloidal flow: irrotational in the horizontal plane), and rotational flow in the horizontal plane around the slab edge (toroidal flow) induced by slab rollback (e.g., Kincaid and Griffiths, 2003; Funiciello et al., 2003a; Schellart,

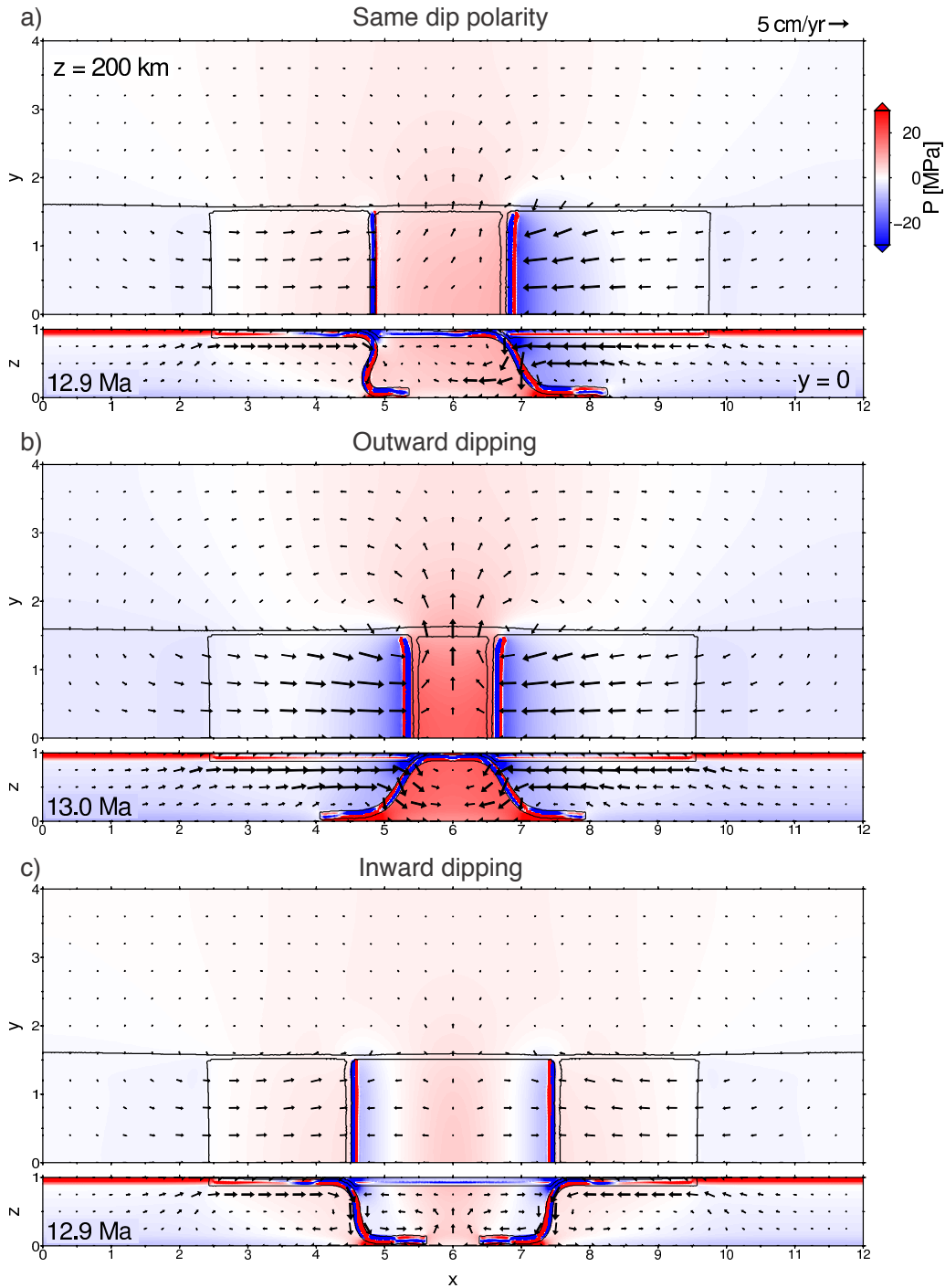


Figure 4.6: Dynamic pressure fields for the three double slab models, plotted at equivalent numerical time-steps ($t' = 700$). For each model, a sub-lithospheric horizontal slice ($z = 200$ km), and a vertical slice through the model mid-plane ($y = 0$) is shown. Overlain are velocity vectors, and an iso-viscosity contour is plotted in the horizontal slice panels to show the surface position of the plate boundaries.

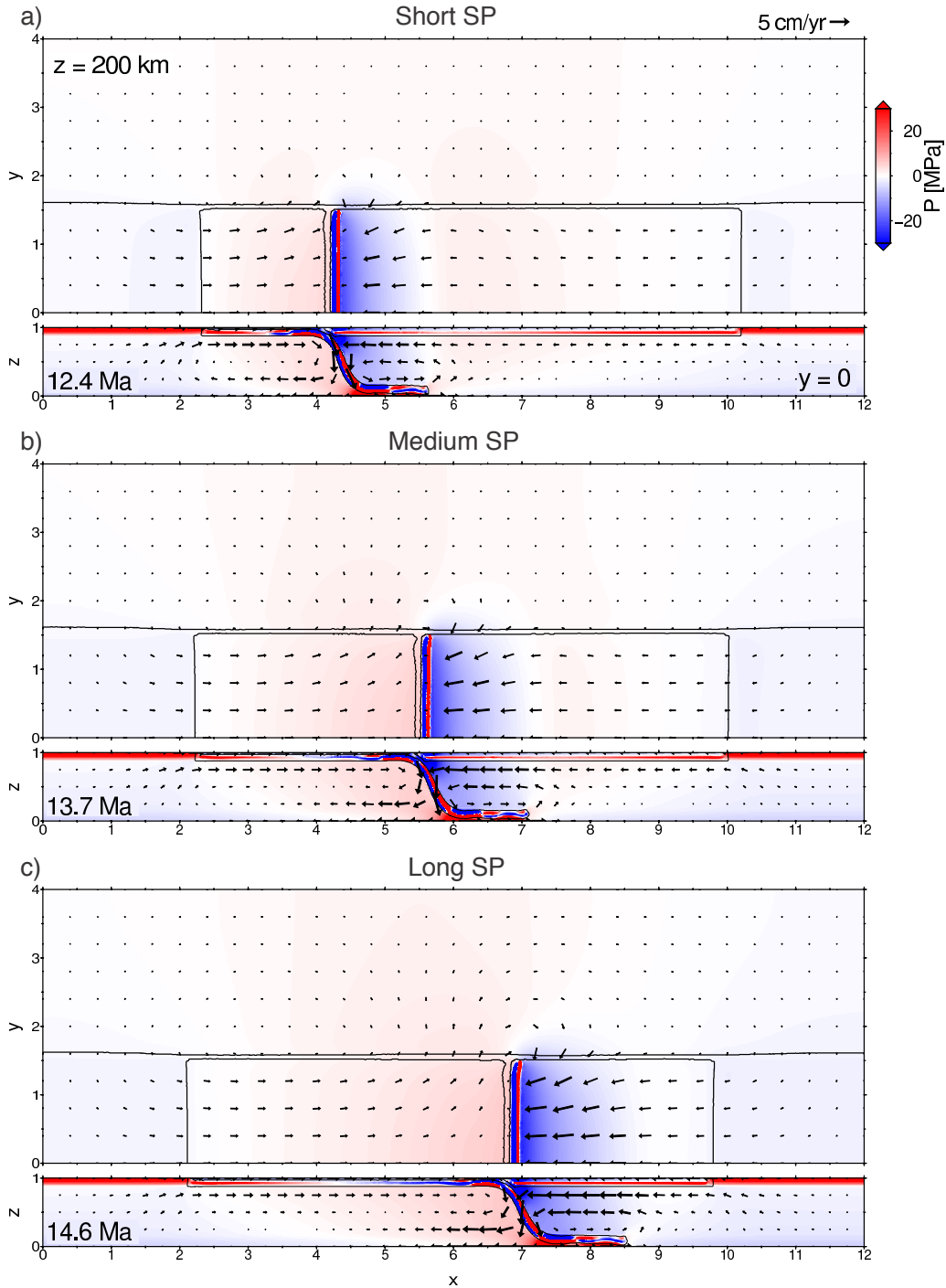


Figure 4.7: Dynamic pressure fields for the single slab models with various initial subducting plate lengths ($L_{SP} = 2000$ km, 3000 km, 4000 km). Quantities plotted are equivalent to those presented in Figure 4.6.

2004; Piromallo et al., 2006; Stegman et al., 2006). Relative to this, the addition of another subducting slab induces dynamic pressure and mantle flow fields that are significantly distinct (Fig. 4.6) to those observed in such single slab analog and numerical experiments (Fig. 4.7).

Figure 4.6 shows the dynamic pressure field, and associated mantle flow, at equivalent numerical time-steps for the three double subduction models. A key feature in these systems is that an additional slab isolates the central/intra-arc asthenospheric region, giving rise to three distinct asthenospheric regions with variable dynamic pressures. Adjacent to the subduction zones, and beneath the passive side plate, dynamic pressure gradually reduces to near-zero. Thus, pressure gradients develop in the trench-parallel direction which drive asthenospheric flow in the horizontal plane (e.g., Fig. 4.6b). The magnitudes of such pressure gradients, and the associated flow, depend on the magnitude of the intra-arc pressure which is a function of the rate, and direction (i.e., towards or away from the intra-arc region), at which the surrounding slabs are migrating. In addition to exerting a strong control on the degree of trench-parallel mantle flow, the positive pressures that develop in the intra-arc region induce highly variable pressure gradients across the slab (i.e., in trench-perpendicular direction) relative to the single slab case, which are discussed in subsequent sections.

Mantle flow partitioning

In order to quantify the mantle flow characteristics, we examine the temporal evolution of the flow partitioning between toroidal and poloidal components (e.g., Bercovici, 1993). Following Tackley (2000) and Piromallo et al. (2006), the 2-D velocities on each horizontal layer are decomposed into irrotational (i.e., poloidal) and divergence-free (i.e., toroidal) vector fields. This decomposition is conducted by solving Poisson's equations for both the poloidal and toroidal potentials (Tackley (2000): Equation 14), and then using the resultant scalar potentials to construct the vector fields (Tackley (2000): Equation 13). The vertical velocity field is added to the poloidal field. We quantify the flow partitioning by computing the RMS of the poloidal and toroidal velocity fields for each of the 10 horizontal layers ($\Delta z = 66$ km) extracted. The total toroidal/poloidal ratio (TPR) is then taken as the sum of the toroidal RMS velocities divided by the sum of the poloidal RMS velocities.

We compare double slab subduction-induced mantle flow fields with single slab models with various initial subducting plate lengths ($L_{SP} = 2000$ km, 3000 km, 4000 km),

in order to span the range of upper and subducting plate lengths in the double slab models. As shown in Figure 4.5, the three single slab models have near-constant convergence rates (≈ 7 cm/yr), but variable rates of trench motion ($L_{SP} = 2000$ km: $v_T \approx 2$ cm/yr, $L_{SP} = 4000$ km: $v_T \approx 3.5$ cm/yr). As the convergence does not vary as a function of plate length, the poloidal flow amplitude also exhibits only minor variation (Fig. 4.8a). In contrast, the toroidal flow amplitude increases with slab rollback velocity, which is proportional to subducting plate length (Fig. 4.8b). This occurs because the horizontal motion of the stiff slab, in the form of slab rollback/trench retreat, preferentially induces rotational flow in the horizontal plane. Despite this variability in TPR for single slab models, we find that for our chosen lithosphere-asthenosphere viscosity contrast ($\eta'_{lith.} = 500$), all of the single slab numerical models induce mantle flow that has a poloidal component stronger than the toroidal component (i.e., $TPR < 1$).

However, the double slab subduction-induced mantle flow fields are not dominantly restricted to $TPR < 1$. The outward dipping model has a high TPR of ≈ 2.4 , after the initial phase and prior to arc-collision (Fig. 4.8c). The same dip polarity model has a TPR in excess of any of the single slab models for the vast majority of the model evolution, and $TPR > 1$ for $t < 15$ Myr. The inward dipping model has a TPR which largely falls within the range observed for the various single slab models. For both the inward and outward dipping cases, despite the presence of two subducting slabs with total convergence rates ($\propto$ volume flux of subducting slab) elevated relative to the single slab models, the RMS amplitude of the poloidal flow is comparable to that of the single slab models. This follows from the dominance of lateral slab motion (i.e., rollback) in the outward case, but is less intuitive to understand in the inward dipping model, where trench motions are negligible. The same dip polarity model does, however, have a poloidal RMS velocity elevated by ~ 0.2 cm/yr relative to the single slabs.

The amplitude of the toroidal flow (v_{tor}) components are, in contrast, much more variable. The outward dipping model induces dominantly toroidal flow because subduction occurs almost entirely via slab rollback. Relative to single slabs, the same dip polarity model also exhibits high v_{tor} , albeit much lower than the outward dipping case. This is likely due to both of the two trenches/slabs exhibiting significant horizontal motions, with the rear trench advancing and the frontal trench retreating (Fig. 4.4b). This results in the elevated TPR ob-

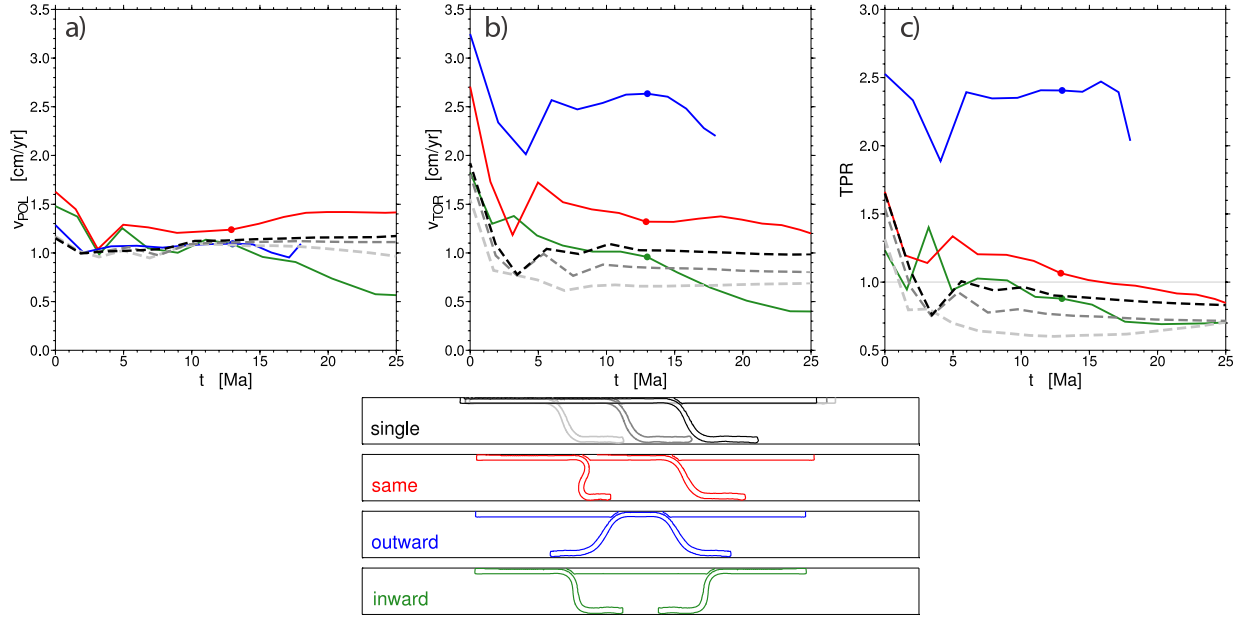


Figure 4.8: Temporal evolution of mantle flow partitioning into poloidal ($\nabla_h \times \vec{v}_{pol} = 0$) and toroidal ($\nabla_h \cdot \vec{v}_{tor} = 0$) components for double slab models with variable dip polarities, and single slab models with variable plate lengths ($L_{SP} = 2000$ km, 3000 km, 4000 km). Plotted is, a), the total RMS of the poloidal velocity field (summed over 10 horizontal layers), b), the total RMS of toroidal velocity field (summed over 10 horizontal layers), and, c), the ratio of the total toroidal to total poloidal RMS velocities (e.g., Piromallo et al., 2006).

served in both the outward dipping case and, to a lesser, extent the same dip polarity case. For inward dipping slabs, v_{tor} falls within the single slab range until ≈ 17 Myr (when convergence rates reduce due to slab-tip interactions at the center of the box: Fig. 4.4a), which results in a TPR comparable to that computed in the single slab models, despite the trenches being stationary here and mobile in the single slab models. This occurs because the additional slab presents a lateral boundary to the plate convergence-induced return flow, which appears to induce significant toroidal flow without lateral trench motions. Thus, double slab systems appear to promote toroidal mantle flow, either through rapid trench motions or the presence of additional lateral viscosity variations.

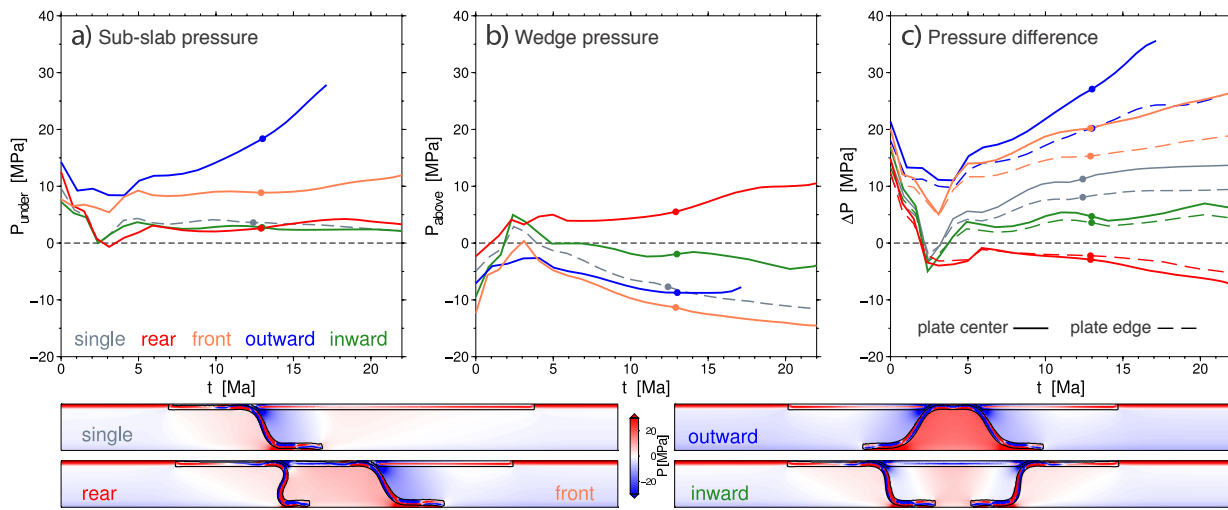


Figure 4.9: Asthenospheric pressure evolution for each of the double slab models, with the addition of a single slab model for reference. Color scale equivalent to that in Fig. 4.4. Panels show, a), Asthenospheric pressure extracted at a depth of 330 km (i.e. mid-box), 300 km away from the under-side of the slab (i.e. on the oceanic side for a typical ocean-continent system), and at the mid-plane ($y = 0$), b), Asthenospheric pressure extracted at an equivalent depth and slab distance but on the opposite, wedge side of the slab edge, and, (c), The wedge-directed pressure drop/gradient taken between these two pressure points (i.e., $b - a$). In addition to the mid-plane (solid curves), the pressure drop near to the slab edge, at $y = 750$ km, is shown (dashed line).

Slab pressure gradients

For each of the slabs in the double slab models (and a single slab reference, with $L_{SP} = 2000$ km), Figure 4.9 shows the temporal evolution of the asthenospheric pressure extracted at mid-box depths ($z = 330$ km) at either side of the slab (330 km from the slab edge). In addition, we compute the pressure difference between these two points (Fig. 4.9c). This pressure difference, ΔP , is an approximate metric for the force on the slab due to mantle flow which, for single subduction systems, has a horizontal component opposing trench retreat (i.e., right-directed for our single slab setup) and a vertical component opposing slab sinking/convergence (i.e., upwards-directed). For our sign convention, such a typical ΔP is positive, and is reached after the initial transient period ($t < 5$ Myr). For our single slab reference, an average pressure difference of 10 MPa and an average dip angle of 70° gives a slab normal (wedge-directed) force of approximately 7.0 TN/m, which corresponds to $\sim 16\%$ of the slab pull (44 TN/m) for our selected lithosphere-asthenosphere density contrast (85 kg/m^3).

However, for the various double slab configurations, ΔP is highly variable. In the outward dipping case, the slabs are both rolling back into the intra-arc region which induces very high positive dynamic pressure in this region (~ 20 MPa) which, because the slabs are dipping away from this pressurized region, results in very high ΔP relative to the single slab reference. In contrast, the slabs in the inward dipping case dip towards the central region, which reduces ΔP relative to that observed for the single slab. This occurs because the mantle wedges of these slabs are located within the intra-arc region and so the lateral restriction of the mantle wedge flow produces wedge pressure less negative than that typically observed. We have additionally examined the slab pressure gradients and associated kinematics for models with lithospheric plates that are longer ($L_{SP} = 3000$ km), in order to verify that the observed systematics are not simply a product of one slab obstructing the other, particularly for the inward dipping case where the slab-tips are almost touching at the end of the run (Fig. 4.3c). While the pressure drop magnitudes are less variable relative to the single slab case, due to a larger/less constrained central region, the systematic variation of both the slab kinematics (Fig. 4.10a-c) and slab pressures (Fig. 4.10d-f) are broadly equivalent.

In the same dip polarity configuration the frontal slab dips away from the pressurized central region resulting in elevated ΔP , as for the outward dipping slabs (Fig. 4.9c).

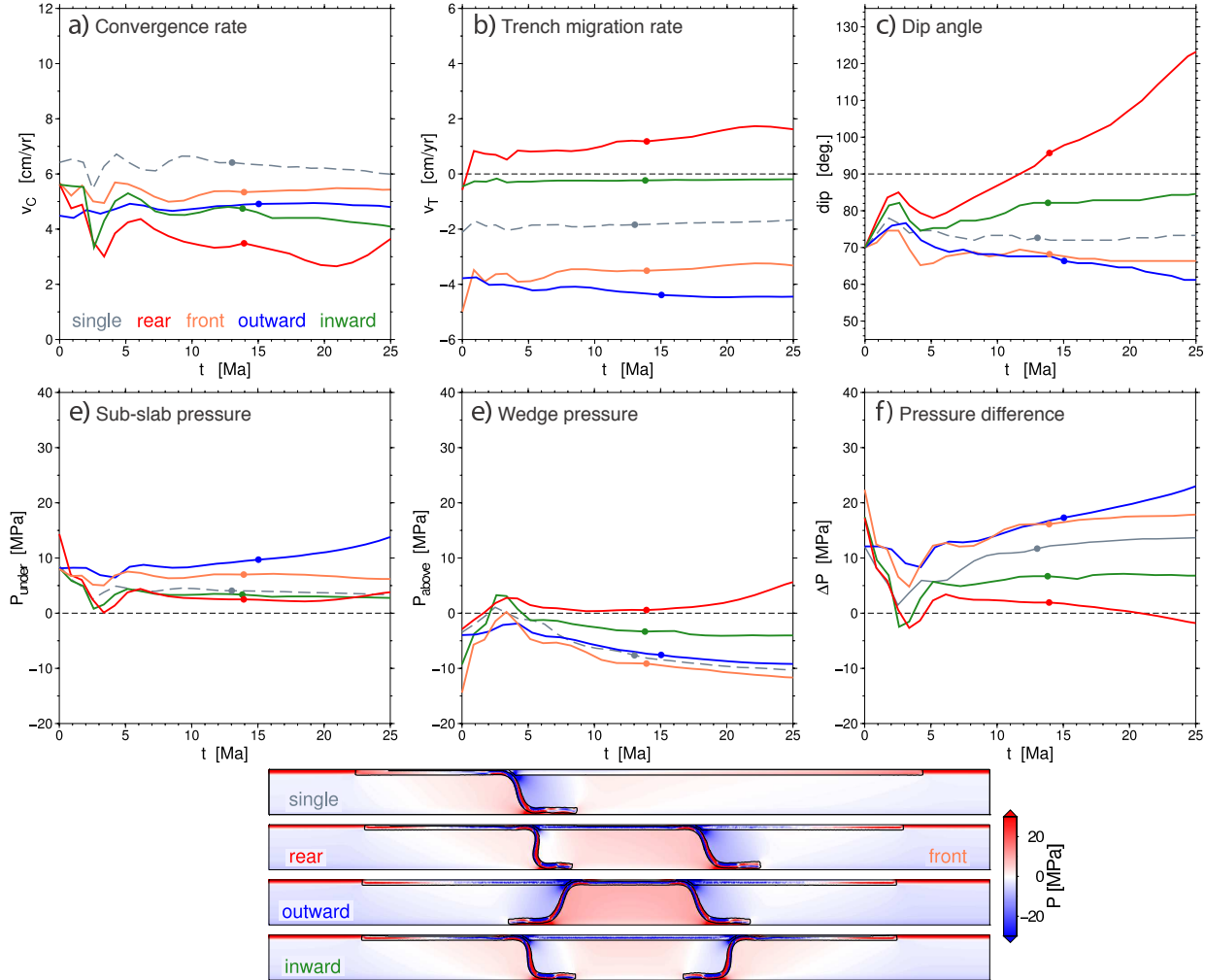


Figure 4.10: Slab kinematics (a-c), as in Figure 4.4a-c, and temporal evolution of mantle pressure (d-f), as in Figure 4.9a-c, for the double slab models with longer lithospheric plates. In these models, the plates have initial trench-perpendicular lengths of 3000 km (i.e., total plate length = 9000 km), while the reference models have initial plate lengths of 2000 km. Below the panel plots, snapshots of the dynamic pressure fields are shown for each of the models along the mid-plane, $y = 0$ (time-steps plotted are shown as points on the panel plots).

Relative to the outward dipping model, the intra-arc region is less pressurized (~ 10 MPa) due to the presence of only one slab (frontal) that is rapidly rolling back into this region. Thus, despite being elevated with respect to the single slab case, the frontal slab ΔP is less than the outward dipping ΔP . In contrast, the rear slab dips into the central region which results in reduced ΔP , as for inward dipping slabs. However, because the positive intra-arc pressure is elevated relative to that in the inward dipping model, ΔP is further reduced and becomes negative (Fig. 4.9c). Such a negative ΔP is strikingly different to that expected for a typical subduction zone, and that observed for the frontal slab within the same model, and corresponds to a horizontal force component directed left-wards, toward the underside of the subducting slab.

4.4.3 Mantle pressure control on slab kinematics

The subduction-resisting lifting force produced by across-slab mantle pressure gradients (ΔP) is typically a significant proportion of the subduction-driving slab pull force (e.g., for single slab: $\sim 16\%$), and so exerts a first order control on convergence rates, v_C . To first order, the convergence rates for slabs with mobile, or “free”, upper plates are dictated by the magnitude of this force induced by asthenospheric flow. In contrast, the convergence rates of slabs that have upper plates that are attached to either another subducting plate (the rear slab in the same dip polarity model), or an upper plate of another subducting plate (the inward dipping slabs) do not appear to be dictated primarily by this asthenospheric pressure gradient. The slabs with free upper plates are the single slab models, the outward dipping slabs, and the frontal (/rightmost) slab in the same dip polarity model. For these slabs, as the magnitude of the ΔP increases (Fig. 4.9c), the convergence rate decreases (Fig. 4.4a): The single slab has the lowest magnitude ΔP of the free upper plate models and, therefore, the highest convergence rates. The outward dipping slabs have the lowest individual trench convergence rates of the free upper plate models, owing to their high pressure drops.

For double slabs with free upper plates, the relationship between slab kinematics and ΔP can be largely captured by the analytical models described in section 4.3.2. In Figure 4.11, we plot the ΔP and slab dips extracted from the numerical models (after the initial transient phase), and underlay these points by the analytical solutions for two different convergence rates (5 cm/yr, 7.5 cm/yr). The analytical solutions predict that as the magnitude

of the pressure drop, and thus vertical lifting force, decreases, the slab dip angle increases. In addition, for a fixed ΔP , as the convergence rate increases, the slab dip angle decreases. The points extracted from the various single plate models with free upper plates (Fig. 4.5) track the analytical solution for $v_C = 7.5$ cm/yr. For these models, the actual average convergence rate in the numerical models is ~ 7 cm/yr. The points extracted from the double slabs with free upper plates appear to also follow the functional form of the analytically-derived curves, despite locating at analytical convergence rates (≤ 5 cm/yr) reduced relative to those observed in the numerical models (outward dipping: ~ 5.5 cm/yr, frontal slab in same dip polarity model: ~ 6.5 cm/yr). A direct correspondence between the double slab numerical models and the analytical expressions derived based on single slab mantle flow geometries is not expected as the mantle flow induced in the double slab systems is significantly more complicated than that observed for the single slabs, which is assumed in the analytical force balances. That said, this approximate correspondence suggests that the double slabs, while having asthenospheric pressure drops greater in magnitude than single slab systems, have kinematics dictated by mantle pressure in a similar manner to the isolated single slab systems.

In contrast, the double slab subduction zones with upper plates that are either fixed to center of the domain (inward dipping) or “pulled” by an additional subduction zone (frontal slab in same polarity model) do not display a relationship between slab kinematics and mantle pressure that is captured by the analytical force balances (Fig. 4.11). Both of these slabs have dip angles that are for the most part higher than that predicted by the analytical expressions, and have large dip variability for near-constant ΔP . This lack of analytical-numerical correspondence occurs firstly because the dips of these slabs are either greater than (rear slab) or near to (inward) vertical for significant proportions of the model evolution, while the analytical force balances are setup for slabs with shallower, more typical dip angles. Additionally, these slabs exhibit non-standard behavior because the lack of a free upper plate prohibits the subduction system from moving towards (e.g., via slab/trench rollback) the kinematic parameters expected for single slab systems for a given ΔP . Beyond restricting trench motion, the upper plate of the rear slab (same dip polarity model) transmits slab pull-related forces from the frontal slab to the rear subduction zone which, by modifying the rear slab force balance, actively pulls the rear slab away from the kinematics envisaged for

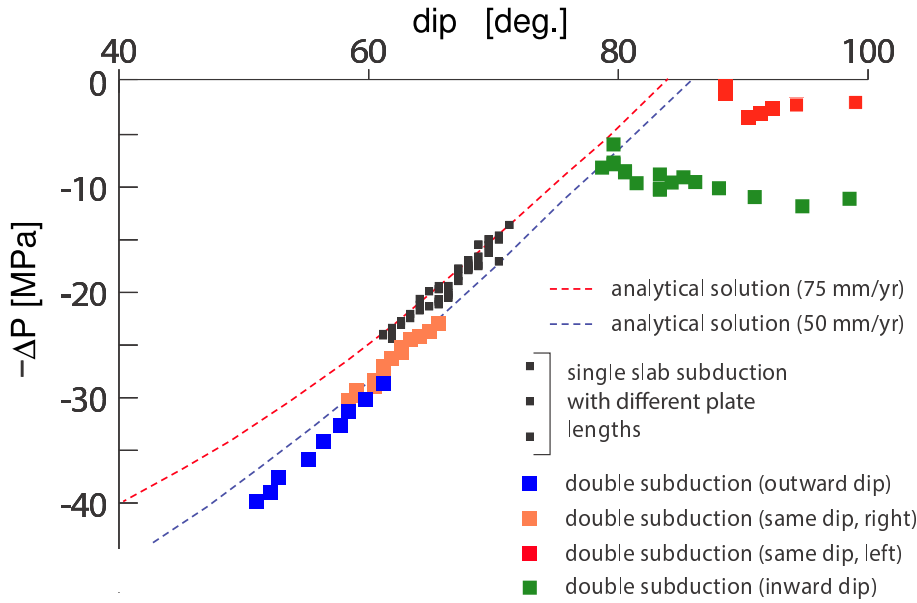


Figure 4.11: Slab dip angle (i.e., Figure 4.4c) vs. asthenospheric pressure drop (i.e., Figure 4.9c) for both the double slab subduction models, and single slab models. For all slabs except the front/right slab in the “same dip polarity” model, which does not reach a clear steady state geometry, the points are extracted after the slab evolution has reached an apparent steady state (near-constant convergence rate). The curves underlain show the analytic relationships for variable convergence rates, computed using *FASTER* (Section 4.3.2).

standard subduction systems. As proposed by Čížková and Bina (2015), this additional pull provides a potential mechanism for the trench advance observed at the rear subduction zone (Fig. 4.4b). In the next section, we therefore compute the stresses transmitted between these two slabs in order to examine the degree to which the kinematics of the rear slab are coupled to the frontal slab in this way.

4.4.4 Slab coupling

Dynamic model

In order to investigate the magnitude of the forces transmitted from the frontal to the rear subducting slabs in the same dip polarity model, we directly compute average shear and normal stresses in the weak, isoviscous subduction zone channels for the same dip polarity model (Fig. 4.12). While we focus on the double slab configuration, we also present the

equivalent coupling stresses for a single slab reference model, with $L_{SP} = 2000$ km (Fig. 4.13). For simplicity, we compute coupling stresses for a model which does not contain a strong lithospheric core. This is done because, while the slab evolution of this model is almost indistinguishable from the reference case, the lithospheric pressure field is more continuous and so stresses computed at a particular depth are more representative of the whole lithospheric channel, which simplifies the calculation. The shear stress, parallel to the interface, is computed as,

$$\tau_{shear} = 2\eta_c \dot{\epsilon}_{shear} \approx \eta_c \frac{\Delta v_{parr.}}{h_c \sin(\theta)}, \quad (4.1)$$

where η_c is the viscosity of the crust/coupling channel (2.8×10^{20} Pa s), $\dot{\epsilon}_{shear}$ is the shear strain rate, $\Delta v_{parr.}$ is the difference between the channel-parallel velocity component either side of the crustal channel ($\Delta v_{parr.} = v_{parr., UP} - v_{parr., SP}$), h_c is the horizontal thickness of the crustal channel, and θ is the average dip angle of the channel. For simplicity the geometrical properties (h_c and θ), and velocities either side of the channel, are extracted at a mid-lithospheric depth of 50 km. Our simplified shear stress computation assumes that down-dip gradients in the channel-perpendicular velocity are negligible. Our chosen velocity sign convention, which gives positive $v_{parr.}$ in the channel up-dip direction and positive $v_{norm.}$ in the subducting-to-subducting plate direction, results in positive shear stress values for both trenches (Fig. 4.12a). This corresponds to a shear force on the subducting plate with a horizontal component directed towards the subducting plate, and a vertical component directed upwards.

Similarly the deviatoric normal stress on the crustal interface, is computed as,

$$\tau_{norm.} = 2\eta_c \dot{\epsilon}_{norm.} \approx 2\eta_c \frac{\Delta v_{norm.}}{h_c \sin(\theta)}, \quad (4.2)$$

where $\Delta v_{norm.}$ is the difference between the channel-perpendicular velocity component either side of the crustal channel ($\Delta v_{norm.} = v_{norm., UP} - v_{norm., SP}$). We compute positive deviatoric normal stresses (Fig. 4.12b), which, for our chosen sign convention, correspond to compressive deviatoric normal stresses (Fig. 4.12d). Such a compressive deviatoric normal stress exerts a force on the subducting plate with a horizontal component directed towards the subducting plate, and a vertical component directed downwards. The deviatoric normal

stress can then be combined with the dynamic pressure P from our numerical solution (average value within the channel at $z = 50$ km), to give the full normal stress ($\sigma_{norm.} = \tau_{norm.} + P$).

After computing the channel width-averaged normal ($\sigma_{norm.}$) and shear stresses (τ_{shear}), we then integrate the stress components along a geometrical approximation of the subduction zone boundary and rotate the resulting forces to retrieve the effective horizontal and vertical coupling forces, F_X and F_Z , acting on the subducting slab (Fig. 4.12c). For simplicity, we assume that the dip of the subduction is constant at the value computed at mid-lithospheric depths. The horizontal and vertical forces can then be approximately computed by multiplying the corresponding stress components by the length of the plate interface,

$$F_X = (\tau_{shear} \cos(\theta) + \sigma_{norm} \sin(\theta)) \frac{h_l}{\sin(\theta)}, \quad (4.3)$$

and,

$$F_Z = (\tau_{shear} \sin(\theta) - \sigma_{norm} \cos(\theta)) \frac{h_l}{\sin(\theta)}, \quad (4.4)$$

where h_l is the lithospheric thickness (80 km) and θ is the average dip angle of the channel. Positive F_X is directed to the left (i.e., upper-to-subducting plate direction) and positive F_Z is directed upwards.

The coupling stresses for the frontal subduction zone, which has a free upper plate, are very similar in magnitude and temporal evolution to those observed for a single slab with a free upper plate (cf. Fig. 4.12, 4.13). After the initial transient phase ($t < 5$ Myr), where the stresses are dictated both by the initially imposed geometry and also slab folding, the shear stress (Fig. 4.12a) and deviatoric normal stress at the frontal subduction zone (dashed brown line: Fig. 4.12b) are near-constant, at values of ≈ 15 MPa. The across-channel velocity gradients, both channel-parallel (shear) and channel-perpendicular (normal), are dominated by horizontal plate convergence and so these stresses scale approximately with the convergence velocity, v_C . The near-constant deviatoric stresses for the frontal subduction zone can therefore be attributed to the near constant v_C observed (Fig. 4.4a). For the rear slab, v_C is more temporally variable and also reduced in magnitude, and so the deviatoric stresses have lower average values (~ 10 MPa) with more variability. Despite this, the channel-averaged deviatoric stresses for both channels are still reasonable constant and generally < 20 MPa.

The key difference between the two subduction zone channels, in the double slab model, lies in the dynamic pressure, P . Figure 4.12b shows the temporal variation in the average channel P , and the full P field (zoomed into the plate boundaries) is shown in Figure 4.12d for two illustrative time-steps. After the initial transient phase, the pressure evolution in the channels of the two subduction zones is very different. At the frontal subduction zone, P gradually transitions from highly negative (i.e., extensional) to positive (compressional) as the model evolves. As for the single slab models shown in Figure 4.13, we attribute the switch from negative to positive channel pressure to a transition in style of return flow below the right-most upper plate (Fig. 4.3a). During the later stages of the model, the return flow cell develops a greater lateral extent, resulting in greater integrated basal traction on the upper plate and. This causes the upper plate to be dragged into the subduction plate hinge more rapidly than slab roll-back. The positive coupling P is, therefore, co-existent with fore-arc compression in the upper plate (e.g., Tagawa et al., 2007b; Capitanio et al., 2010a), as also demonstrated in chapter 1 (Holt et al., 2015a).

In contrast, the pressure in the rear subduction zone channel remains highly negative throughout. While the coupling stresses have amplitudes of 10–20 MPa, rear slab P has an average value of approximately -70 MPa. We attribute such strongly extensional coupling P at the rear subduction zone channel to the transmission of slab pull from the frontal slab. Unlike the frontal subduction zone, where the full normal stress becomes compressional at ~ 13 Myr, the full normal stress is extensional for the entire evolution of the rear subduction zone. Due the high magnitude of the channel P , the coupling forces acting on the rear slab are largely dictated by P . While the positive τ_{shear} and $\tau_{norm.}$ give a horizontal coupling force (F_X) component directed towards the subducting plate, the negative P results in a F_X (≈ 5.5 TN/m) directed towards the upper plate (Fig. 4.12c), i.e., in the direction of the observed trench advance (Fig. 4.4b). Similarly, F_Z is dictated chiefly by P , resulting in an upwards directed F_Z . In contrast, for both the frontal trench and the single slab model, the transition in P , from extensional to compressional, results in an F_Z that transitions from upwards to downwards directed, and an F_X that transitions from upper plate to subducting plate directed (Fig. 4.12c, 4.13c). Thus, the slab-pull induced strongly negative P modifying the force balance acting on the rear slab, relative to a slab with a free upper plate. As proposed by Čížková and Bina (2015), the computed rear-slab F_X indeed provides an explanation for

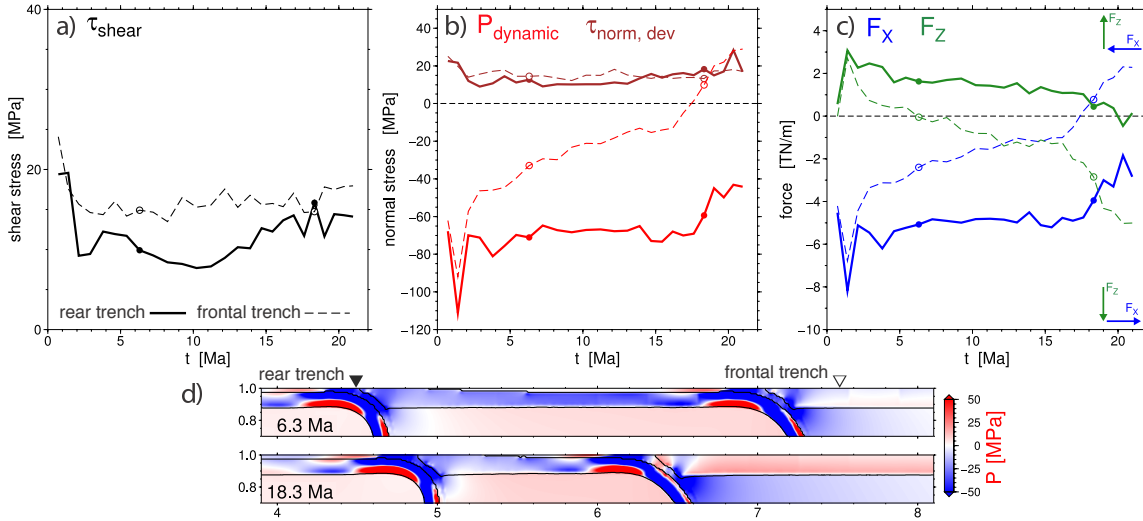


Figure 4.12: Plate coupling, as a function of time, for the trenches in the “same dip polarity” model, without a strong core. As outlined in the main text, deviatoric stresses within the weak, “crustal” channel are computed using the velocities, rotated by the channel dip angle, extracted at $z = 40$ km and the channel viscosity ($= 2.8 \times 10^{20}$ Pa s). Panels show, a), channel shear stress (positive corresponds to an up-dip directed force on the subducting plate), b), channel normal stress (positive = compression) and dynamic pressure (Stokes solution), and (c), horizontal (blue) and vertical (green) forces acting on the subducting slab (i.e., horizontal, vertical components of the stress components integrated approximately over subduction interface down to the base of the lithosphere). Arrows in the right-hand corners show the force directions. The lower plot, (d), shows the dynamic pressure field, zoomed into the trenches, at two time-steps (points on panels a, b, and c). Inverted triangles indicate initial trench locations.

the trench-advance observed (Fig. 4.4b). For the rear slab, considering also the horizontal component of the force derived from the across-slab pressure difference (directed left-wards), it is apparent that a clock-wise torque acts on this slab. This sense of torque is in accordance with the slab evolution observed, i.e., trench advance and slab steepening.

Instantaneous models

Our dynamic models assume that the upper-lower mantle boundary is an impermeable, free slip boundary. Therefore, our slabs cannot penetrate below a depth of 660 km, and readily move laterally atop the lower mantle. While this assumption is made for simplicity, tomographic images do appear to show many slabs unable to sink into the lower mantle, and

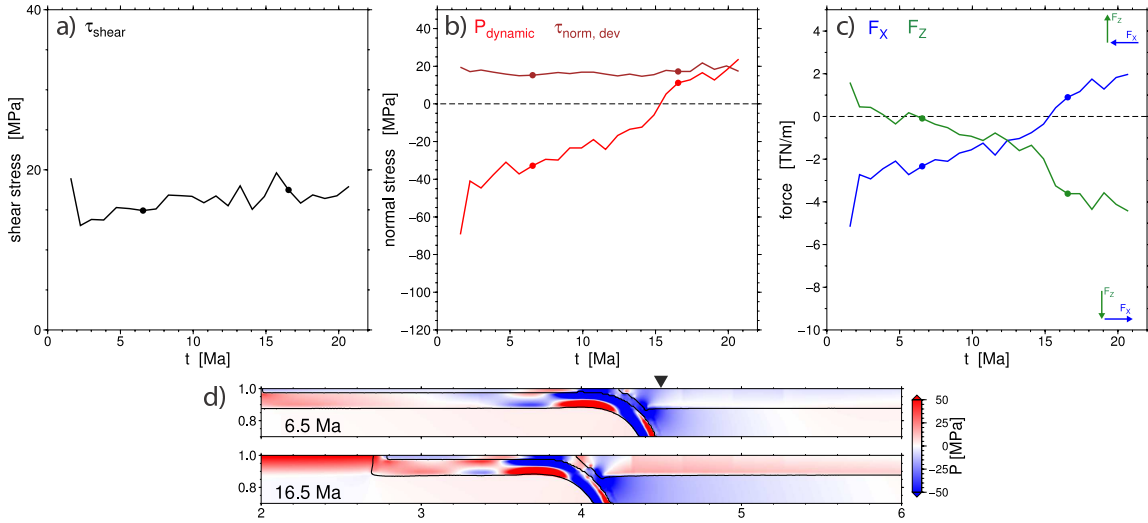


Figure 4.13: Plate coupling, as a function of time, for the trench in an example single slab model ($L_{SP} = 2000$ km), with a strong core. Quantities plotted are equivalent to those presented for the double slab model in Figure 4.12.

able to propagate significant lateral distances above the upper-lower mantle boundary (e.g., Li et al., 2008). Particularly, this appears to be the case for the two trenches, the Izu-Bonin-Marianas and Ryuku, that make up the present day example of a same dip polarity double subduction system (e.g., Takeuchi et al., 2014). That said, many slabs also appear to propagate through the upper-lower mantle boundary, and so here we use instantaneous models to examine the effect of a viscous lower mantle on the magnitude of the rear slab coupling stresses/forces discussed in the previous section. In addition to the strength of the lower mantle, such instantaneous models allow us to examine the control that the width of the subducting and upper plates (i.e., the 3-D geometry) has on the inter-slab coupling (Fig. 4.15), and the crustal and lithospheric strengths (Fig. 4.16).

Our instantaneous models use evolved subducting slab morphologies taken from our dynamic same dip polarity model without a strong plate core, at two different time-steps ($t' = 400$ and 900). Two different time-steps are examined in order to verify that the systematics observed are not a direct result of a particular stage of dynamic model evolution. We take the dynamic slab geometry at the center of the trench ($y = 0$), and use this slab geometry for the whole lateral extent of the subduction system (i.e., we assume linear trenches

for instantaneous computations). The model setup (e.g., box dimensions, upper mantle resolution), and parameters (other than those systematically varied), are equivalent to the dynamic model, aside from the addition of a lower mantle (box height increases to 1320 km). A caveat of this instantaneous modeling approach is the inherent modeling assumption that the morphological evolution of the subducting slabs with a viscous lower mantle is equivalent to that observed in the dynamic models with an impermeable boundary. While the slab evolution would likely be different if the models with a viscous lower mantle evolved dynamically, these instantaneous models nonetheless allow us to develop first-order intuition for how forces can be approximated between two subduction zones in a 3-D geometry.

The instantaneous modeling strategy is illustrated in Figure 4.14. Shown is an example instantaneous computation with the reference parameters (i.e., table 4.1) and an intermediate mantle viscosity of $\eta'_{660 \text{ km}} = 25$. Dynamic pressure, P , profiles extracted through the center of the lithospheric plates are compared for the instantaneous model, and the dynamic model at the stage used for the slab shapes in the instantaneous model. As expected, the pressure profiles are qualitatively very similar, with average channel P within 5 MPa of each other for both frontal and rear subduction zones. The pressures in the subducting plate hinge region, associated with slab bending, are higher amplitude in the instantaneous model. This occurs because the slab pull, which drives slab bending, is less strongly supported by the viscous lower mantle (instantaneous model) than an impermeable lower boundary (dynamic model).

Figure 4.15 shows the effect of lower mantle viscosity, and plate width, on the rear slab channel shear stress (τ_{shear}) and dynamic pressure (P). Also presented is the integrated horizontal force (F_X : eq. 4.3) which, for all instantaneous models, is directed towards the upper plate, i.e., in the direction of trench advance. The positive τ_{shear} and normal deviatoric stress (not shown), which are proportional to the convergence rate (V_C), are typically much lower magnitude (< 35 MPa) than the negative/extensional P (50-150 MPa). Thus, the variability in P , as a function of the varied parameters, largely dictates the variability in F_X . Increasing the strength of the lower mantle ($\eta'_{660 \text{ km}}$) firstly reduces the rear subduction zone channel τ_{shear} (Fig. 4.15a), due to the increasing viscous resistance/drag on the slab tip reducing the rear slab V_C . Additionally, increasing $\eta'_{660 \text{ km}}$ reduces P , because increasing viscous support of the frontal subducting slab reduces the slab pull force available to be transmitted

through the center plate and to the rear subduction zone. Thus, as the strength of the lower mantle increases, the inter-slab coupling reduces. This suggests that the coupling forces computed in our dynamic models, with impermeable lower boundaries and so strong slab support, may be underestimating this coupling effect. As for all instantaneous computations, the magnitude of the trench-advance directed F_X is greater for the more evolved slab geometry (i.e., as the slabs become close). For the later slab geometry (Fig. 4.15: red points/lines), F_X varies from ~ 9.5 TN/m to ~ 4.5 TN/m as $\eta'_{660\text{ km}}$ increases from 1 to 1000.

The width of the subduction system has a comparably large effect on the horizontal coupling stress between the two slabs (Fig. 4.15d-f). While the plate width (w) has only a minor effect on the rear slab τ_{shear} , there is a strong decrease in the magnitude of P as w is increased. For the less evolved slab morphology (black points/lines), the rear slab coupling P decreases from ~ 120 MPa to ~ 60 MPa as w increases from 500 km to the full lateral extent of the domain (i.e., infinite plate width). This reduction in P for increasing w results in a strong reduction in the trench advance-directed F_X , for both stages of slab evolution. Our simplified instantaneous models therefore suggest that the width of the subduction system is an important component in dictating horizontal slab-slab coupling, with narrow slabs capable of transmitting approximately twice the slab pull force as very wide subduction systems. Shown in Figure 4.16, we also test the control that the lithospheric ($\eta'_{lith.} = \eta_{lith.} / \eta_{mant.}$) and crustal/channel strength ($\eta'_{crust} = \eta_{crust} / \eta_{mant.}$) have on horizontal slab coupling, as these parameters are poorly constrained, and likely exhibit significant variation between individual subduction systems. Both of these mechanical parameters have a large effect on slab coupling. Increasing $\eta'_{lith.}$ from 5 to 5000 (reference $\eta'_{lith.} = 500$) results in an increase in F_X by a factor of ~ 3 because the stiffer (mid-)plate acts as a more efficient stress guide for the forces transmitted from the frontal slab. Increasing η'_{crust} from 0.01 to 10 (reference $\eta'_{crust} = 1$) causes an increase in F_X by a factor of 2-3, because a stronger crust produces more efficient force transmission across the coupling channel. To summarize, our instantaneous models suggest that narrow subduction zones, strong lithospheric plates, and strong crustal channels promote strong horizontal coupling between two subduction zones dipping in the same direction.

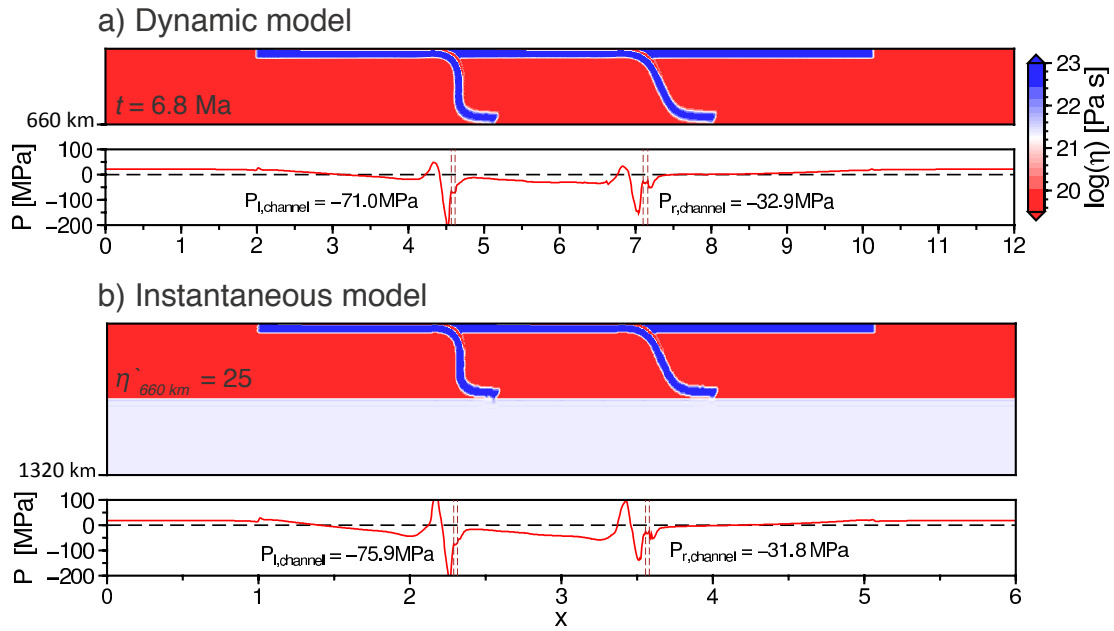


Figure 4.14: Comparison of pressure profiles for single time-step of dynamic model (a), and an example instantaneous model with slab geometry taken from the dynamic model (b). Shown for both models is the viscosity field through the mid-plane of the trench ($y = 0$), and pressure profiles taken at the depth for which the channel pressure is computed ($z = 50$ km). In addition, the channel pressures at the two subduction zones are displayed. All model parameters remain fixed (mechanical and rheological properties, slab/plate geometry, lateral box size), except the box depth is doubled for the instantaneous model (i.e., $z = 1320$ km, instead of 660 km), and a stronger lower mantle is added ($\eta_{lower\ mant.}/\eta_{upper\ mant.} = 25$).

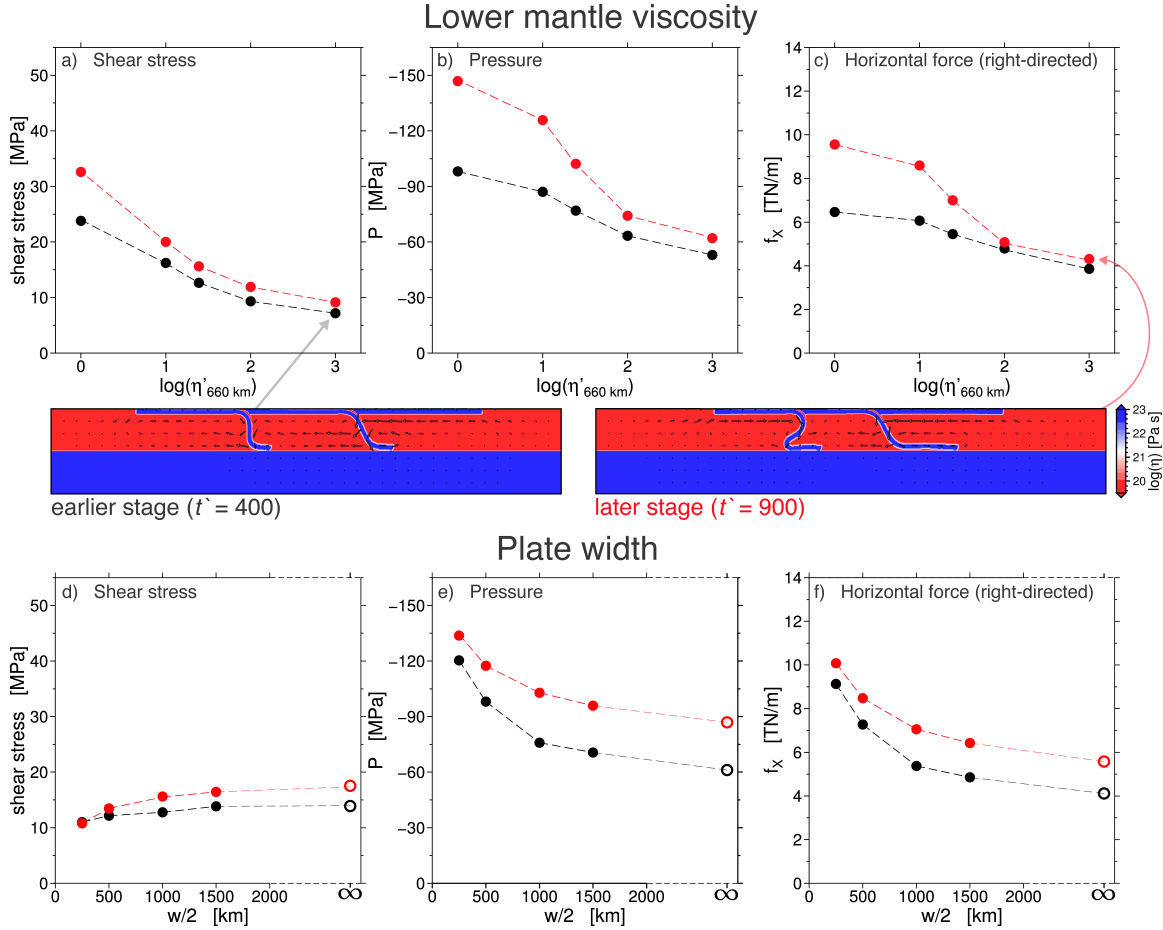


Figure 4.15: Rear slab coupling stresses, and associated horizontal force on the rear slab, as a function of lower mantle viscosity (a-c), and subducting and upper plate width (d-f) for instantaneous models with slab geometries taken from the same dip polarity dynamic model (Fig. 4.3a). The top row shows the rear slab coupling stress (a), the average channel pressure (b), and the horizontal force (positive = upper-plate directed) as a function of the lower/upper mantle viscosity contrast, for an earlier (black points: $t = 6.3$ Myr) and later (red points: $t = 18.3$ Myr) dynamic model geometry. The central row shows example instantaneous velocity solutions for both of the slab morphologies ($\eta_{\text{lower mant.}}/\eta_{\text{upper mant.}} = 1000$). The bottom row shows the same rear slab stress/force quantities plotted as a function of plate half-width (dynamic model $w/2 = 1000$ km). In addition to plates with finite widths, we present an infinite width/2-D case, where the subducting/upper plates span the lateral-extent (y) of the computational domain.)

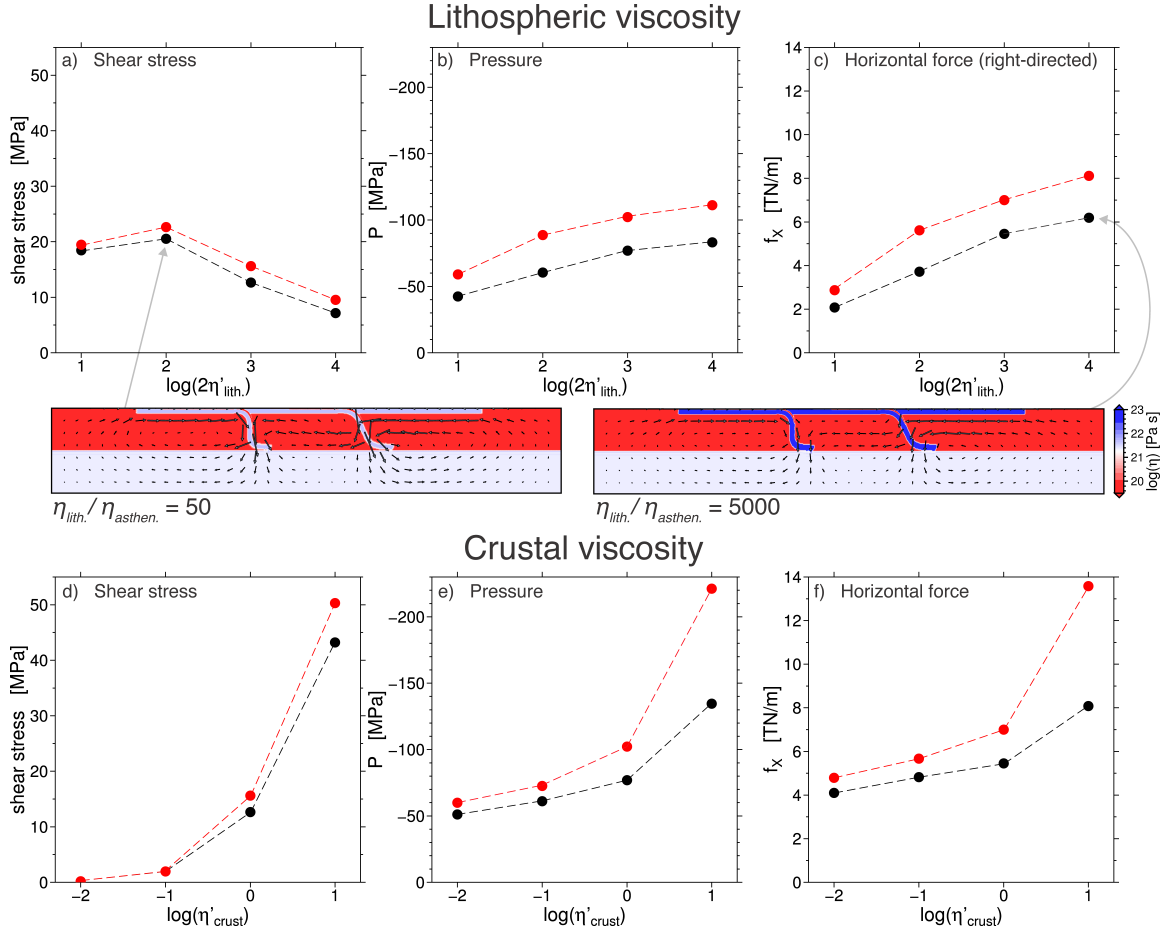


Figure 4.16: Rear slab coupling stresses, and associated horizontal force on the rear slab, as a function of lithospheric plate viscosity (a-c), and crustal layer viscosity (d-f) for instantaneous models with slab geometries taken from the same dip polarity dynamic model (Fig. 4.3a). The top row shows the rear slab coupling stress (a), the average channel pressure (b), and the horizontal force (positive = upper-plate directed) as a function of the lithosphere/upper mantle viscosity contrast, $\eta'_{lith.}$ (reference $\eta'_{lith.} = 500$), for an earlier (black points: $t = 6.3$ Myr) and later (red points: $t = 18.3$ Myr) dynamic model geometry. The central row shows example instantaneous velocity solutions for the earlier slab morphology with different lithospheric viscosities (left: $\eta'_{lith.} = 50$, right: $\eta'_{lith.} = 5000$). The bottom row shows the same rear slab stress/force quantities plotted as a function of crustal/upper mantle viscosity contrast, η'_{crust} (reference $\eta'_{crust} = 1$).

4.5 Conclusions

We have used dynamically consistent, numerical and analytical models to show that the kinematics of a subducting slab, particularly convergence rate and trench motion, can be significantly modified by the presence of an additional slab within reasonably close proximity (i.e., within a few 1000 kms). For instance, relative to single slab setups, all of our double slab systems exhibit accelerated plate convergence between the two side plates (e.g., Jagoutz et al., 2015). Plate kinematics are strongly variable, as a function of the slab dip direction combinations, due to i), the variable mobilities of the subducting/upper plates which in some cases are free to move, and in other cases attached to subducting/upper plates of the other subduction system and, ii), the wide variability of across-slab mantle pressure gradients (i.e., lifting forces/torques), relative to standard single slab models, which occur to the isolation of an asthenospheric compartment inbetween the two slabs. For plates with finite widths (i.e., 3-D models), we find that such double slab pressure fields, in conjunction with an additional slab “barrier” and often enhanced slab rollback, preferentially promote horizontal, rotational flow around the slab edges (i.e., toroidal flow). We therefore predict that toroidal flow in regions with multiple proximal subduction zone, potentially detectable with shear-wave splitting (e.g., Di Leo et al., 2012, 2014), to be more vigorous than that for single slab systems with finite trench widths (e.g., Kincaid and Griffiths, 2003; Piromallo et al., 2006).

Despite highly variable slab pressure gradients, we find that much of the variability in slab kinematics can be captured by our single slab analytical force balances (Royden et al., *in prep.*), for the slabs which have upper plates that are mobile (i.e., “free” upper plates). However, this correspondence breaks down for slabs with upper plates that are either fixed in place, or attached to another subducting plate. A pertinent example being the rear slab in the model where the slabs dip in the same direction. Here, slab pull from the slab in front exerts a horizontal force on the rear slab, which induces trench advance and an over-vertical slab (e.g., Izu-Bonin-Marianas: Čížková & Bina, 2015). A horizontal, trench advance-directed, force that is $\sim 10\%$ of the slab pull force (i.e., comparable to ridge push) is transmitted from the frontal to the rear subduction in our reference model. We use instantaneous computations to investigate the first order controls on horizontal slab coupling between two slabs that dip in the same direction. We find that the degree to which multiple slabs in this configuration can

“talk” to each other depends not only on how close the slabs are together, but is a sensitive function of the lateral extent of such subduction systems in 3-D, and the mechanical properties of the lithospheric plates and crustal channel. Thus, in order to accurately capture the slab coupling at a specific subduction zone settings, further work is required to develop geodynamic models which accurately capture the specific 3-D geometry, and lithospheric strengths (e.g., oceanic plate ages).

Outlook

We have used dynamic subduction models to investigate the dynamic controls on trench migration and slab bending, with a particular focus on the role of the overriding plate and asthenospheric rheology. While we have made significant inroads on developing a framework for understanding the how an overriding plate (chapter 1), a strong lower mantle (chapter 1), and power-law mantle rheology (chapter 3) can act to modify trench retreat rates relative to those observed in more simplified models, a more evolved modeling strategy that combines the modeling properties of these two studies would be an obvious next step to further refine our understanding of the dynamics, and elevate our works direct applicability to Earth. This would allow us to consider subduction-induced flow that can both flow through the lower mantle, and also around the edges of the slab, within one subduction system, which would facilitate a more complete assessment of the mantle flow style, in order to inform interpretations of shear wave splitting measurements, and assess the magnitude of the subduction-resisting forces acting on the slab. Deciphering the dynamic controls on the full subduction-induced flow field in a regional setting is a fundamental step in moving towards a deep understanding of the role of regional subduction within the global convective system, which constitutes our long-term goal.

Regarding our investigation into slab curvature (chapter 2), more refined observational constraints on lithospheric thickness would undoubtedly improve the robustness of our curvature correlations. However, it is not immediately obvious where such refinements will come from on a global scale, aside from iterative improvements in models of seismic tomography. While we are confident in our assertion that overriding plate structure exerts a strong control on slab curvature, initial work in chapter 1 suggests that overriding plate density may also be of importance. Therefore, assessing whether there is strong correlation between slab curvature and a metric for overriding plate density (e.g., geological province) is also of interest. Another aspect that we touched on during this study is the implications that this work has for plate bending energetics. Particularly, an important question is whether this additional, overriding plate control on slab bending induces more or less bending dissipation, relative to previous parameterizations which neglect the overriding plate. The relative importance of bending dissipation in mantle convection remains contentious, and so deciphering this is of great interest. We would like to explore this further using analytical models, as in Buffett and Becker (2012), within which we additionally include a parameterization of the

pressure gradients associated with the presence, and mechanical properties, of the overriding plate.

Implicitly assumed in the first three, regional subduction chapters, is that individual subducting slabs are somewhat isolated from global tectonics, and the global convective system. Chapter 4 illustrates an extreme case of where this assumption breaks down, in that here we consider multiple, interacting subduction zones. This is a small step along the path of understanding how the global subduction system organizes and evolves through time. As done in this study, we believe that augmenting our dynamically consistent numerical models with analytical models (e.g. Royden and Husson, 2006; Jagoutz et al., 2015) is a powerful strategy for distilling the key, yet often confusticated, physics of subduction, and we will continue along this path. Once benchmarked against the numerical models, as is currently underway, such analytic formulations can be scaled up to larger systems (e.g., multiple subduction systems) with minimal computational cost, and the simplified physics (i.e., subduction “rules”) can be used as an additional constraint on the validity of plate tectonic reconstructions. In the numerical models, a more refined rheological implementation of the subduction boundary in geodynamic models, as is currently the subject of much effort in research groups focused on modeling the mega-thrust cycle, is required for a more rigorous assessment of the efficiency of force transmission between adjacent subducting slabs.

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